2026 SUMMER SCHOOL
Transforming Technology, Modern Practice, and Clinical Impact
JUNE 16–20 | UNIVERSITY OF MICHIGAN

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AAPM
SUMMER SCHOOL
EDUCATING MEDICAL PHYSICISTS WORLDWIDE

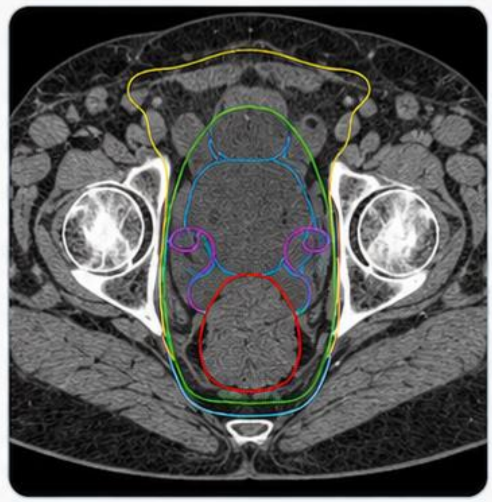
PATIENT CASE FILE • PEL-01

Mr. Blaise McFillington

“The rectum has opinions, and they change daily.”

1 PATIENT SNAPSHOT

72-year-old male	Prostate bed	Post-prostatectomy	Fraction 14
Prescription: 66 Gy / 33 fx	Prep in use: Bladder-filling + rectal-emptying	Today's wildcard: Rectal filling & bowel gas vary	Allowable time: 30 minutes



2 WHY HE'S ON THE ADAPTIVE COUCH

His prostate-bed target drifts with bladder and rectal/bowel filling. Daily adaptation lets the team hold a tighter CTV-to-PTV margin and keep rectal dose down — as long as someone makes the daily call about what wins when coverage and rectal sparing collide.

3 THE ROOM TODAY

Mature daily-adaptive pelvic program. You (physicist) are on the console, the physician is remote but reachable by phone, and two RTTs round out the team. Team-based decisions are encouraged — nobody is a lone hero here.

4 THE CLOCK IS REAL

You get 30 minutes. The slot is back to back — run over and you push the next patient late AND give his own bladder time to fill and rectum time to re-load before delivery. Slow is not safe here.

SAFETY IS ADAPTIVE. TEAMWORK MAKES IT EFFECTIVE.

COMMUNICATE · IDENTIFY RISK · EVALUATE · MITIGATE · LEARN & IMPROVE

»» TURN THE PAGE TO MEET HIS SIX DECISIONS. ««

1 REFERENCE PLANNING

PLAYER CARD

PEL-01 | Mr. Blaise McFillington

Let's Adapt!

Scenario: Undefined Daily Priority

Clinical Situation: Before the fraction starts, your team reviews the reference directive for Mr. McFillington. It lists prescription, CTV-to-PTV margins, prostate-bed target goals, and rectal, bladder, and bowel goals. What it does not state is what should happen on a day when rectal filling and bowel gas push the anterior rectal wall closer to the prostate-bed CTV. In other words, the directive tells you the goals, but it does not define the daily tradeoff hierarchy when coverage and rectal sparing compete. You review the physician notes but do not find anything about goal priorities.

The Reference plan meets the listed goals, so there is no obvious failure yet. Still, the team recognizes that on a fuller-rectum day, plan selection could hinge on whether the clinic prioritizes strict prostate-bed CTV intent, improved rectal sparing, or physician review of the tradeoff.

Decision Prompt: Do you proceed today, and what minimum documentation or escalation is needed before this ambiguity reaches plan selection?

Choices

A. Hold today's fraction until a physician-signed priority hierarchy is added.

B. Proceed on team judgment; goals are met in the reference plan and no conflict exists yet.

C. Proceed, document the assumption, and have the physician confirm the hierarchy before next fraction.

D. Proceed, note the priority assumption used, and flag the directive for clarification before next fraction.

Decision Log: Why did your team choose this option?

Notes:

2 ON-SESSION IMAGING

PLAYER CARD

Let's Adapt!

PEL-01 | Mr. Blaise McFillington

Scenario: Good Image, Wrong Place

Clinical Situation: On-session CBCT for Mr. McFillington looks good on a quick overview and the team accepts the image.

When the team looks closely, though, bowel gas is distorting the anterior rectal wall in exactly the slices where that wall abuts the high-dose region. Global registration looks reasonable but the one area you cannot fully trust is the interface that will likely drive the rectal-dose decision.

The team has to decide whether an image that is acceptable overall is acceptable for the specific decision it now has to support.

Decision Prompt: Is a globally acceptable image good enough here, or does the degraded rectal interface need to be addressed first?

Choices

A. Restart the adaptive session and re-image after rectal-emptying prep.

B. Accept but manually review the rectal interface and note the image limitation there.

C. Accept the image; global quality is fine and the plan should calculate without issue.

D. Attempt brief bowel-gas mitigation on table, reacquire the CBCT, then review the interface.

Decision Log: Why did your team choose this option?

Notes:

3 OAR CONTOURING

PLAYER CARD

PEL-01 | Mr. Blaise McFillington

Let's Adapt!

Scenario: The Contour Looks Mostly Right

Clinical Situation: The generated rectum contour for Mr. McFillington is correct over most of its length. However, on the high-dose slices flagged during image review, the anterior rectal wall appears under-segmented near the prostate-bed CTV. The contour looks acceptable if reviewed quickly in the axial plane, but the area that is least reliable is also the area most likely to matter later for target/OAR tradeoff and plan selection. At this step, the team is not yet looking at the final plan DVH. The decision is whether the rectum contour is accurate enough to approve as an OAR/influencer for downstream contour propagation, optimization, and later plan evaluation.

Decision Prompt: Do you approve the mostly correct rectum contour, or correct the anterior-wall region before allowing the adaptive workflow to proceed?

Choices

A. Accept the generated rectum contour because it is accurate over most slices and the workflow is time-sensitive.

B. Fully review and correct the rectum across all involved slices before approving the OAR contour set.

C. Edit the anterior rectal-wall slices closest to the prostate-bed CTV, then approve the contour set for downstream planning.

D. Request a second-reviewer check before approving the rectum contour, because the uncertainty is in the decision-driving region.

Decision Log: Why did your team choose this option?

Notes:

4 TARGET CONTOURING

PLAYER CARD

Let's Adapt!

PEL-01 | Mr. Blaise McFillington

Scenario: Where Does the Bed End?

Clinical Situation: Target propagation for Mr. McFillington places the prostate-bed CTV well inferiorly, but the superior border near the bladder neck is ambiguous. The reference plan never fixed whether the superior extent should track today's anatomy or hold to the original reference intent, so the propagated border looks plausible while quietly varying from fraction to fraction.

The PTV still covers the region, which makes it easy to accept the contour without resolving what the superior target intent actually is.

Decision Prompt: Do you accept the propagated superior border, anchor it to a defined intent, or escalate before planning?

Choices

A. Reconcile the superior border to the reference image intent and document the rule.

B. Hold for physician target review before planning.

C. Accept propagation as-is; the PTV covers the superior region.

D. Match the superior border to the most recent prior fraction and note the convention.

Decision Log: Why did your team choose this option?

Notes:

5 PLAN SELECTION

PLAYER CARD

PEL-01 | Mr. Blaise McFillington

Let’s Adapt!

Scenario: Both Plans Pass - Now What?

Clinical Situation: Mr. McFillington's adapted plan spares rectum but trims superior bed coverage; the scheduled/on-session recalculated plan over-covers the bed but raises rectal dose. Both meet the listed coverage goals, so the choice between them comes down to a tie-breaker the directive never defined.

The adapted plan's rectal numbers look better on screen, which makes selecting it feel like the natural 'adaptive' choice even though the reason for choosing it has not been written down anywhere.

Decision Prompt: On what documented basis do you break the tie between the scheduled and adapted plans?

Choices

A. Call the physician to choose and co-sign the selection rationale before delivery.

B. Use the team’s documented tie-breaker: preserve superior prostate-bed CTV intent unless rectal dose exceeds the action threshold, then record the rationale.

C. Select the plan that better preserves superior prostate-bed intent from prior fractions, and document why.

D. Select the adapted plan because rectal metrics look better; it feels like the natural adaptive choice.

Decision Log: Why did your team choose this option?

Notes:

6 DELIVERY / CONFIRMATION IMAGING

PLAYER CARD

Let’s Adapt!

PEL-01 | Mr. Blaise McFillington

Scenario: A Small, In-Tolerance Drift

Clinical Situation: Confirmation imaging for Mr. McFillington shows the rectum closest to the target has filled slightly since adaptation resulting in the rectal wall being closer to the high dose area. On inspection, the anterior-wall position has moved by a small amount (<2 mm) and that sits clearly within the program's predefined confirmation-drift threshold and PTV. Time in the slot is nearly used up. Because a threshold exists and today's change is clearly inside it, the team has a documented basis to proceed.

Decision Prompt: With the drift clearly inside your documented threshold, do you treat, or do more work the threshold does not require?

Choices

A. Restart the adaptive session on current anatomy.

B. Apply a positioning correction, then re-check before treating.

C. Re-adapt the plan to today's slightly fuller rectum to be safe.

D. Treat now: the measured drift is clearly within the documented threshold; record the check.

Decision Log: Why did your team choose this option?

Notes:

PEL-01 Explanation Summary

Mr. Blaise McFillington • Prostate Bed / Pelvic CTgART

Patient Context

Clinical Summary	72M, prostate bed (post-prostatectomy), 66 Gy/33 fx, fraction 14. Bladder-filling and rectal-emptying prep in use; rectal filling and bowel gas vary across fractions.
Clinic / Coverage Model	Mature daily-adaptive pelvic program. Physicist on console, physician remote with phone availability, two RTTs. Team-based decisions encouraged.
Adaptive Rationale	Prostate-bed CTV position tracks bladder filling and rectal/bowel filling; daily adaptation supports a reduced CTV-to-PTV margin and lower rectal dose.
Allowable Time	30 minutes
Time Consequence	The slot is back-to-back; overage pushes the next patient later and risks bladder-filling drift and rectal re-filling for this patient before delivery.

Core Lesson

When the treatment directive does not define daily adaptation intent, every downstream decision inherits ambiguity. Teams need a documented tie-breaker when target intent and OAR sparing compete.

Overall Handling Approach

Make uncertainty visible, verify the anatomy that drives the decision, use documented thresholds or priority rules, and record the rationale before delivery.

1. Reference Planning — Undefined Daily Priority

Why it matters: The reference/treatment directive issue sets the decision rule before the patient is deep into the adaptive session.
General handling approach: Apply or define the missing rule early, document the working basis, and escalate only when the ambiguity affects today’s treatment decision.

2. On-Session Imaging — Good Image, Wrong Place

Why it matters: Image adequacy must be judged against the specific decision the image must support, not only global image quality.
General handling approach: Review the decision-driving anatomy directly. Reacquire or mitigate only when the limitation prevents reliable contouring or plan evaluation.

3. OAR Contouring — The Contour Looks Mostly Right

Why it matters: At this stage the team is approving OAR/influencer anatomy for downstream optimization and plan comparison, not relying on a final DVH.
General handling approach: Focus edits on the OAR/influencer regions that can change optimization, dose calculation, or plan comparison; use second review when uncertainty remains clinically material.

4. Target Contouring — Where Does the Bed End?

Why it matters: A plausible propagated target does not by itself prove that clinical target intent has been preserved.
General handling approach: Reconcile propagated targets to reference intent, diagnostic anatomy, prior fractions, or a documented rule before planning.

5. Plan Selection — Both Plans Pass - Now What?

Why it matters: Scheduled/on-session recalculated and adapted plans can both appear acceptable while resolving different clinical risks.
General handling approach: Select based on documented target/OAR priorities, composite checks, sensitivity tests, and secondary dose review rather than the most attractive single metric.

6. Delivery / Confirmation Imaging — A Small, In-Tolerance Drift

Why it matters: Confirmation imaging should be governed by predefined thresholds, stability checks, or re-verification rather than visual impression or time pressure alone.
General handling approach: Treat only when the approved geometry remains within threshold or can be restored and re-verified; otherwise re-adapt, restart, delay, or cancel according to policy.

PEL-01 Scoring Sheet

Mr. Blaise McFillington • Allowable time: 30 minutes

Workflow step	Choice	Safety penalty	Time penalty	Whammy?	Notes
1. Reference Planning	____	____	____	☐ Yes ☐ No	
2. On-Session Imaging	____	____	____	☐ Yes ☐ No	
3. OAR Contouring	____	____	____	☐ Yes ☐ No	
4. Target Contouring	____	____	____	☐ Yes ☐ No	
5. Plan Selection	____	____	____	☐ Yes ☐ No	
6. Delivery / Confirmation Imaging	____	____	____	☐ Yes ☐ No	

Total safety penalties	__________	**Total time used**	__________ min
Time overage	max(0, time - allowable) = ________	**Whammy resolution**	☐ +10 min choose again ☐ +15 safety
Final score	100 - safety penalties - time overage = ________		
Team	________________________	**Facilitator**	________________________

Scoring reminder: Final score = 100 - safety penalties - time overage. If a whammy is selected, add 10 minutes and choose again, or keep the choice and add 15 safety penalty.

PEL-01 Answer Key • 1. Reference Planning

Mr. Blaise McFillington

Scenario: Undefined Daily Priority

Choice A: Hold today's fraction until a physician-signed priority hierarchy is added.

Scoring: Safety 1; Time 13 min; no cascade; Whammy: No; S/O/D/RPN 2/2/1/4

Explanation: Safest in principle but over-escalates a treatable fraction; the delay lands on this patient (bladder/rectal drift) and the next.

Penalty rationale: Very high time; maximal escalation is not free.

Mitigation concept: Thresholds separating 'clarify before next fraction' from 'stop now'.

Choice B: Proceed on team judgment; goals are met in the reference plan and no conflict exists yet.

Scoring: Safety 6; Time 1 min; no cascade; Whammy: No; S/O/D/RPN 3/3/3/27

Explanation: Reasonable on a day with no active conflict: goals are met and nothing forces the tradeoff yet. Residual risk: the gap resurfaces unprepared on a full-rectum day.

Penalty rationale: Low time, moderate safety: defensible today but leaves the root ambiguity.

Mitigation concept: Predefined priority hierarchy in the directive.

Choice C: Proceed, document the assumption, and have the physician confirm the hierarchy before next fraction.

Scoring: Safety 2; Time 5 min; no cascade; Whammy: No; S/O/D/RPN 2/2/1/4

Explanation: Strong, but routing physician confirmation mid-session costs more time for marginal added safety over B today.

Penalty rationale: Higher time for confirmation B already sets in motion.

Mitigation concept: Physician-confirmed hierarchy before next fraction.

Choice D: Proceed, note the priority assumption used, and flag the directive for clarification before next fraction.

Scoring: Safety 2; Time 2 min; no cascade; Whammy: No; S/O/D/RPN 2/2/2/8

Explanation: Best: treats today, records the working assumption so the decision is reconstructable, and starts the fix without halting care.

Penalty rationale: Lowest combined penalty: small time cost buys traceability.

Mitigation concept: On-session documentation + clarification loop.

PEL-01 Answer Key • 2. On-Session Imaging

Mr. Blaise McFillington

Scenario: Good Image, Wrong Place

Choice A: Restart the adaptive session and re-image after rectal-emptying prep.

Scoring: Safety 2; Time 12 min; +2 min cascade; Whammy: No; S/O/D/RPN 2/2/2/8

Explanation: Safest image but a full restart for one localized artifact is disproportionate and risks the slot.

Penalty rationale: Highest time; reserve for non-mitigable degradation.

Mitigation concept: Graded reacquisition before full restart.

Choice B: Accept but manually review the rectal interface and note the image limitation there.

Scoring: Safety 3; Time 3 min; no cascade; Whammy: No; S/O/D/RPN 2/3/2/12

Explanation: Best: keeps the session moving while explicitly bounding the unreliable region; targeted scrutiny fits a localized artifact.

Penalty rationale: Low combined penalty: cheap insurance where it matters.

Mitigation concept: Targeted manual review of the degraded interface.

Choice C: Accept the image; global quality is fine and the plan should calculate without issue.

Scoring: Safety 10; Time 1 min; +8 safety cascade; Whammy: No; S/O/D/RPN 4/4/5/80

Explanation: Tempting because the image looks fine globally, but accepting it sends an unreliable rectal wall into contouring undetected.

Penalty rationale: High safety, minimal time: artifact propagates into the OAR decision.

Mitigation concept: Decision-region image-adequacy check, not just global.

Choice D: Attempt brief bowel-gas mitigation on table, reacquire the CBCT, then review the interface.

Scoring: Safety 3; Time 8 min; no cascade; Whammy: No; S/O/D/RPN 2/2/2/8

Explanation: Reasonable but reacquiring for a localized, often-transient gas artifact usually costs more time than focused review needs.

Penalty rationale: More time for an image B may not require.

Mitigation concept: Gas mitigation + targeted reacquisition.

Ad\dennisstanley Goworkout4them!

PEL-01 Answer Key • 3. OAR Contouring

Mr. Blaise McFillington

Scenario: The Contour Looks Mostly Right

Choice A: Accept the generated rectum contour because it is accurate over most slices and the workflow is time-sensitive.

Scoring: Safety 7; Time 1 min; +2 safety cascade, +3 min cascade; Whammy: No; S/O/D/RPN 3/3/3/27

Explanation: This is plausible under time pressure because the contour is mostly correct and the team is not yet evaluating a final DVH. The risk is that the missed anterior rectal wall is exactly the region most likely to influence the downstream target/OAR tradeoff.

Penalty rationale: High safety penalty because the decision-driving OAR region remains incomplete. Low time penalty because the team proceeds quickly, but the unresolved contour issue creates downstream risk and review burden.

Mitigation concept: Region-prioritized OAR contour review

Choice B: Fully review and correct the rectum across all involved slices before approving the OAR contour set.

Scoring: Safety 2; Time 7 min; no cascade; Whammy: No; S/O/D/RPN 3/1/2/6

Explanation: This is safe and defensible, especially if the contour error may extend beyond the obvious anterior-wall region. The tradeoff is time: a full rectum review may not be necessary if the clinically relevant uncertainty is localized.

Penalty rationale: Low safety penalty because the contour is comprehensively reviewed before approval. Time penalty is higher because full OAR review slows the online adaptive session.

Mitigation concept: Comprehensive OAR contour review

Choice C: Edit the anterior rectal-wall slices closest to the prostate-bed CTV, then approve the contour set for downstream planning.

Scoring: Safety 1; Time 4 min; no cascade; Whammy: No; S/O/D/RPN 3/1/2/6

Explanation: This is the best balanced response. It targets the clinically relevant contour defect without spending time on low-impact slices. The team corrects the region that could affect downstream optimization, OAR evaluation, and scheduled-versus-adapted plan selection.

Penalty rationale: Very low safety penalty because the decision-driving anatomy is corrected. Moderate time penalty reflects focused contour editing while preserving workflow efficiency.

Mitigation concept: Focused high-risk-slice correction

Choice D: Request a second-reviewer check before approving the rectum contour, because the uncertainty is in the decision-driving region.

Scoring: Safety 2; Time 6 min; +1 min cascade; Whammy: No; S/O/D/RPN 3/1/2/6

Explanation: This is reasonable if the team is uncertain whether the anterior rectal wall has been correctly identified or if local policy requires escalation for contour uncertainty near the target. It improves confidence but may cost more time than a focused edit by the primary reviewer.

Penalty rationale: Low safety penalty because the uncertainty is escalated before planning. Time penalty is higher because it requires another reviewer and may delay plan generation.

Mitigation concept: Second-reviewer escalation threshold

PEL-01 Answer Key • 4. Target Contouring

Mr. Blaise McFillington

Scenario: Where Does the Bed End?

Choice A: Reconcile the superior border to the reference image intent and document the rule.

Scoring: Safety 2; Time 5 min; no cascade; Whammy: No; S/O/D/RPN 2/2/2/8

Explanation: Best: anchors to reference intent and records the rule, giving the lowest residual target uncertainty.

Penalty rationale: Modest time for a defensible, reconstructable target.

Mitigation concept: Reconcile-to-reference + documentation.

Choice B: Hold for physician target review before planning.

Scoring: Safety 3; Time 9 min; +2 min cascade; Whammy: No; S/O/D/RPN 2/2/2/8

Explanation: Justified only if intent is genuinely unknown; routine bed borders should be protocolized, not escalated each fraction.

Penalty rationale: High time; daily routing is not scalable.

Mitigation concept: Protocolized target intent to reduce escalations.

Choice C: Accept propagation as-is; the PTV covers the superior region.

Scoring: Safety 8; Time 1 min; +6 safety cascade; Whammy: No; S/O/D/RPN 4/3/4/48

Explanation: Coverage masks an undefined intent; the superior border drifts undetected. Tempting because the PTV looks fine.

Penalty rationale: Moderate-high safety, low time: ambiguity persists into selection.

Mitigation concept: Define superior-border propagation intent in the directive.

Choice D: Match the superior border to the most recent prior fraction and note the convention.

Scoring: Safety 4; Time 3 min; no cascade; Whammy: No; S/O/D/RPN 3/3/3/27

Explanation: Reasonable stopgap: consistency with the prior fraction is defensible if recorded, though it does not anchor to original intent.

Penalty rationale: Low-moderate: reproducible without halting.

Mitigation concept: Documented consistency convention.

PEL-01 Answer Key • 5. Plan Selection

Mr. Blaise McFillington

Scenario: Both Plans Pass - Now What?

Choice A: Call the physician to choose and co-sign the selection rationale before delivery.

Scoring: Safety 2; Time 6 min; +1 min cascade; Whammy: No; S/O/D/RPN 3/1/2/6

Explanation: This is safe and defensible because the treatment directive did not define the tie-breaker between superior prostate-bed intent and improved rectal sparing. Physician clarification before delivery creates a traceable clinical decision, but it relies on real-time physician availability rather than a predefined plan-selection process.

Penalty rationale: Low safety penalty because the ambiguity is escalated before treatment. Time penalty is moderate because the team must pause, contact the physician, explain the tradeoff, and document the decision while the patient remains on the table.

Mitigation concept: Real-time clinical intent clarification

Choice B: Use the team's documented tie-breaker: preserve superior prostate-bed CTV intent unless rectal dose exceeds the action threshold, then record the rationale.

Scoring: Safety 1; Time 3 min; no cascade; Whammy: No; S/O/D/RPN 3/1/1/3

Explanation: This is the best balanced option. The team uses a documented, clinically meaningful rule rather than choosing based on whichever metric looks better. It preserves prostate-bed CTV intent unless the rectal dose crosses a predefined threshold that justifies changing the tradeoff.

Penalty rationale: Very low safety penalty because the choice is based on a documented decision rule that includes both target intent and OAR action threshold. Time penalty is low-to-moderate because the team still needs to verify the threshold and record the rationale.

Mitigation concept: Documented plan-selection tie-breaker

Choice C: Select the plan that better preserves superior prostate-bed intent from prior fractions, and document why.

Scoring: Safety 2; Time 4 min; no cascade; Whammy: No; S/O/D/RPN 3/1/2/6

Explanation: This is clinically reasonable because maintaining superior prostate-bed intent across fractions may be more important than improving rectal metrics on a single day. The limitation is that the choice does not explicitly state whether rectal dose crossed a predefined action threshold.

Penalty rationale: Low safety penalty because target intent is protected and the rationale is documented. Time penalty is moderate because the team must compare prior-fraction intent, evaluate the tradeoff, and document why rectal sparing was not the deciding factor.

Mitigation concept: Prior-fraction intent consistency

Choice D: Select the adapted plan because rectal metrics look better; it feels like the natural adaptive choice.

Scoring: Safety 8; Time 1 min; +3 safety cascade, +2 min cascade; Whammy: YES - Add 10 minutes and choose again, or keep this choice and add 15 safety penalty.; S/O/D/RPN 4/3/3/36

Explanation: This is tempting because the adapted plan improves the visible rectal metrics and feels aligned with the purpose of daily adaptation. The risk is that the team is selecting based on the adaptive label and a favorable OAR metric without a documented tie-breaker, while accepting reduced superior prostate-bed coverage relative to prior intent.

Penalty rationale: Safety penalty is high because favorable rectal metrics are allowed to substitute for documented clinical intent. Time penalty is low because the team proceeds quickly, but the unresolved rationale creates downstream risk at confirmation and post-fraction review.

Mitigation concept: Do not equate adapted with automatically better

PEL-01 Answer Key • 6. Delivery / Confirmation Imaging

Mr. Blaise McFillington

Scenario: A Small, In-Tolerance Drift

Choice A: Restart the adaptive session on current anatomy.

Scoring: Safety 6; Time 12 min; no cascade; Whammy: No; S/O/D/RPN 3/2/2/12

Explanation: Clear over-escalation: a full restart for a small, within-tolerance change over-runs the slot and is hard to justify when the threshold is met.

Penalty rationale: Highest time: disproportionate to a within-tolerance finding.

Mitigation concept: Restart only when drift exceeds threshold.

Choice B: Apply a positioning correction, then re-check before treating.

Scoring: Safety 3; Time 4 min; no cascade; Whammy: No; S/O/D/RPN 3/3/3/27

Explanation: Reasonable but more than needed: a reassessment and positioning correction is sensible when drift is borderline, but here it is clearly within tolerance, so the extra step mostly costs time.

Penalty rationale: Low-moderate: proportionate only if drift were borderline.

Mitigation concept: Reserve correction for borderline drift.

Choice C: Re-adapt the plan to today's slightly fuller rectum to be safe.

Scoring: Safety 5; Time 8 min; no cascade; Whammy: No; S/O/D/RPN 3/3/3/27

Explanation: Over-reaction: re-adapting for a change that is clearly within the documented threshold spends significant time and re-opens contouring/selection risk for no safety gain.

Penalty rationale: Moderate-high time: unnecessary re-work for within-tolerance drift.

Mitigation concept: Do not re-adapt for within-tolerance change.

Choice D: Treat now: the measured drift is clearly within the documented threshold; record the check.

Scoring: Safety 1; Time 2 min; no cascade; Whammy: No; S/O/D/RPN 2/2/2/8

Explanation: Best here: a predefined threshold exists and the measured drift is clearly within it, so treating and recording the check is the correct, efficient call -- over-reacting to a within-tolerance change wastes the slot and patient tolerance.

Penalty rationale: Lowest combined penalty: thresholds are met and documented.

Mitigation concept: Treat-through when drift is clearly within a documented threshold.

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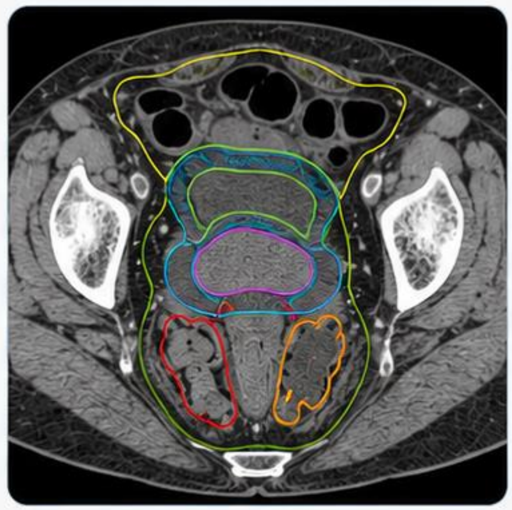
PATIENT CASE FILE • PEL-02

Ms. Nora Nodalboost

“At the cuff-nodal seam, small gaps can make big problems.”

1 PATIENT SNAPSHOT

59-year-old female	Post-op endometrial cancer	Fraction 16	Nodal CTV + vaginal-cuff boost
Prescription: 45 Gy / 25 fx + cuff boost	Prep in use: Bladder/bowel prep	Today’s wildcard: Sigmoid is limiting OAR	Allowable time: 30 minutes



2 WHY SHE’S ON THE ADAPTIVE COUCH

Bladder and bowel filling change the vaginal cuff and nodal relationship. Daily adaptation helps preserve cuff coverage while controlling sigmoid dose — but seam logic must be explicit.

3 THE ROOM TODAY

Busy general adaptive program. Rotating physicists, physician covering two units, and two RTTs with one newer to adaptive workflows. Cross-checking between team members is part of the routine.

4 THE CLOCK IS REAL

You get 30 minutes. Run over and you compress the afternoon nodal cohort and increase the chance the next bladder-prep window is missed.

SAFETY IS ADAPTIVE. TEAMWORK MAKES IT EFFECTIVE.

COMMUNICATE · IDENTIFY RISK · EVALUATE · MITIGATE · LEARN & IMPROVE

TURN THE PAGE TO MEET HER SIX DECISIONS.

1 REFERENCE PLANNING

PLAYER CARD

PEL-02 | Ms. Nora Nodalboost

Let's Adapt!

Scenario: A Seam With No Rule

Clinical Situation: Ms. Nodalboost is on a post-op endometrial course with a nodal CTV plus a vaginal-cuff boost region.

The reference plan propagates the high-dose cuff region deformably but propagates the elective nodal CTV rigidly, and it never says how to reconcile the seam between them when bladder and bowel filling move the anatomy.

On stable days the two volumes nest cleanly and the mixed scheme looks settled. On a filling-change day, the boundary between them can gap or overlap, and nothing in the workflow currently prompts anyone to check that seam.

Decision Prompt: Do you proceed on the mixed scheme as designed, add a seam check, or revise the reference approach first?

Choices

A. Add a per-fraction cuff/nodal seam check this session and request a reference amendment defining reconciliation.

B. Accept the mixed scheme as designed; it has worked on prior fractions.

C. Stop and revise the reference plan offline to use one consistent propagation method before treating.

D. Proceed today and request a reference-plan amendment defining seam reconciliation before next fraction.

Decision Log: Why did your team choose this option?

Notes:

2 ON-SESSION IMAGING

PLAYER CARD

Let’s Adapt!

PEL-02 | Ms. Nora Nodalboost

Scenario: Clean Where It Doesn't Count

Clinical Situation: On-session CBCT for Ms. Nodalboost images the cuff region cleanly, but moderate bowel gas crosses the elective nodal CTV region - the volume that is already the more fragile one but it is also rigidly propagated. The team's main attention is naturally on the cuff, which looks fine.

The degraded region and the weakest part of the propagation scheme are the same place, so an image that looks acceptable where the team is looking is least reliable.

Decision Prompt: Do you accept the image, flag the nodal region as lower-confidence for review, or re-image to try and improve the image where confidence is lowest?

Choices

A. Attempt bowel-gas mitigation and reacquire, focusing on the nodal region.

B. Restart the session after bowel-prep.

C. Accept but flag the nodal region as lower-confidence for the propagation review.

D. Accept; the cuff (the main target) images well.

Decision Log: Why did your team choose this option?

Notes:

3 OAR CONTOURING

PLAYER CARD

PEL-02 | Ms. Nora Nodalboost

Let's Adapt!

Scenario: A Limiting OAR You Can't Quite See

Clinical Situation: The generated sigmoid contour for Ms. Nodalboost looks plausible overall, but its course through the cuff-to-nodal seam is uncertain on the gassy slices. This is the same region where the cuff boost volume and elective nodal CTV may gap or overlap because they are being propagated using different methods.

At this step, the team is not yet judging the final sigmoid DVH. The decision is whether the sigmoid contour is reliable enough to approve as an OAR/influencer for downstream propagation, optimization, and later plan comparison. The contour uncertainty is localized, but it sits exactly where sigmoid proximity could influence the target/OAR tradeoff.

Decision Prompt: Do you approve the mostly plausible sigmoid contour, or resolve the uncertain seam-region slices before allowing the adaptive workflow to proceed?

Choices

A. Correct the sigmoid through the cuff-to-nodal seam slices before approving the OAR contour set.

B. Correct the sigmoid through the seam slices and cross-check its course against the prior fraction for consistency.

C. Request a second-reviewer check before approving the sigmoid contour, because the uncertainty is in the decision-driving region.

D. Accept the generated sigmoid contour because its overall course looks reasonable and the session is time-sensitive.

Decision Log: Why did your team choose this option?

Notes:

4 TARGET CONTOURING

PLAYER CARD

PEL-02 | Ms. Nora Nodalboost

Let’s Adapt!

Scenario: A Gap Only the Boundary Sees

Clinical Situation: At the cuff-to-nodal transition, the deformed cuff region and the rigid nodal CTV leave a small unintended gap over part of Ms. Nodalboost's target. Each volume passes its own coverage goal, so no single PTV DVH flags the problem. The gap lives only at the boundary between the two volumes, which means it is invisible to per-structure metrics and will stay invisible unless someone looks directly at the seam.

Decision Prompt: Do you accept two passing volumes, bridge the seam, or add a composite check before planning?

Choices

A. Manually bridge the seam gap and recompute combined coverage.

B. Accept; both volumes meet their individual coverage goals.

C. Bridge the gap and add a standing composite-coverage check to the workflow.

D. Return to contour editing to re-derive both volumes with one method, then regenerate the plan.

Decision Log: Why did your team choose this option?

Notes:

5 PLAN SELECTION

PLAYER CARD

Let’s Adapt!

PEL-02 | Ms. Nora Nodalboost

Scenario: Sigmoid Numbers vs. an Unseen Gap

Clinical Situation: Ms. Nodalboost's adapted plan covers the bridged seam but pushes the sigmoid toward its limit; the scheduled/on-session recalculated plan spares the sigmoid but reopens the seam gap. Both pass the listed goals.

The scheduled plan's sigmoid numbers look the safest on screen, which makes it the easy pick if the team does not keep the reopened, hard-to-see seam gap in mind while deciding.

Decision Prompt: How do you weigh a sigmoid dose you can see against a coverage gap you cannot, and what do you record?

Choices

A. Choose the adapted plan and document that seam coverage was the deciding factor.

B. Choose the scheduled plan; its sigmoid numbers look safest.

C. Choose based on the composite-check result and record both the coverage and sigmoid rationale.

D. Return to contour editing and regenerate the adapted plan if neither resolves both; revise offline if still unacceptable.

Decision Log: Why did your team choose this option?

Notes:

6 DELIVERY / CONFIRMATION IMAGING

PLAYER CARD

Let’s Adapt!

PEL-02 | Ms. Nora Nodalboost

Scenario: Late Sigmoid Shift

Clinical Situation: Confirmation imaging for Ms. Nodalboost shows bowel has shifted since approval, slightly changing the sigmoid-to-target relationship near the seam. This time slot is tight, and re-adapting risks running over your time and delaying other patients.

The shift looks small and the plan is already approved, so treating immediately is tempting - but the sigmoid is the limiting OAR, and whether this shift is within tolerance has not yet been checked against a defined threshold.

Decision Prompt: Do you treat on the small-looking shift, correct and re-verify, or apply a defined threshold before delivery?

Choices

A. Treat now; the shift looks minor and time is short.

B. Compare to a predefined OAR-drift threshold; treat if within, re-adapt if not, document either way.

C. Restart the adaptive session on current bowel anatomy.

D. Apply a positioning correction and re-verify the sigmoid relationship before treating.

Decision Log: Why did your team choose this option?

Notes:

PEL-02 Explanation Summary

Ms. Nora Nodalboost • Post-Operative Endometrial / Pelvic Nodal CTgART

Patient Context

Clinical Summary	59F, post-op endometrial cancer, 45 Gy/25 fx nodal CTV + vaginal-cuff boost region, fraction 16. Bladder/bowel filling drives cuff position; sigmoid is the limiting OAR.
Clinic / Coverage Model	Busy general adaptive program; rotating physicists, physician covering two units, two RTTs, with one newer to adaptive workflows. Cross-checking between team members is part of the routine.
Adaptive Rationale	Vaginal cuff and nodal CTV shift with bladder/bowel filling; adaptation maintains cuff coverage while controlling sigmoid dose.
Allowable Time	30 minutes
Time Consequence	Overage compresses the afternoon nodal cohort and raises the chance the next bladder-prep window is missed.

Core Lesson

A mixed rigid/deformable propagation strategy can create a seam that gaps, overlaps, or becomes invisible unless the workflow explicitly checks composite coverage and limiting-OAR proximity.

Overall Handling Approach

Make uncertainty visible, verify the anatomy that drives the decision, use documented thresholds or priority rules, and record the rationale before delivery.

1. Reference Planning — A Seam With No Rule

Why it matters: The reference/treatment directive issue sets the decision rule before the patient is deep into the adaptive session.
General handling approach: Apply or define the missing rule early, document the working basis, and escalate only when the ambiguity affects today's treatment decision.

2. On-Session Imaging — Clean Where It Doesn't Count

Why it matters: Image adequacy must be judged against the specific decision the image must support, not only global image quality.
General handling approach: Review the decision-driving anatomy directly. Reacquire or mitigate only when the limitation prevents reliable contouring or plan evaluation.

3. OAR Contouring — A Limiting OAR You Can't Quite See

Why it matters: At this stage the team is approving OAR/influencer anatomy for downstream optimization and plan comparison, not relying on a final DVH.
General handling approach: Focus edits on the OAR/influencer regions that can change optimization, dose calculation, or plan comparison; use second review when uncertainty remains clinically material.

4. Target Contouring — A Gap Only the Boundary Sees

Why it matters: A plausible propagated target does not by itself prove that clinical target intent has been preserved.
General handling approach: Reconcile propagated targets to reference intent, diagnostic anatomy, prior fractions, or a documented rule before planning.

5. Plan Selection — Sigmoid Numbers vs. an Unseen Gap

Why it matters: Scheduled/on-session recalculated and adapted plans can both appear acceptable while resolving different clinical risks.
General handling approach: Select based on documented target/OAR priorities, composite checks, sensitivity tests, and secondary dose review rather than the most attractive single metric.

6. Delivery / Confirmation Imaging — Late Sigmoid Shift

Why it matters: Confirmation imaging should be governed by predefined thresholds, stability checks, or re-verification rather than visual impression or time pressure alone.
General handling approach: Treat only when the approved geometry remains within threshold or can be restored and re-verified; ot herwise re-adapt, restart, delay, or cancel according to policy.

PEL-02 Scoring Sheet

Ms. Nora Nodalboost • Allowable time: 30 minutes

Workflow step	Choice	Safety penalty	Time penalty	Whammy?	Notes
1. Reference Planning	____	____	____	◻ Yes ◻ No	
2. On-Session Imaging	____	____	____	◻ Yes ◻ No	
3. OAR Contouring	____	____	____	◻ Yes ◻ No	
4. Target Contouring	____	____	____	◻ Yes ◻ No	
5. Plan Selection	____	____	____	◻ Yes ◻ No	
6. Delivery / Confirmation Imaging	____	____	____	◻ Yes ◻ No	

Total safety penalties	__________	**Total time used**	__________ min
Time overage	max(0, time - allowable) = ________	**Whammy resolution**	◻ +10 min choose again ◻ +15 safety
Final score	100 - safety penalties - time overage = ________		
Team	________________________	**Facilitator**	________________________

Scoring reminder: Final score = 100 - safety penalties - time overage. If a whammy is selected, add 10 minutes and choose again, or keep the choice and add 15 safety penalty.

PEL-02 Answer Key • 1. Reference Planning

Ms. Nora Nodalboost

Scenario: A Seam With No Rule

Choice A: Add a per-fraction cuff/nodal seam check this session and request a reference amendment defining reconciliation.

Scoring: Safety 2; Time 3 min; no cascade; Whammy: No; S/O/D/RPN 2/2/2/8

Explanation: Best: a standing seam check guards today and the amendment request starts the durable fix -- both at once.

Penalty rationale: Lowest combined penalty: small recurring time plus the fix.

Mitigation concept: Per-fraction seam verification + reference amendment.

Choice B: Accept the mixed scheme as designed; it has worked on prior fractions.

Scoring: Safety 9; Time 1 min; +8 safety cascade; Whammy: No; S/O/D/RPN 4/4/4/64

Explanation: Tempting because it has worked before, but only until filling changes; the seam gap then appears with nothing prompting anyone to look.

Penalty rationale: High safety, low time: the latent mismatch waits for a bad-anatomy day.

Mitigation concept: Consistent propagation or a defined seam rule.

Choice C: Stop and revise the reference plan offline to use one consistent propagation method before treating.

Scoring: Safety 2; Time 13 min; +2 min cascade; Whammy: No; S/O/D/RPN 2/2/2/8

Explanation: Revising offline is cleanest geometry but stops a treatable fraction for a manageable seam.

Penalty rationale: Highest time; over-correction relative to the finding.

Mitigation concept: Revise reference offline only if seam errors recur.

Choice D: Proceed today and request a reference-plan amendment defining seam reconciliation before next fraction.

Scoring: Safety 3; Time 4 min; no cascade; Whammy: No; S/O/D/RPN 2/2/2/8

Explanation: Strong, but requesting the amendment without adding today's seam check leaves this fraction on the unguarded seam.

Penalty rationale: Modest time; durable fix but today less protected than B.

Mitigation concept: Reference-plan amendment defining reconciliation.

PEL-02 Answer Key • 2. On-Session Imaging

Ms. Nora Nodalboost

Scenario: Clean Where It Doesn't Count

Choice A: Attempt bowel-gas mitigation and reacquire, focusing on the nodal region.

Scoring: Safety 2; Time 7 min; no cascade; Whammy: No; S/O/D/RPN 2/3/2/12

Explanation: Best: cleaner nodal imaging directly de-risks the rigid-propagation step where confidence is lowest; worth the time.

Penalty rationale: Moderate time for a better image where it matters.

Mitigation concept: Targeted reacquisition at the vulnerable region.

Choice B: Restart the session after bowel-prep.

Scoring: Safety 2; Time 12 min; +2 min cascade; Whammy: No; S/O/D/RPN 2/2/2/8

Explanation: A full restart for regional gas is usually disproportionate when targeted reacquisition would do.

Penalty rationale: Highest time; reserve for non-mitigable degradation.

Mitigation concept: Graded reacquisition first.

Choice C: Accept but flag the nodal region as lower-confidence for the propagation review.

Scoring: Safety 4; Time 2 min; no cascade; Whammy: No; S/O/D/RPN 3/3/3/27

Explanation: Reasonable: a confidence flag routes scrutiny to the weakest region cheaply, though it does not improve the image.

Penalty rationale: Low: improves detection at minimal cost.

Mitigation concept: Confidence flagging tied to propagation method.

Choice D: Accept; the cuff (the main target) images well.

Scoring: Safety 10; Time 1 min; +7 safety cascade; Whammy: No; S/O/D/RPN 4/4/4/64

Explanation: Judging the image by the clean cuff ignores that the nodal region drives the riskier rigid propagation. Tempting but misdirected.

Penalty rationale: High safety, low time: degraded region feeds the seam.

Mitigation concept: Per-structure image-adequacy assessment.

PEL-02 Answer Key • 3. OAR Contouring

Ms. Nora Nodalboost

Scenario: A Limiting OAR You Can't Quite See

Choice A: Correct the sigmoid through the cuff-to-nodal seam slices before approving the OAR contour set.

Scoring: Safety 2; Time 4 min; no cascade; Whammy: No; S/O/D/RPN 4/1/2/8

Explanation: This is a strong focused mitigation. The team corrects the localized sigmoid uncertainty in the same region where the cuff boost volume and elective nodal CTV may gap or overlap. It avoids approving an uncertain limiting OAR contour before downstream planning and plan comparison.

Penalty rationale: Low safety penalty because the decision-driving sigmoid region is corrected before planning. Moderate time penalty reflects focused contour editing without adding a full second-review or broader contour audit.

Mitigation concept: Focused correction of decision-driving OAR region

Choice B: Correct the sigmoid through the seam slices and cross-check its course against the prior fraction for consistency.

Scoring: Safety 1; Time 5 min; no cascade; Whammy: No; S/O/D/RPN 4/1/1/4

Explanation: This is the best balanced choice when prior-fraction anatomy is available and comparable. It corrects the uncertain sigmoid slices and adds a consistency check so the team does not silently accept a one-day contour deviation at the cuff-to-nodal seam.

Penalty rationale: Very low safety penalty because the clinically relevant OAR uncertainty is corrected and checked for consistency. Time penalty is slightly higher than focused editing alone because the team must compare against the prior fraction before approval.

Mitigation concept: Focused correction plus prior-fraction consistency check

Choice C: Request a second-reviewer check before approving the sigmoid contour, because the uncertainty is in the decision-driving region.

Scoring: Safety 2; Time 7 min; +1 min cascade; Whammy: No; S/O/D/RPN 4/1/2/8

Explanation: This is safe and defensible if the team is uncertain about the sigmoid course or if local policy requires escalation for limiting OAR uncertainty near target overlap regions. The downside is time, especially if the second reviewer repeats work that could have been resolved with a focused edit and prior-fraction check.

Penalty rationale: Low safety penalty because the uncertainty is escalated before plan generation. Higher time penalty reflects waiting for another reviewer and potentially repeating contour review while the patient remains on the table.

Mitigation concept: Second-reviewer escalation for limiting OAR uncertainty

Choice D: Accept the generated sigmoid contour because its overall course looks reasonable and the session is time-sensitive.

Scoring: Safety 8; Time 1 min; +3 safety cascade, +3 min cascade; Whammy: YES - Add 10 minutes and choose again, or keep this choice and add 15 safety penalty.; S/O/D/RPN 4/3/3/36

Explanation: This is the tempting shortcut. The sigmoid contour looks plausible overall, but the uncertain portion lies exactly in the cuff-to-nodal seam zone where target propagation and later plan selection are most sensitive. Approving it can make the downstream adapted-plan comparison appear more reliable than it is.

Penalty rationale: High safety penalty because the limiting OAR is accepted despite localized uncertainty in the decision-driving region. Low time penalty because the team proceeds quickly, but the unresolved contour uncertainty can create downstream plan-selection and confirmation risk.

Mitigation concept: Do not approve a limiting OAR based only on overall plausibility

PEL-02 Answer Key • 4. Target Contouring

Ms. Nora Nodalboost

Scenario: A Gap Only the Boundary Sees

Choice A: Manually bridge the seam gap and recompute combined coverage.

Scoring: Safety 3; Time 4 min; no cascade; Whammy: No; S/O/D/RPN 2/3/2/12

Explanation: Best: bridging the gap restores transition coverage now with minimal time; the standing check is good but not required to treat safely today.

Penalty rationale: Low combined penalty: fixes the immediate geometry.

Mitigation concept: Manual seam bridge + recompute.

Choice B: Accept; both volumes meet their individual coverage goals.

Scoring: Safety 12; Time 1 min; +6 safety cascade; Whammy: No; S/O/D/RPN 5/4/5/100

Explanation: Tempting because both DVHs pass, but the gap is undetectable per-volume; accepting it can underdose the transition.

Penalty rationale: Highest safety: the classic invisible-to-metrics geometric failure.

Mitigation concept: Composite/overlap coverage check at the seam.

Choice C: Bridge the gap and add a standing composite-coverage check to the workflow.

Scoring: Safety 2; Time 6 min; no cascade; Whammy: No; S/O/D/RPN 2/2/2/8

Explanation: Also strong: adds a reusable composite check so the seam cannot silently gap again; slightly more setup than B.

Penalty rationale: Modest time for a durable safeguard.

Mitigation concept: Composite-target verification step.

Choice D: Return to contour editing to re-derive both volumes with one method, then regenerate the plan.

Scoring: Safety 3; Time 11 min; +2 min cascade; Whammy: No; S/O/D/RPN 2/2/2/8

Explanation: Single-method re-derivation removes the root cause but is heavy mid-session for one fraction.

Penalty rationale: High time; durable but disproportionate day-to-day.

Mitigation concept: Adopt single-method propagation at offline revision.

PEL-02 Answer Key • 5. Plan Selection

Ms. Nora Nodalboost

Scenario: Sigmoid Numbers vs. an Unseen Gap

Choice A: Choose the adapted plan and document that seam coverage was the deciding factor.

Scoring: Safety 4; Time 3 min; no cascade; Whammy: No; S/O/D/RPN 3/3/3/27

Explanation: Reasonable: naming the seam as decisive makes the tradeoff explicit, though it does not formally weigh sigmoid headroom.

Penalty rationale: Low-moderate: documented clinical reasoning.

Mitigation concept: Explicit seam-coverage rationale.

Choice B: Choose the scheduled plan; its sigmoid numbers look safest.

Scoring: Safety 9; Time 1 min; +5 safety cascade; Whammy: No; S/O/D/RPN 4/4/4/64

Explanation: Picking on sigmoid numbers alone can silently reaccept the invisible seam gap; tempting because those numbers look best.

Penalty rationale: High safety: the unseen risk drives the wrong choice.

Mitigation concept: Selection criteria that include composite coverage.

Choice C: Choose based on the composite-check result and record both the coverage and sigmoid rationale.

Scoring: Safety 2; Time 5 min; no cascade; Whammy: No; S/O/D/RPN 2/2/2/8

Explanation: Best: uses the composite check as the objective basis and records both sides -- the most defensible, reconstructable choice.

Penalty rationale: Modest time; coherent decision trail.

Mitigation concept: Composite-informed selection + dual rationale.

Choice D: Return to contour editing and regenerate the adapted plan if neither resolves both; revise offline if still unacceptable.

Scoring: Safety 3; Time 9 min; +2 min cascade; Whammy: No; S/O/D/RPN 2/2/2/8

Explanation: Operationally valid when neither plan is acceptable: return to editing and regenerate, or revise offline; costs time so reserve for true conflict.

Penalty rationale: High time; the right path only when both plans fail.

Mitigation concept: Return-to-editing / offline revision for genuine conflicts.

PEL-02 Answer Key • 6. Delivery / Confirmation Imaging

Ms. Nora Nodalboost

Scenario: Late Sigmoid Shift

Choice A: Treat now; the shift looks minor and time is short.

Scoring: Safety 10; Time 1 min; no cascade; Whammy: No; S/O/D/RPN 4/3/4/48

Explanation: 'Looks minor' without a threshold is how limiting-OAR drift slips through; tempting under time pressure.

Penalty rationale: High safety, low time: subjective acceptance of OAR drift.

Mitigation concept: Predefined OAR-drift threshold at confirmation.

Choice B: Compare to a predefined OAR-drift threshold; treat if within, re-adapt if not, document either way.

Scoring: Safety 2; Time 6 min; no cascade; Whammy: No; S/O/D/RPN 2/2/2/8

Explanation: Also strong: threshold-based and documented; marginally more time than B for a formal record.

Penalty rationale: Moderate time; balanced and reconstructable.

Mitigation concept: Documented OAR-drift threshold.

Choice C: Restart the adaptive session on current bowel anatomy.

Scoring: Safety 2; Time 12 min; no cascade; Whammy: No; S/O/D/RPN 2/2/2/8

Explanation: Justified for large drift; excessive for a minor sigmoid shift and risks the slot.

Penalty rationale: Highest time; reserve for threshold breaches.

Mitigation concept: Restart only beyond threshold.

Choice D: Apply a positioning correction and re-verify the sigmoid relationship before treating.

Scoring: Safety 3; Time 4 min; no cascade; Whammy: No; S/O/D/RPN 3/3/3/27

Explanation: Best: a positioning correction and targeted re-verify can restore the intended sigmoid relationship without full re-adaptation, at low cost.

Penalty rationale: Low combined penalty: proportionate response.

Mitigation concept: Positioning correction + targeted re-verify.

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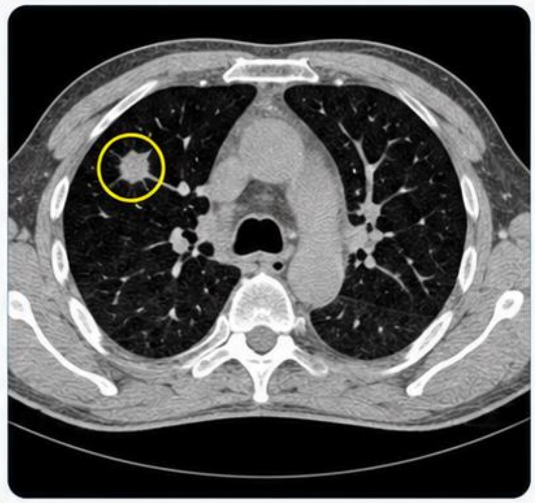
PATIENT CASE FILE • LUNG-01

Mr. Reggie Regression

“The tumor got smaller. Now what?”

1 PATIENT SNAPSHOT

72-year-old	Peripheral RUL NSCLC	SBRT · on supplemental O_2	Fraction 3
Prescription: 50 Gy / 5 fx	Today: Fraction 3	Approach: ITV-based	The daily wildcard: Tumor regressed vs. simulation



2 WHY HE'S ON THE ADAPTIVE COUCH

His tumor has visibly regressed and the daily baseline shifts. Adaptation can preserve coverage while limiting chest-wall and lung dose — but regression is an observation, not a target policy, and the reference plan never said how to handle it.

3 THE ROOM TODAY

Established lung SBRT program with a respiratory-managed workflow. Physicist and physician on site at the start; the physician steps out mid-session, leaving you and two RTTs to run the adaptive decisions.

4 THE CLOCK IS REAL

You get 35 minutes. Overage delays a second SBRT patient whose 4D setup is time-critical — and rushing that case risks motion-management errors downstream.

SAFETY IS ADAPTIVE. TEAMWORK MAKES IT EFFECTIVE.

COMMUNICATE · IDENTIFY RISK · EVALUATE · MITIGATE · LEARN & IMPROVE

TURN THE PAGE TO MEET HIS SIX DECISIONS.

1 REFERENCE PLANNING

PLAYER CARD

Let's Adapt!

LUNG-01 | Mr. Reggie Regression

Scenario: Yesterday's Choice, Today's Unclear Rule

Clinical Situation: Mr. Regression is on a peripheral lung SBRT course, and his tumor has visibly regressed since simulation. The reference plan documents the original simulation ITV/PTV approach, but the chart does not clearly state how interval regression should be handled during the course.

On the previous fraction, the team appears to have treated using a smaller visible GTV-based high-dose target while maintaining a rigidly propagated lower-dose residual-disease envelope. A physician note says to "chase the visible GTV," but it does not define what to do when the GTV edge is blurred, how much uncertainty to include, or how the lower-dose residual-disease envelope should be reconciled with the regressed target. The prior fraction was treated, but the plan-selection rationale and target-editing rule were not documented well enough for today's team to reproduce with confidence.

Decision Prompt: Do you repeat yesterday's undocumented approach, hold the validated ITV-based intent, or define the regression rule before changing today's target?

Choices

A. Apply a documented regression rule if one exists; if not, hold the validated ITV-based intent today and request clarification of the GTV-chasing / residual-envelope rule.

B. Repeat what appears to have been done yesterday: chase the visible GTV and rigidly propagate the lower-dose residual-disease envelope.

C. Pause SBRT delivery pending physician clarification of the regression policy and how to handle blurred GTV edge uncertainty.

D. Hold the original ITV-based target today, document the regression concern, and request physician review before the next fraction.

Decision Log: Why did your team choose this option?

Notes:

2 ON-SESSION IMAGING

PLAYER CARD

Let's Adapt!

LUNG-01 | Mr. Reggie Regression

Scenario: The Edge You Need Is the Edge You Don't Trust

Clinical Situation: On-session CBCT for Mr. Regression shows the treated lesion is smaller than at simulation, consistent with interval regression. There is also mild motion blur and partial-volume uncertainty at the visible GTV edge. The rest of the thoracic image is adequate, and there is a faint new hazy pulmonary finding away from the target region that does not appear to affect today's dose distribution.

The team's problem is not simply whether the CBCT is "good enough." The problem is that the physician's prior instruction was to chase the visible GTV, but today the visible edge is the least reliable part of the image. The incidental lung finding may need documentation or follow-up, but the blurred GTV edge is what directly affects today's adaptive target decision.

Decision Prompt: Which finding affects today's target decision, and what image review is needed before contouring?

Choices

A. Proceed; the incidental haziness is not in the treatment region, and the GTV is visible enough.

B. Repeat the CBCT or perform a brief motion assessment before finalizing the GTV edge.

C. Hold for physician review of the new pulmonary haziness before treating.

D. Document the incidental haziness separately and tighten review of the blurred GTV edge before proceeding.

Decision Log: Why did your team choose this option?

Notes:

3 OAR CONTOURING

PLAYER CARD

LUNG-01 | Mr. Reggie Regression

Let's Adapt!

Scenario: A Serial OAR Near a Moving Target

Clinical Situation: The generated chest-wall contour and propagated proximal bronchial tree look reasonable overall.

However, a few bronchial tree slices near the regressed tumor appear under-segmented. This matters because if the team tightens or shifts the high-dose GTV-based target, the dose gradient near the bronchial structure could become more important. At this step, the team is not yet judging the final DVH. The decision is whether the serial-OAR contour is complete enough to approve for downstream optimization, secondary review, and later plan comparison. The contour uncertainty is localized, but it sits near the anatomy that could influence the target/OAR tradeoff.

Decision Prompt: Do you approve the mostly plausible serial-OAR contours, or complete the bronchial contour where dose could rise before proceeding?

Choices

A. Complete the bronchial contour on the affected slices and verify it against the reference/prior-fraction bronchial structure.

B. Request a serial-OAR second check before approving the bronchial contour for planning.

C. Complete only the visibly missing bronchial slices, then approve the OAR contour set.

D. Accept the generated chest-wall and bronchial contours because they look reasonable overall and the session is time-sensitive.

Decision Log: Why did your team choose this option?

Notes:

4 TARGET CONTOURING

PLAYER CARD

Let's Adapt!

LUNG-01 | Mr. Reggie Regression

Scenario: To Tighten or Not to Tighten

Clinical Situation: Mr. Regression's visible tumor is smaller than at simulation. The physician's prior instruction was to chase the visible GTV for the high-dose adaptive target while rigidly maintaining a lower-dose residual-disease envelope. That strategy may be clinically reasonable, but today the visible GTV edge is blurred by motion and partial-volume uncertainty.

Tightening the GTV to the smallest visible disease would spare more lung, but it risks excluding uncertain residual disease at the edge. Holding the entire original ITV is more robust but may ignore the physician's intent to adapt to regression. The team needs to apply a documented regression rule, not invent a new target policy at the console.

Decision Prompt: How do you contour today's target when the instruction says to chase the GTV, but the GTV edge is uncertain?

Choices

A. Tighten the high-dose GTV to the smallest visible tumor edge to maximize lung sparing.

B. Hold for physician target review before planning because the GTV-chasing rule does not define blurred-edge uncertainty.

C. Apply the documented rule: chase the confident visible GTV while preserving the rigidly propagated lower-dose residual-disease envelope, and document the basis.

D. Keep the original ITV/PTV envelope for the high-dose target today and note regression for physician review.

Decision Log: Why did your team choose this option?

Notes:

5 PLAN SELECTION

PLAYER CARD

LUNG-01 | Mr. Reggie Regression

Let's Adapt!

Scenario: Lung Sparing vs. SBRT Robustness

Clinical Situation: Mr. Regression's adapted plan follows the smaller high-dose GTV-based target and lowers lung dose, while maintaining the lower-dose residual-disease envelope. The scheduled/on-session recalculated plan holds the original ITV-based high-dose coverage and is more robust to edge uncertainty, but it delivers more lung dose. Both appear to meet the listed goals. The adapted plan's lower lung dose looks attractive on screen. The real question is whether today's target definition followed a documented regression rule and whether the blurry edge uncertainty was handled consistently.

Decision Prompt: What objective basis separates these two acceptable plans for an ablative target?

Choices

A. Return to target review, re-confirm the GTV edge and residual-disease envelope, then regenerate the adapted plan before selecting.

B. Pick the scheduled/on-session recalculated plan for SBRT robustness and document the conservative rationale.

C. Select according to the documented regression rule and secondary dose check, recording the target/OAR rationale.

D. Pick the adapted plan because the lung dose is lower and the tumor has regressed.

Decision Log: Why did your team choose this option?

Notes:

6 DELIVERY / CONFIRMATION IMAGING

PLAYER CARD

Let's Adapt!

LUNG-01 | Mr. Reggie Regression

Scenario: Baseline Drift at the Gate

Clinical Situation: In this clinic's respiratory-managed SBRT workflow, confirmation imaging for Mr. Regression shows the GTV baseline has shifted relative to the expected breathing/motion envelope after approval, and his breathing has become irregular compared with the motion state used for adaptation.

The plan is approved and the slot is short, so delivering now is tempting. However, the plan-selection rationale depended on how much the team trusted the visible GTV edge and residual-disease envelope. If the breathing baseline has shifted, the target/motion relationship assumed by the approved plan may no longer hold.

Decision Prompt: Do you deliver now, re-coach and re-verify the motion envelope, or apply a defined motion tolerance before delivery?

Choices

A. Re-coach breathing and re-verify the motion envelope before delivery.

B. Deliver now; the plan was approved and time is short.

C. Check drift against a predefined motion tolerance; re-coach, re-image, or re-adapt according to threshold, and document.

D. Restart the adaptive session on the current breathing/motion baseline.

Decision Log: Why did your team choose this option?

Notes:

LUNG-01 Explanation Summary

Mr. Reggie Regression • Peripheral Lung SBRT / Regression During Treatment

Patient Context

Clinical Summary	72M, peripheral RUL NSCLC, SBRT 50 Gy/5 fx, fraction 3. On supplemental O2; tumor regression noted vs. simulation. ITV-based approach.
Clinic / Coverage Model	Established lung SBRT adaptive program with a respiratory-managed (motion-managed) SBRT workflow; physicist + physician on site at start, physician steps out mid-session, two RTTs.
Adaptive Rationale	Tumor regression and daily baseline shift change the ITV-to-target relationship; adaptation preserves coverage while limiting chest-wall and lung dose.
Allowable Time	35 minutes
Time Consequence	Overage delays a second SBRT patient whose 4D setup is time-critical; rushing that case risks motion-management errors downstream.

Core Lesson

Tumor regression is an observation, not a target policy. Chasing a smaller visible GTV requires documented rules for edge uncertainty, residual-disease coverage, and motion management.

Overall Handling Approach

Make uncertainty visible, verify the anatomy that drives the decision, use documented thresholds or priority rules, and record the rationale before delivery.

1. Reference Planning — Yesterday's Choice, Today's Unclear Rule

Why it matters: The reference/treatment directive issue sets the decision rule before the patient is deep into the adaptive session.
General handling approach: Apply or define the missing rule early, document the working basis, and escalate only when the ambiguity affects today's treatment decision.

2. On-Session Imaging — The Edge You Need Is the Edge You Don't Trust

Why it matters: Image adequacy must be judged against the specific decision the image must support, not only global image quality.
General handling approach: Review the decision-driving anatomy directly. Reacquire or mitigate only when the limitation prevents reliable contouring or plan evaluation.

3. OAR Contouring — A Serial OAR Near a Moving Target

Why it matters: At this stage the team is approving OAR/influencer anatomy for downstream optimization and plan comparison, not relying on a final DVH.
General handling approach: Focus edits on the OAR/influencer regions that can change optimization, dose calculation, or plan comparison; use second review when uncertainty remains clinically material.

4. Target Contouring — To Tighten or Not to Tighten

Why it matters: A plausible propagated target does not by itself prove that clinical target intent has been preserved.
General handling approach: Reconcile propagated targets to reference intent, diagnostic anatomy, prior fractions, or a documented rule before planning.

5. Plan Selection — Lung Sparing vs. SBRT Robustness

Why it matters: Scheduled/on-session recalculated and adapted plans can both appear acceptable while resolving different clinical risks.
General handling approach: Select based on documented target/OAR priorities, composite checks, sensitivity tests, and secondary dose review rather than the most attractive single metric.

6. Delivery / Confirmation Imaging — Baseline Drift at the Gate

Why it matters: Confirmation imaging should be governed by predefined thresholds, stability checks, or re-verification rather than visual impression or time pressure alone.
General handling approach: Treat only when the approved geometry remains within threshold or can be restored and re-verified; otherwise re-adapt, restart, delay, or cancel according to policy.

LUNG-01 Scoring Sheet

Mr. Reggie Regression • Allowable time: 35 minutes

Workflow step	Choice	Safety penalty	Time penalty	Whammy?	Notes
1. Reference Planning	____	____	____	□ Yes □ No	
2. On-Session Imaging	____	____	____	□ Yes □ No	
3. OAR Contouring	____	____	____	□ Yes □ No	
4. Target Contouring	____	____	____	□ Yes □ No	
5. Plan Selection	____	____	____	□ Yes □ No	
6. Delivery / Confirmation Imaging	____	____	____	□ Yes □ No	

Total safety penalties	__________	**Total time used**	__________ min
Time overage	max(0, time - allowable) = ________	**Whammy resolution**	□ +10 min choose again □ +15 safety
Final score	100 - safety penalties - time overage = ________		
Team	________________________	**Facilitator**	________________________

Scoring reminder: Final score = 100 - safety penalties - time overage. If a whammy is selected, add 10 minutes and choose again, or keep the choice and add 15 safety penalty.

LUNG-01 Answer Key • 1. Reference Planning

Mr. Reggie Regression

Scenario: Yesterday's Choice, Today's Unclear Rule

Choice A: Apply a documented regression rule if one exists; if not, hold the validated ITV-based intent today and request clarification of the GTV-chasing / residual-envelope rule.

Scoring: Safety 1; Time 4 min; no cascade; Whammy: No; S/O/D/RPN 4/1/1/4

Explanation: This is the best balanced choice. It respects prior physician intent if it has been documented, but it does not let an undocumented prior-fraction action become today's target policy. If no rule exists, holding the validated ITV-based intent protects SBRT robustness while prompting clarification of the GTV-chasing and residual-envelope approach.

Penalty rationale: Very low safety penalty because the team avoids inventing a regression policy on the table. Moderate time penalty reflects the need to review documentation and request clarification for subsequent fractions.

Mitigation concept: Documented regression rule before target change

Choice B: Repeat what appears to have been done yesterday: chase the visible GTV and rigidly propagate the lower-dose residual-disease envelope.

Scoring: Safety 6; Time 2 min; +2 safety cascade, +2 min cascade; Whammy: No; S/O/D/RPN 4/3/3/36

Explanation: This is tempting because it appears consistent with the prior fraction and with the physician phrase to chase the visible GTV. The risk is that the team is reproducing an undocumented action without knowing how blurred-edge uncertainty or residual-disease coverage was handled.

Penalty rationale: Moderate-high safety penalty because the prior approach may be reasonable but is not reproducible without documentation. Low time penalty because the team proceeds quickly, but uncertainty propagates to contouring and plan selection.

Mitigation concept: Do not convert undocumented prior action into policy

Choice C: Pause SBRT delivery pending physician clarification of the regression policy and how to handle blurred GTV edge uncertainty.

Scoring: Safety 2; Time 9 min; no cascade; Whammy: No; S/O/D/RPN 4/1/2/8

Explanation: This is safe and may be appropriate if today's target cannot be defined without physician input. The tradeoff is that it may be more disruptive than necessary if the validated ITV-based intent remains clinically acceptable for today.

Penalty rationale: Low safety penalty because the ambiguity is escalated before treatment. High time penalty reflects interruption of an SBRT slot and possible delay while the patient remains in position.

Mitigation concept: Escalate unresolved regression policy

Choice D: Hold the original ITV-based target today, document the regression concern, and request physician review before the next fraction.

Scoring: Safety 2; Time 3 min; +1 min cascade; Whammy: No; S/O/D/RPN 4/1/2/8

Explanation: This is conservative and defensible because it preserves the validated simulation-based intent for today. The limitation is that it may not fully address the physician's stated desire to chase the visible GTV if that intent should apply now.

Penalty rationale: Low safety penalty because SBRT robustness is protected. Low-moderate time penalty because the team proceeds with documentation and follow-up, though some uncertainty persists about physician intent.

Mitigation concept: Conservative validated-intent fallback

LUNG-01 Answer Key • 2. On-Session Imaging

Mr. Reggie Regression

Scenario: The Edge You Need Is the Edge You Don’t Trust

Choice A: Proceed; the incidental haziness is not in the treatment region, and the GTV is visible enough.

Scoring: Safety 6; Time 1 min; +2 safety cascade, +2 min cascade; Whammy: No; S/O/D/RPN 4/3/3/36

Explanation: This is plausible because the incidental haziness does not appear to affect today’s dose region. The problem is that the team also waves through the blurred GTV edge, which is the decision-driving uncertainty for a GTV-chasing workflow.

Penalty rationale: Moderate-high safety penalty because the edge uncertainty is not addressed before target contouring. Low time penalty because the workflow continues quickly, but the unresolved image uncertainty carries forward.

Mitigation concept: Separate incidental finding from target-edge uncertainty

Choice B: Repeat the CBCT or perform a brief motion assessment before finalizing the GTV edge.

Scoring: Safety 2; Time 6 min; no cascade; Whammy: No; S/O/D/RPN 4/1/2/8

Explanation: This is safe and appropriate if motion blur prevents confident GTV-edge definition. It directly addresses the image uncertainty that matters for today’s target decision.

Penalty rationale: Low safety penalty because the team improves confidence before contouring. Time penalty is moderate because repeat imaging or motion assessment consumes on-table time.

Mitigation concept: Re-image or assess motion when target edge is unreliable

Choice C: Hold for physician review of the new pulmonary haziness before treating.

Scoring: Safety 4; Time 7 min; +1 min cascade; Whammy: No; S/O/D/RPN 3/2/2/12

Explanation: This may be reasonable if the new haziness is clinically concerning, but it focuses on the less relevant finding for today’s adaptive decision. The blurred GTV edge remains the more immediate target-contouring issue.

Penalty rationale: Moderate safety penalty because attention is diverted from the decision-driving uncertainty. Higher time penalty reflects physician review for a finding that may not affect today’s plan.

Mitigation concept: Triage incidental findings without missing target uncertainty

Choice D: Document the incidental haziness separately and tighten review of the blurred GTV edge before proceeding.

Scoring: Safety 1; Time 4 min; no cascade; Whammy: No; S/O/D/RPN 4/1/1/4

Explanation: This is the best balanced choice when the GTV edge remains interpretable with focused review. It separates the incidental pulmonary finding from the target-driving uncertainty and ensures the edge is scrutinized before contouring.

Penalty rationale: Very low safety penalty because both findings are handled appropriately. Moderate time penalty reflects focused edge review and documentation without automatically restarting the session.

Mitigation concept: Triage findings by relevance to today’s decision

LUNG-01 Answer Key • 3. OAR Contouring

Mr. Reggie Regression

Scenario: A Serial OAR Near a Moving Target

Choice A: Complete the bronchial contour on the affected slices and verify it against the reference/prior-fraction bronchial structure.

Scoring: Safety 1; Time 5 min; no cascade; Whammy: No; S/O/D/RPN 4/1/1/4

Explanation: This is the best balanced choice. It corrects the serial-OAR contour where the adapted target could change the dose relationship and checks consistency against prior/reference anatomy.

Penalty rationale: Very low safety penalty because the decision-driving OAR region is corrected and verified. Moderate time penalty reflects focused editing plus comparison to reference or prior-fraction structure.

Mitigation concept: Focused serial-OAR correction with consistency check

Choice B: Request a serial-OAR second check before approving the bronchial contour for planning.

Scoring: Safety 2; Time 7 min; +1 min cascade; Whammy: No; S/O/D/RPN 4/1/2/8

Explanation: This is safe and defensible if the bronchial anatomy is difficult to identify or local policy requires escalation for serial-OAR uncertainty. It may be slower than necessary if the missing slices are straightforward to correct.

Penalty rationale: Low safety penalty because the uncertainty is escalated before planning. Higher time penalty reflects waiting for another reviewer and potential repeated review.

Mitigation concept: Second check for serial-OAR uncertainty

Choice C: Complete only the visibly missing bronchial slices, then approve the OAR contour set.

Scoring: Safety 2; Time 4 min; no cascade; Whammy: No; S/O/D/RPN 4/1/2/8

Explanation: This is reasonable and efficient because it fixes the visible defect before planning. The limitation is that it lacks the consistency check against reference or prior-fraction anatomy for a serial OAR near a changing target.

Penalty rationale: Low safety penalty because the missing contour is corrected. Moderate-low time penalty because the edit is focused and no second check is added.

Mitigation concept: Focused correction of affected slices

Choice D: Accept the generated chest-wall and bronchial contours because they look reasonable overall and the session is time-sensitive.

Scoring: Safety 7; Time 1 min; +2 safety cascade, +3 min cascade; Whammy: No; S/O/D/RPN 4/3/3/36

Explanation: This is the tempting shortcut. The contours look plausible overall, but the bronchial under-segmentation is near the region where a tightened high-dose target could alter the serial-OAR relationship.

Penalty rationale: High safety penalty because a serial OAR is approved despite localized uncertainty near the decision-driving anatomy. Low time penalty because the team proceeds quickly, but downstream plan comparison may be unreliable.

Mitigation concept: Do not approve serial-OAR contour based only on overall plausibility

LUNG-01 Answer Key • 4. Target Contouring

Mr. Reggie Regression

Scenario: To Tighten or Not to Tighten

Choice A: Tighten the high-dose GTV to the smallest visible tumor edge to maximize lung sparing.

Scoring: Safety 9; Time 1 min; +3 safety cascade, +3 min cascade; Whammy: No; S/O/D/RPN 5/3/3/45

Explanation: This is unsafe because it treats the smallest visible edge as truth despite motion blur and undefined residual-disease uncertainty. It may spare lung but risks under-covering uncertain tumor extent in an ablative SBRT course.

Penalty rationale: High safety penalty because target tightening is based on an uncertain edge. Low time penalty because the team proceeds quickly, but target-miss risk carries into plan selection and delivery.

Mitigation concept: Do not chase an uncertain GTV edge

Choice B: Hold for physician target review before planning because the GTV-chasing rule does not define blurred-edge uncertainty.

Scoring: Safety 2; Time 8 min; no cascade; Whammy: No; S/O/D/RPN 5/1/2/10

Explanation: This is safe and appropriate if the rule truly does not define how to handle blurred-edge uncertainty. The drawback is time and reliance on real-time physician availability.

Penalty rationale: Low safety penalty because target ambiguity is escalated before planning. High time penalty reflects interruption for physician target review during SBRT setup.

Mitigation concept: Escalate undefined target-edge policy

Choice C: Apply the documented rule: chase the confident visible GTV while preserving the rigidly propagated lower-dose residual-disease envelope, and document the basis.

Scoring: Safety 1; Time 5 min; no cascade; Whammy: No; S/O/D/RPN 5/1/1/5

Explanation: This is the best balanced choice if the regression rule is documented. It follows physician intent to adapt to regression while preserving residual-disease coverage and avoiding over-tightening to an uncertain edge.

Penalty rationale: Very low safety penalty because the target is adapted using a defined rule that handles uncertainty. Moderate time penalty reflects careful target review and documentation.

Mitigation concept: Defined GTV-chasing rule with residual-envelope preservation

Choice D: Keep the original ITV/PTV envelope for the high-dose target today and note regression for physician review.

Scoring: Safety 3; Time 3 min; +1 min cascade; Whammy: No; S/O/D/RPN 5/1/2/10

Explanation: This is conservative and robust for SBRT, but it may miss the physician’s intended adaptive strategy if the documented rule supports GTV chasing with residual-envelope preservation.

Penalty rationale: Moderate-low safety penalty because target robustness is maintained but documented adaptive intent may not be followed. Time penalty is low-moderate.

Mitigation concept: Validated ITV fallback when regression rule is unclear

LUNG-01 Answer Key • 5. Plan Selection

Mr. Reggie Regression

Scenario: Lung Sparing vs. SBRT Robustness

Choice A: Return to target review, re-confirm the GTV edge and residual-disease envelope, then regenerate the adapted plan before selecting.

Scoring: Safety 2; Time 8 min; no cascade; Whammy: No; S/O/D/RPN 5/1/2/10

Explanation: This is safe if target uncertainty remains unresolved after contouring. It prevents selection between plans based on a questionable target definition, but it is slower than proceeding when the documented rule was already applied correctly.

Penalty rationale: Low safety penalty because target uncertainty is addressed before plan selection. High time penalty reflects re-review and regeneration of the adapted plan.

Mitigation concept: Reconfirm target basis before plan selection

Choice B: Pick the scheduled/on-session recalculated plan for SBRT robustness and document the conservative rationale.

Scoring: Safety 3; Time 3 min; +1 min cascade; Whammy: No; S/O/D/RPN 5/1/2/10

Explanation: This is defensible when edge confidence is limited or regression policy is not clear. It protects ablative robustness but may be more conservative than necessary if the documented regression rule was applied successfully.

Penalty rationale: Moderate-low safety penalty because robustness is preserved. Time penalty is low-moderate because the team documents the conservative choice without additional planning.

Mitigation concept: Conservative SBRT robustness

Choice C: Select according to the documented regression rule and secondary dose check, recording the target/OAR rationale.

Scoring: Safety 1; Time 4 min; no cascade; Whammy: No; S/O/D/RPN 5/1/1/5

Explanation: This is the best balanced plan-selection choice. It ties the lower lung dose benefit to a documented target rule, residual-disease envelope, and secondary dose check rather than selecting the adapted plan simply because it looks better dosimetrically.

Penalty rationale: Very low safety penalty because plan selection is based on documented target intent and dose verification. Moderate time penalty reflects review, secondary check, and rationale documentation.

Mitigation concept: Documented regression-basis plan selection

Choice D: Pick the adapted plan because the lung dose is lower and the tumor has regressed.

Scoring: Safety 8; Time 1 min; +3 safety cascade, +2 min cascade; Whammy: No; S/O/D/RPN 5/3/3/45

Explanation: This is tempting because lower lung dose is attractive and the tumor has visibly regressed. The risk is that the team lets the favorable lung metric substitute for documented target robustness and residual-disease coverage.

Penalty rationale: High safety penalty because the adapted plan is selected for an attractive metric without verifying target-rule validity. Low time penalty because the team proceeds quickly, but unresolved target uncertainty persists.

Mitigation concept: Do not let lung sparing override target-rule documentation

LUNG-01 Answer Key • 6. Delivery / Confirmation Imaging

Mr. Reggie Regression

Scenario: Baseline Drift at the Gate

Choice A: Re-coach breathing and re-verify the motion envelope before delivery.

Scoring: Safety 2; Time 6 min; no cascade; Whammy: No; S/O/D/RPN 5/1/2/10

Explanation: This is safe and reasonable when breathing irregularity appears after approval. It may be sufficient if re-coaching restores the expected baseline and the motion envelope is verified before treatment.

Penalty rationale: Low safety penalty because motion uncertainty is addressed before delivery. Moderate time penalty reflects re-coaching and repeat verification.

Mitigation concept: Re-coach and verify respiratory baseline

Choice B: Deliver now; the plan was approved and time is short.

Scoring: Safety 10; Time 1 min; no cascade; Whammy: YES - Add 10 minutes and choose again, or keep this choice and add 15 safety penalty.; S/O/D/RPN 5/3/3/45

Explanation: This is unsafe because the plan's validity depends on the target/motion relationship used during adaptation. Approval does not remain valid if the breathing baseline has shifted outside tolerance.

Penalty rationale: Very high safety penalty because motion uncertainty is ignored immediately before SBRT delivery. Low time penalty because treatment proceeds quickly, but the risk is high.

Mitigation concept: Do not deliver through unverified motion drift

Choice C: Check drift against a predefined motion tolerance; re-coach, re-image, or re-adapt according to threshold, and document.

Scoring: Safety 1; Time 5 min; no cascade; Whammy: No; S/O/D/RPN 5/1/1/5

Explanation: This is the best balanced choice. It uses predefined motion tolerance to decide whether treatment can proceed, whether re-coaching/re-imaging is enough, or whether re-adaptation is needed.

Penalty rationale: Very low safety penalty because the response is threshold-based and documented. Moderate time penalty reflects motion assessment and possible corrective action.

Mitigation concept: Threshold-based confirmation motion management

Choice D: Restart the adaptive session on the current breathing/motion baseline.

Scoring: Safety 2; Time 10 min; no cascade; Whammy: No; S/O/D/RPN 5/1/2/10

Explanation: This is safe and may be necessary if the baseline shift persists outside tolerance. It is slower than first applying the predefined motion threshold and attempting re-coaching or re-verification.

Penalty rationale: Low safety penalty because the plan is rebuilt on current motion anatomy. Very high time penalty because restarting the adaptive session is disruptive and may worsen patient fatigue.

Mitigation concept: Restart only when motion tolerance requires it

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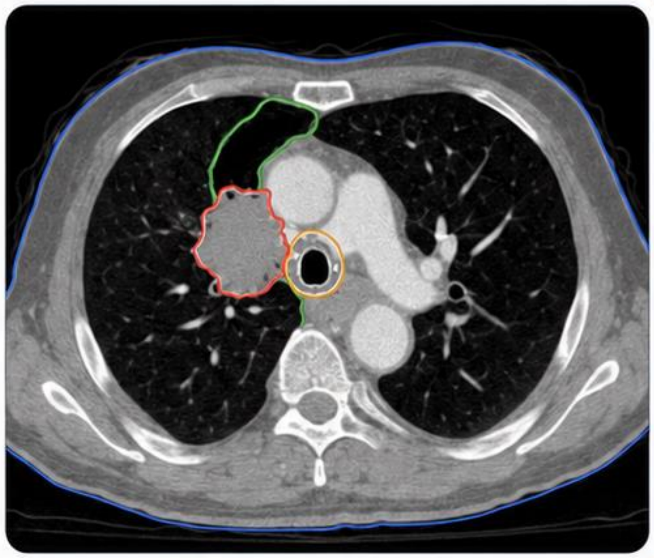
PATIENT CASE FILE • LUNG-02

Ms. Mona Motion

“The lung re-inflated under the beams.”

1 PATIENT SNAPSHOT

64-year-old	Central NSCLC
Adjacent to proximal bronchial tree	Fraction 7
Prescription: 60 Gy / 15 fx	Limiting OAR: Proximal bronchial tree
Today’s wildcard: Atelectasis re-expanding the lung	Allowable time: 35 minutes



2 WHY SHE’S ON THE ADAPTIVE COUCH

Her atelectasis is resolving, re-expanding lung and moving the target away from the mediastinum. Adaptation prevents treating collapsed-lung geometry — but the density change affects the dose calculation that BOTH the scheduled and adapted plans depend on.

3 THE ROOM TODAY

Academic adaptive program: a trainee physicist supervised remotely, an attending physician in clinic, and experienced RTTs. Good support — but the console calls are yours to make.

4 THE CLOCK IS REAL

You get 35 minutes. Overage cascades into a central-tumor patient whose proximal-bronchial constraints demand careful review — rushing the next case is the real risk.

SAFETY IS ADAPTIVE.
TEAMWORK MAKES IT EFFECTIVE.

COMMUNICATE | IDENTIFY RISK | EVALUATE | MITIGATE | LEARN & IMPROVE

TURN THE PAGE TO MEET HER SIX DECISIONS.

1 REFERENCE PLANNING

PLAYER CARD

LUNG-02 | Ms. Mona Motion

Let's Adapt!

Scenario: Density That Changed Under the Beams

Clinical Situation: Ms. Motion has a central lung tumor adjacent to the proximal bronchial tree. Since simulation, her atelectasis has partially resolved, re-expanding lung along several beam paths. The reference plan was built on the simulation anatomy, when part of that lung was collapsed. The treatment directive documents the prescription, target goals, and serial-OAR goals, but it does not define what degree of interval re-expansion or density change should trigger additional review before trusting on-session dose calculation.

This matters because today's scheduled/on-session recalculated plan is the reference fluence recalculated on the current on-session anatomy. An adapted plan would also be calculated on the same current anatomy. So the first question is not "scheduled versus adapted." The first question is whether today's external contour, lung density representation, and re-expanded anatomy are reliable enough for either dose calculation to be trusted.

Decision Prompt: Before plan comparison, what must the team confirm about the interval density and anatomy change?

Choices

A. Proceed; the scheduled plan is recalculated on today's anatomy, so the density change is already handled.

B. Stop and revise the reference plan offline for the new lung geometry before treating today.

C. Confirm that the on-session image, external contour, and density representation are adequate for dose calculation before trusting either plan, then document the interval re-expansion.

D. Compare the scheduled/on-session recalculated plan and adapted plan first, then investigate the density change only if one plan fails.

Decision Log: Why did your team choose this option?

Notes:

2 ON-SESSION IMAGING

PLAYER CARD

Let's Adapt!

LUNG-02 | Ms. Mona Motion

Scenario: Truncated Contour in the Beam Path

Clinical Situation: On-session CBCT for Ms. Motion confirms the re-expansion, but the inferior body contour is truncated by the field of view in a beam-entry region for several of the central beams. The target itself images well.

Because the truncation sits in the entry path of central beams, neither the recalculated scheduled plan nor the adapted plan can be trusted in that region until the external contour represents today's anatomy - and whether the platform even allows a reliable external-contour correction here is itself uncertain.

Decision Prompt: Can either plan be trusted, can the contour be corrected at all, and if not, what do you do before evaluating dose?

Choices

A. Verify whether the platform allows a reliable external-contour correction; if not, treat plan evaluation as unreliable.

B. Accept; the truncation is peripheral and the target images well.

C. Confirm the external-contour limitation, then proceed only if it does not affect the beam-entry path.

D. Reacquire with an adequate field of view (or re-setup) so the external contour represents today's anatomy before trusting either plan.

Decision Log: Why did your team choose this option?

Notes:

3 OAR CONTOURING

PLAYER CARD

LUNG-02 | Ms. Mona Motion

Let's Adapt!

Scenario: Right Shape, Moved Relationship

Clinical Situation: Ms. Motion's propagated proximal bronchial tree and esophagus contours are correct in shape, but re-expansion has shifted their position relative to the target, and the high-dose abutment point has moved.

The contours look anatomically right, which is reassuring, but the prior high-dose relationship no longer matches today's geometry, so the old max-dose figures may no longer describe the real situation.

Decision Prompt: Do you accept the correct-looking contours or re-evaluate the OAR-target relationship on today's geometry?

Choices

A. Re-evaluate the abutment and recompute serial-OAR max-dose on today's anatomy.

B. Re-evaluate the bronchial/esophageal abutment on today's geometry before planning.

C. Accept; the serial-OAR contours are anatomically correct.

D. Return to contour editing for a serial-OAR review before regenerating the plan.

Decision Log: Why did your team choose this option?

Notes:

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4 TARGET CONTOURING

PLAYER CARD

Let's Adapt!

LUNG-02 | Ms. Mona Motion

Scenario: Tumor or Re-Aerated Lung?

Clinical Situation: Target propagation onto Ms. Motion's re-expanded lung is plausible, but it is unclear whether the propagated GTV reflects true tumor or includes previously collapsed tissue that is now re-aerated. On CBCT, distinguishing the two is genuinely hard.

The propagation tracks today's density and looks reasonable, so accepting it is tempting even though the target's identity - tumor versus re-aerated parenchyma - has not been confirmed.

Decision Prompt: Do you accept the propagated GTV or reconcile it against diagnostic imaging before planning?

Choices

A. Compare to simulation/diagnostic anatomy and adjust the GTV, noting the basis.

B. Accept propagation; it tracks the visible density.

C. Reconcile the GTV to reference/diagnostic imaging and document the re-aeration reasoning.

D. Hold for physician target review before planning.

Decision Log: Why did your team choose this option?

Notes:

5 PLAN SELECTION

PLAYER CARD

LUNG-02 | Ms. Mona Motion

Let's Adapt!

Scenario: Two Plans, Same Anatomy

Clinical Situation: Both of Ms. Motion's plans now evaluate on today's anatomy: the adapted plan improves the serial-OAR relationship, while the recalculated scheduled plan is simpler to verify. The adapted plan's OAR numbers look better. Because both use current anatomy, the choice is really about the OAR-target relationship and verifiability - and about whether the team confirmed the target reconciliation held before being swayed by the better-looking numbers.

Decision Prompt: With both plans on current anatomy, what basis legitimately separates them, and was the target confirmed first?

Choices

A. Pick the adapted plan; its serial-OAR numbers look better.

B. Pick the adapted plan after confirming the target reconciliation held, and document why.

C. Return to contour editing to reconcile the target, then regenerate the adapted plan before selecting.

D. Select per the documented target/OAR review and a secondary dose check, recording rationale.

Decision Log: Why did your team choose this option?

Notes:

6 DELIVERY / CONFIRMATION IMAGING

PLAYER CARD

Let's Adapt!

LUNG-02 | Ms. Mona Motion

Scenario: Still Re-Expanding

Clinical Situation: Confirmation imaging for Ms. Motion shows further partial re-expansion since the adapted plan was approved minutes ago, again shifting the target-to-bronchial relationship. The anatomy is, in other words, still actively changing. Treating now risks delivering to a moving target; re-adapting immediately risks chasing a geometry that has not settled. The team needs to decide what 'stable enough to treat' means here.

Decision Prompt: What defines 'stable enough to treat,' and do you treat, wait and re-image, or restart?

Choices

A. Apply a predefined stability/drift threshold; treat if settled, re-adapt if still moving, document.

B. Restart the adaptive session on the current (latest) anatomy.

C. Wait briefly for stabilization and re-image to confirm the geometry has settled.

D. Treat now; further waiting just delays and anatomy may keep changing.

Decision Log: Why did your team choose this option?

Notes:

LUNG-02 Explanation Summary

Ms. Mona Motion • Central Lung CTgART / Atelectasis Re-Expansion

Patient Context

Clinical Summary	64F, central NSCLC adjacent to proximal bronchial tree, 60 Gy/15 fx adaptive, fraction 7. Atelectasis has partially resolved since simulation, re-expanding lung near the target.
Clinic / Coverage Model	Academic adaptive program; trainee physicist supervised remotely, attending physician in clinic, experienced RTTs.
Adaptive Rationale	Resolving atelectasis re-expands lung and moves the target away from the mediastinum; adaptation prevents treating collapsed-lung geometry.
Allowable Time	35 minutes
Time Consequence	Overage cascades into a central-tumor patient whose proximal-bronchial constraints demand careful review; rushing the next case is the real risk.

Core Lesson

Current-anatomy recalculation is only trustworthy if the current anatomy is represented correctly. Density, external contour, target identity, and serial-OAR geometry all change together.

Overall Handling Approach

Make uncertainty visible, verify the anatomy that drives the decision, use documented thresholds or priority rules, and record the rationale before delivery.

1. Reference Planning — Density That Changed Under the Beams

Why it matters: The reference/treatment directive issue sets the decision rule before the patient is deep into the adaptive session.
General handling approach: Apply or define the missing rule early, document the working basis, and escalate only when the ambiguity affects today's treatment decision.

2. On-Session Imaging — Truncated Contour in the Beam Path

Why it matters: Image adequacy must be judged against the specific decision the image must support, not only global image quality.
General handling approach: Review the decision-driving anatomy directly. Reacquire or mitigate only when the limitation prevents reliable contouring or plan evaluation.

3. OAR Contouring — Right Shape, Moved Relationship

Why it matters: At this stage the team is approving OAR/influencer anatomy for downstream optimization and plan comparison, not relying on a final DVH.
General handling approach: Focus edits on the OAR/influencer regions that can change optimization, dose calculation, or plan comparison; use second review when uncertainty remains clinically material.

4. Target Contouring — Tumor or Re-Aerated Lung?

Why it matters: A plausible propagated target does not by itself prove that clinical target intent has been preserved.
General handling approach: Reconcile propagated targets to reference intent, diagnostic anatomy, prior fractions, or a documented rule before planning.

5. Plan Selection — Two Plans, Same Anatomy

Why it matters: Scheduled/on-session recalculated and adapted plans can both appear acceptable while resolving different clinical risks.
General handling approach: Select based on documented target/OAR priorities, composite checks, sensitivity tests, and secondary dose review rather than the most attractive single metric.

6. Delivery / Confirmation Imaging — Still Re-Expanding

Why it matters: Confirmation imaging should be governed by predefined thresholds, stability checks, or re-verification rather than visual impression or time pressure alone.
General handling approach: Treat only when the approved geometry remains within threshold or can be restored and re-verified; otherwise re-adapt, restart, delay, or cancel according to policy.

LUNG-02 Scoring Sheet

Ms. Mona Motion • Allowable time: 35 minutes

Workflow step	Choice	Safety penalty	Time penalty	Whammy?	Notes
1. Reference Planning	____	____	____	◻ Yes ◻ No	
2. On-Session Imaging	____	____	____	◻ Yes ◻ No	
3. OAR Contouring	____	____	____	◻ Yes ◻ No	
4. Target Contouring	____	____	____	◻ Yes ◻ No	
5. Plan Selection	____	____	____	◻ Yes ◻ No	
6. Delivery / Confirmation Imaging	____	____	____	◻ Yes ◻ No	

Total safety penalties	__________	**Total time used**	__________ min
Time overage	max(0, time - allowable) = ________	**Whammy resolution**	◻ +10 min choose again ◻ +15 safety
Final score	100 - safety penalties - time overage = ________		
Team	________________________	**Facilitator**	________________________

Scoring reminder: Final score = 100 - safety penalties - time overage. If a whammy is selected, add 10 minutes and choose again, or keep the choice and add 15 safety penalty.

LUNG-02 Answer Key • 1. Reference Planning

Ms. Mona Motion

Scenario: Density That Changed Under the Beams

Choice A: Proceed; the scheduled plan is recalculated on today's anatomy, so the density change is already handled.

Scoring: Safety 8; Time 1 min; +3 safety cascade, +2 min cascade; Whammy: No; S/O/D/RPN 4/3/3/36

Explanation: This is the tempting shortcut. Recalculation on today's anatomy only helps if the on-session image, external contour, and density representation are adequate for dose calculation. A large interval density change from atelectasis resolution can affect both the scheduled/on-session recalculated plan and any adapted plan.

Penalty rationale: High safety penalty because the team assumes recalculation solves the density problem without verifying whether today's anatomy is represented correctly. Low time penalty because the team proceeds quickly, but unreliable calculation anatomy can compromise every downstream plan comparison.

Mitigation concept: Do not trust recalculation before verifying calculation anatomy

Choice B: Stop and revise the reference plan offline for the new lung geometry before treating today.

Scoring: Safety 3; Time 10 min; no cascade; Whammy: No; S/O/D/RPN 4/1/2/8

Explanation: This is safe but may be more disruptive than necessary. Offline revision may be appropriate if the new lung geometry cannot be represented reliably on-session or reflects a durable anatomic change requiring replanning. However, the first decision should be whether today's on-session anatomy and density representation are adequate for calculation.

Penalty rationale: Moderate-low safety penalty because the team avoids treating on unverified anatomy. Very high time penalty because offline revision likely delays or cancels today's adaptive session even though a verified on-session calculation may have been adequate.

Mitigation concept: Escalate to offline revision only when current-anatomy calculation is not reliable

Choice C: Confirm that the on-session image, external contour, and density representation are adequate for dose calculation before trusting either plan, then document the interval re-expansion.

Scoring: Safety 1; Time 4 min; no cascade; Whammy: No; S/O/D/RPN 4/1/1/4

Explanation: This is the best balanced choice. It recognizes that both the scheduled/on-session recalculated plan and the adapted plan depend on the same current-anatomy representation. Before comparing plans, the team confirms that the image, external contour, and density representation are adequate for dose calculation and documents the interval re-expansion.

Penalty rationale: Very low safety penalty because the team verifies the calculation anatomy before trusting either plan pathway. Moderate time penalty reflects focused review and documentation without automatically stopping the fraction.

Mitigation concept: Verify current-anatomy density and external contour before plan comparison

Choice D: Compare the scheduled/on-session recalculated plan and adapted plan first, then investigate the density change only if one plan fails.

Scoring: Safety 6; Time 2 min; +2 safety cascade, +2 min cascade; Whammy: No; S/O/D/RPN 4/3/3/36

Explanation: This is plausible under time pressure because plan comparison is often where teams focus their attention. The problem is sequence: plan comparison is not meaningful until the team knows whether today's density and external contour support reliable dose calculation. If both plans use unreliable anatomy, passing or failing metrics may be misleading.

Penalty rationale: Moderate-high safety penalty because the team may compare two plans before confirming whether either calculation is trustworthy. Low time penalty because the team moves directly to plan comparison, but the density uncertainty remains unresolved.

Mitigation concept: Confirm calculation validity before interpreting plan metrics

LUNG-02 Answer Key • 2. On-Session Imaging

Ms. Mona Motion

Scenario: Truncated Contour in the Beam Path

Choice A: Verify whether the platform allows a reliable external-contour correction; if not, treat plan evaluation as unreliable.

Scoring: Safety 5; Time 4 min; no cascade; Whammy: No; S/O/D/RPN 3/3/3/27

Explanation: Reasonable first move: check whether the platform can correct the external/body representation at all. If it cannot, this correctly flags plan evaluation as unreliable -- but on its own it does not yet restore a trustworthy calculation.

Penalty rationale: Low-moderate: the right question, but not yet a fix.

Mitigation concept: Verify platform correction capability before relying on it.

Choice B: Accept; the truncation is peripheral and the target images well.

Scoring: Safety 11; Time 1 min; +8 safety cascade; Whammy: YES - Add 10 minutes and choose again, or keep this choice and add 15 safety penalty.; S/O/D/RPN 5/4/4/80

Explanation: The tempting shortcut: the target looks fine, but a truncated entry-region contour corrupts the recalculation both plans share; treating on it accepts an unverified dose calculation.

Penalty rationale: Whammy: proceeding when body/external-contour adequacy is questionable in a beam-entry region.

Mitigation concept: Confirm the external contour represents today's anatomy before trusting dose.

Choice C: Confirm the external-contour limitation, then proceed only if it does not affect the beam-entry path.

Scoring: Safety 7; Time 5 min; no cascade; Whammy: No; S/O/D/RPN 4/3/3/36

Explanation: Partly reasonable but risky here: the truncation IS in the beam-entry path for central beams, so 'proceed if it does not affect entry' does not apply, and proceeding under-responds to a beam-path limitation.

Penalty rationale: Moderate: under-responds when the entry path is involved.

Mitigation concept: Match the response to whether the beam-entry path is affected.

Choice D: Reacquire with an adequate field of view (or re-setup) so the external contour represents today's anatomy before trusting either plan.

Scoring: Safety 1; Time 11 min; no cascade; Whammy: No; S/O/D/RPN 2/2/1/4

Explanation: Best here: with the external contour truncated in a beam-entry region and online correction uncertain, reacquiring with an adequate FOV (or re-setting up) is the expensive but correct action; it restores a trustworthy basis for BOTH the recalculated scheduled and adapted plans.

Penalty rationale: Low safety, realistic high time: warranted because the entry-path calculation cannot otherwise be trusted.

Mitigation concept: Reacquire/re-setup for adequate FOV when entry-path contour is truncated.

LUNG-02 Answer Key • 3. OAR Contouring

Ms. Mona Motion

Scenario: Right Shape, Moved Relationship

Choice A: Re-evaluate the abutment and recompute serial-OAR max-dose on today's anatomy.

Scoring: Safety 2; Time 7 min; no cascade; Whammy: No; S/O/D/RPN 2/2/2/8

Explanation: Also strong: adds recomputed serial-OAR max-dose on correct anatomy; slightly more time than B.

Penalty rationale: Modest time; valid serial-OAR dose.

Mitigation concept: Abutment review + recompute.

Choice B: Re-evaluate the bronchial/esophageal abutment on today's geometry before planning.

Scoring: Safety 3; Time 4 min; no cascade; Whammy: No; S/O/D/RPN 3/3/3/27

Explanation: Best: re-evaluating the abutment catches the changed high-dose relationship quickly, which is the decisive check here.

Penalty rationale: Low combined penalty: targets the decisive change.

Mitigation concept: Abutment re-evaluation on today's geometry.

Choice C: Accept; the serial-OAR contours are anatomically correct.

Scoring: Safety 11; Time 1 min; +8 safety cascade; Whammy: No; S/O/D/RPN 5/4/5/100

Explanation: Tempting because the shapes are right, but the moved abutment means the old max-dose is uninformative exactly where it matters.

Penalty rationale: High safety, low time: relationship change undetected.

Mitigation concept: Assess OAR-target relationship, not just shape.

Choice D: Return to contour editing for a serial-OAR review before regenerating the plan.

Scoring: Safety 3; Time 8 min; +2 min cascade; Whammy: No; S/O/D/RPN 2/2/2/8

Explanation: A second-reviewer pass fits a central tumor but can stall if the team can do the re-evaluation.

Penalty rationale: High time; calibrate escalation to need.

Mitigation concept: Second-reviewer rule scoped to genuine uncertainty.

LUNG-02 Answer Key • 4. Target Contouring

Ms. Mona Motion

Scenario: Tumor or Re-Aerated Lung?

Choice A: Compare to simulation/diagnostic anatomy and adjust the GTV, noting the basis.

Scoring: Safety 4; Time 4 min; no cascade; Whammy: No; S/O/D/RPN 3/3/3/27

Explanation: Reasonable: comparing to simulation/diagnostic data grounds the GTV, though documentation is lighter than C.

Penalty rationale: Low-moderate: targeted verification.

Mitigation concept: Diagnostic cross-check + documented adjustment.

Choice B: Accept propagation; it tracks the visible density.

Scoring: Safety 10; Time 1 min; +6 safety cascade; Whammy: No; S/O/D/RPN 5/3/4/60

Explanation: Tempting because it tracks visible density, but density alone can mistake re-aerated lung for tumor (or vice versa), misplacing the target.

Penalty rationale: High safety, low time: target identity unverified.

Mitigation concept: Cross-reference target to diagnostic imaging.

Choice C: Reconcile the GTV to reference/diagnostic imaging and document the re-aeration reasoning.

Scoring: Safety 2; Time 7 min; no cascade; Whammy: No; S/O/D/RPN 2/2/2/8

Explanation: Best: full reconciliation with recorded reasoning gives the most defensible target for a central re-aeration case.

Penalty rationale: Modest time; lowest residual target uncertainty.

Mitigation concept: Reference reconciliation + reasoning.

Choice D: Hold for physician target review before planning.

Scoring: Safety 3; Time 10 min; +2 min cascade; Whammy: No; S/O/D/RPN 2/2/2/8

Explanation: Physician target review is reasonable for genuine re-aeration ambiguity but is not a daily default.

Penalty rationale: High time; reserve for true ambiguity.

Mitigation concept: Pre-agreed re-aeration target rule.

LUNG-02 Answer Key • 5. Plan Selection

Ms. Mona Motion

Scenario: Two Plans, Same Anatomy

Choice A: Pick the adapted plan; its serial-OAR numbers look better.

Scoring: Safety 9; Time 1 min; +5 safety cascade; Whammy: No; S/O/D/RPN 4/4/4/64

Explanation: Tempting because the adapted OAR numbers look best, but picking on optics before confirming the target reconciliation can carry a mis-defined GTV forward.

Penalty rationale: High safety: optics-driven selection without target confirmation.

Mitigation concept: Confirm target before selecting on OAR optics.

Choice B: Pick the adapted plan after confirming the target reconciliation held, and document why.

Scoring: Safety 3; Time 3 min; no cascade; Whammy: No; S/O/D/RPN 3/3/3/27

Explanation: Best: choosing adapted after confirming the reconciliation held is substantively correct and quick to document.

Penalty rationale: Low combined penalty: confirmed basis, recorded.

Mitigation concept: Target-confirmed adapted selection.

Choice C: Return to contour editing to reconcile the target, then regenerate the adapted plan before selecting.

Scoring: Safety 3; Time 11 min; +2 min cascade; Whammy: No; S/O/D/RPN 2/2/2/8

Explanation: Returning to editing to reconcile the target is operationally valid but time-heavy if the reconciliation already held.

Penalty rationale: High time; reserve for unresolved target.

Mitigation concept: Return-to-editing only when target is in doubt.

Choice D: Select per the documented target/OAR review and a secondary dose check, recording rationale.

Scoring: Safety 2; Time 5 min; no cascade; Whammy: No; S/O/D/RPN 2/2/2/8

Explanation: Also strong: adds a secondary dose check and rationale; marginally more process than B.

Penalty rationale: Modest time; verified and reconstructable.

Mitigation concept: Review + secondary check + rationale.

LUNG-02 Answer Key • 6. Delivery / Confirmation Imaging

Ms. Mona Motion

Scenario: Still Re-Expanding

Choice A: Apply a predefined stability/drift threshold; treat if settled, re-adapt if still moving, document.

Scoring: Safety 2; Time 7 min; no cascade; Whammy: No; S/O/D/RPN 2/2/2/8

Explanation: Best: a threshold-based stability decision, documented, is the defensible standard for a moving central target.

Penalty rationale: Moderate time; balanced for a moving target.

Mitigation concept: Documented stability/drift threshold.

Choice B: Restart the adaptive session on the current (latest) anatomy.

Scoring: Safety 2; Time 12 min; no cascade; Whammy: No; S/O/D/RPN 2/2/2/8

Explanation: Restart is right if anatomy will not settle but risks chasing a moving baseline.

Penalty rationale: Highest time; only if no stable window exists.

Mitigation concept: Restart only when stability cannot be reached.

Choice C: Wait briefly for stabilization and re-image to confirm the geometry has settled.

Scoring: Safety 4; Time 5 min; no cascade; Whammy: No; S/O/D/RPN 3/3/3/27

Explanation: Reasonable: a short wait and re-image can confirm the geometry settled, though it lacks a formal threshold.

Penalty rationale: Low-moderate: proportionate to dynamic change.

Mitigation concept: Stabilization wait + confirmatory image.

Choice D: Treat now; further waiting just delays and anatomy may keep changing.

Scoring: Safety 10; Time 1 min; no cascade; Whammy: No; S/O/D/RPN 4/3/4/48

Explanation: Tempting under time pressure, but treating actively changing central anatomy risks a serial-OAR hot point with no stability check.

Penalty rationale: High safety, low time: stability assumed under a moving target.

Mitigation concept: Stability criterion before delivery.

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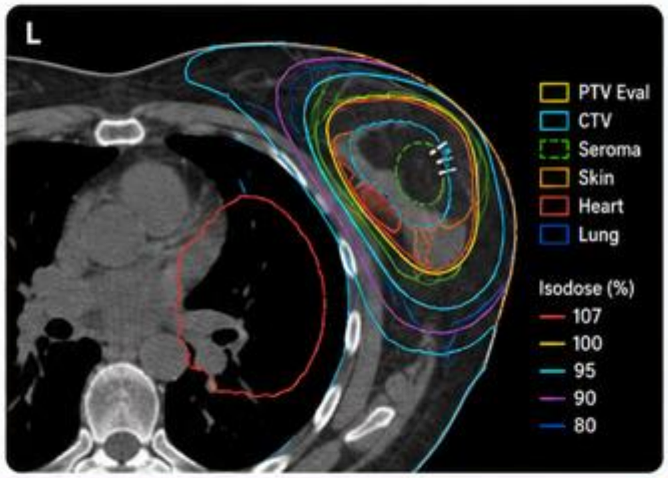
PATIENT CASE FILE • BREAST-01

Ms. Serena Seroma

"Clips say one thing, seroma says another."

1 PATIENT SNAPSHOT

63-year-old	Left breast APBI
Post-lumpectomy	Fraction 4
Prescription: 30 Gy / 5 fx	Binding OAR: Skin
Today's wildcard: Seroma shrank; clips appear shifted	Allowable time: 35 minutes



2 WHY SHE'S ON THE ADAPTIVE COUCH

Her seroma is evolving and the surgical clips appear to have shifted relative to the cavity. Adaptation maintains cavity coverage while limiting skin and heart/lung dose — but clips and seroma are not interchangeable, and no rule says which one defines the target.

3 THE ROOM TODAY

Community adaptive program: physicist on console, physician available by phone, two RTTs. APBI is newly on the adaptive platform here, so the workflow is still settling in.

4 THE CLOCK IS REAL

You get 35 minutes. Overage risks patient positioning tolerance and delays a tangent-setup patient downstream.

SAFETY IS OUR PLAN.
TEAMWORK IS OUR POWER.

SPEAK UP	SHARE THE PLAN	DOUBLE CHECK	SUPPORT EACH OTHER	WE ALL WIN
See something. Say something.	Say it. Confirm it. Own it.	Patient. Plan. Position.	Assume good intent. Deliver safe care.	Every patient. Every time.

TURN THE PAGE TO MEET HER SIX DECISIONS.

1 REFERENCE PLANNING

PLAYER CARD

Let’s Adapt!

BREAST-01 | Ms. Serena Seroma

Scenario: Clips or Seroma?

Clinical Situation: Ms. Seroma is on a post-lumpectomy APBI course. Since simulation, her seroma has shrunk and the surgical clips now appear shifted relative to the seroma/cavity. The reference plan defined the cavity from the simulation seroma but never said whether daily targeting should follow the clips or the soft-tissue seroma when they disagree.

If today the two registration bases point to slightly different target positions, there is no documented rule for which one governs.

Decision Prompt: Which registration basis defines the target today, and what do you document or escalate before planning?

Choices

A. Default to soft-tissue/seroma registration; it is the visible target.

B. Pause and obtain a physician decision on registration basis before treating.

C. Apply a predefined clip-vs-seroma rule, or if none exists, choose the more inclusive basis and flag for a rule.

D. Default to clips today, document the basis, and request a defined clip-vs-seroma rule before next fraction.

Decision Log: Why did your team choose this option?

Notes:

2 ON-SESSION IMAGING

PLAYER CARD

BREAST-01 | Ms. Serena Seroma

Let's Adapt!

Scenario: A Limit Imaging Can't Fix

Clinical Situation: On-session CBCT for Ms. Seroma shows the surgical clips clearly, but the seroma boundary is only faintly visible. The uncertainty appears to be related to limited CBCT soft-tissue contrast and nearby clip artifact, not patient motion or setup error. A repeat CBCT may reproduce the same limitation rather than make the seroma margin clearer.

The team now has to decide what image-based information should drive the adaptive target decision. The clips are easy to see, but the clinical target may not be defined by clips alone. The seroma may be more relevant to the intended target, but it is less reliable on today's image.

Decision Prompt: When the preferred target landmark is poorly visualized and repeat imaging is unlikely to help, how should the team decide the contouring basis?

Choices

A. Reacquire CBCT to try to better define the seroma before target work.

B. Use the clips because they image clearly, and do not consider the faint seroma boundary.

C. Use the clips as reliable anchors, apply window/level review for any visible seroma detail, and document the seroma-visibility limitation.

D. Apply display/contrast optimization and, if the seroma boundary remains uncertain, pause for physician review of the contouring basis.

Decision Log: Why did your team choose this option?

Notes:

3 OAR CONTOURING

PLAYER CARD

BREAST-01 | Ms. Serena Seroma

Let's Adapt!

Scenario: High Dose Creeping Toward Skin

Clinical Situation: Ms. Seroma's generated skin/body and chest-wall contours look reasonable on quick review. However, because the seroma cavity has decreased in size, the high-dose target region is now closer to the skin surface than it was at simulation. The skin has become the decision-driving interface, even though it was not the team's main concern when the reference plan was created.

At this step, the team is not yet relying on the final skin-dose metric. The decision is whether the skin/body contour, buildup region, and chest-wall interface are accurate enough to approve for downstream dose calculation and plan comparison. A small contour or external-surface error may matter more now because the cavity-to-skin distance has narrowed.

Decision Prompt: Do you approve the reasonable-looking skin/body and chest-wall contours, or perform a focused review of the skin interface before allowing planning to proceed?

Choices

A. Perform a focused skin/body-contour review near the cavity and add a max-skin-dose evaluation for downstream plan review.

B. Verify the skin/body contour and buildup-region representation on today's anatomy before approving the OAR contour set.

C. Accept the generated skin and chest-wall contours because they look reasonable overall.

D. Request a second-reviewer check of the skin/chest-wall interface before regenerating the plan.

Decision Log: Why did your team choose this option?

Notes:

4 TARGET CONTOURING

PLAYER CARD

BREAST-01 | Ms. Serena Seroma

Let's Adapt!

Scenario: Committing the Target

Clinical Situation: Ms. Seroma's propagated cavity contour falls between two imperfect landmarks: the clearly visible surgical clips and the smaller, faintly visible seroma cavity. The clips likely represent the surgical bed, but they do not by themselves define the entire cavity. The seroma is smaller and would reduce treated volume, but it may no longer represent the full surgical cavity intended for coverage.

This is the point where the earlier clip-versus-seroma uncertainty becomes a target-definition decision. Choosing the smaller seroma-only contour could spare normal tissue and skin, but it may exclude part of the cavity represented by the clips. Expanding fully to every clip may preserve coverage but could increase dose near the skin if the cavity-to-skin distance has narrowed

Decision Prompt: How should the team define the target when the clips and seroma no longer agree, and what basis should be documented before planning?

Choices

A. Contour to encompass the surgical clips as cavity markers, and document that clip coverage was prioritized.

B. Contour to the visible seroma only because it is smaller and reduces treated volume.

C. Apply the documented target-definition rule, or use an inclusive clip-plus-seroma envelope if no narrower rule exists, and document the reasoning.

D. Hold for physician target review before planning because the clip-versus-seroma basis changes the high-dose volume.

Decision Log: Why did your team choose this option?

Notes:

5 PLAN SELECTION

PLAYER CARD

BREAST-01 | Ms. Serena Seroma

Let's Adapt!

Scenario: Better Coverage, But Why?

Clinical Situation: For today's adaptive session, the team contoured the target primarily to the smaller visible seroma. On plan comparison, the scheduled/on-session recalculated plan shows stronger target coverage metrics, but review reveals that this is partly because it significantly over-covers beyond the seroma-based target. The adapted plan conforms more tightly to today's seroma contour and improves OAR sparing, including skin/chest-wall dose, but its target coverage metrics look less generous.

Both plans can be made to look acceptable depending on what the team values: apparent target coverage, OAR sparing, or consistency with the target-definition basis. The problem is that the contouring basis still has to be clinically justified. A plan should not win simply because it over-covers a possibly under-defined target, and a tighter adapted plan should not win unless the seroma-only basis is defensible.

Decision Prompt: How do you choose between stronger-looking target metrics from over-coverage and better OAR sparing from a tighter adapted plan?

Choices

A. Return to target review to confirm the seroma-based contouring basis before selecting between plans.

B. Pick the scheduled/on-session recalculated plan because the target coverage metrics look better.

C. Select according to the documented target-definition basis and OAR review, recording why over-coverage or tighter adaptation was acceptable.

D. Pick the adapted plan because it better spares skin/chest wall and conforms to today's seroma contour.

Decision Log: Why did your team choose this option?

Notes:

6 DELIVERY / CONFIRMATION IMAGING

PLAYER CARD

Let's Adapt!

BREAST-01 | Ms. Serena Seroma

Scenario: Settling Toward the Skin

Clinical Situation: Confirmation imaging for Ms. Seroma shows the breast and seroma have settled slightly differently after positioning, nudging the high-dose region toward the skin again. The patient is positioned and the change looks small. Re-adapting costs time and patient tolerance, while treating risks a skin hot spot from the settle - and whether this change is within tolerance has not been checked against a defined threshold.

Decision Prompt: Do you treat on the small-looking settle, correct and re-verify, or apply a defined skin-relationship tolerance?

Choices

A. Apply a positioning correction and re-verify the skin relationship before treating.

B. Compare the skin relationship to a predefined tolerance; treat if within, re-adapt if not, document.

C. Restart the adaptive session on the settled anatomy.

D. Treat now; the patient is positioned and the shift looks small.

Decision Log: Why did your team choose this option?

Notes:

BREAST-01 Explanation Summary

Ms. Serena Seroma • APBI / Post-Lumpectomy Adaptive Breast

Patient Context

Clinical Summary	63F, left breast, post-lumpectomy APBI 30 Gy/5 fx, fraction 4. Seroma has shrunk and surgical clips appear shifted relative to the seroma/cavity vs. simulation.
Clinic / Coverage Model	Community adaptive program; physicist on console, physician available by phone, two RTTs. APBI recently brought onto the adaptive platform.
Adaptive Rationale	Seroma evolution and apparent clip shift relative to the cavity change the target between fractions; adaptation maintains cavity coverage while limiting skin and heart/lung dose.
Allowable Time	35 minutes
Time Consequence	Overage risks patient positioning tolerance and delays a tangent-setup patient downstream.

Core Lesson

Target definition must come before plan preference. Clips and seroma are useful but not interchangeable; target basis, OAR sparing, and confirmation thresholds must be documented.

Overall Handling Approach

Make uncertainty visible, verify the anatomy that drives the decision, use documented thresholds or priority rules, and record the rationale before delivery.

1. Reference Planning — Clips or Seroma?

Why it matters: The reference/treatment directive issue sets the decision rule before the patient is deep into the adaptive session.
General handling approach: Apply or define the missing rule early, document the working basis, and escalate only when the ambiguity affects today's treatment decision.

2. On-Session Imaging — A Limit Imaging Can't Fix

Why it matters: Image adequacy must be judged against the specific decision the image must support, not only global image quality.
General handling approach: Review the decision-driving anatomy directly. Reacquire or mitigate only when the limitation prevents reliable contouring or plan evaluation.

3. OAR Contouring — High Dose Creeping Toward Skin

Why it matters: At this stage the team is approving OAR/influencer anatomy for downstream optimization and plan comparison, not relying on a final DVH.
General handling approach: Focus edits on the OAR/influencer regions that can change optimization, dose calculation, or plan comparison; use second review when uncertainty remains clinically material.

4. Target Contouring — Committing the Target

Why it matters: A plausible propagated target does not by itself prove that clinical target intent has been preserved.
General handling approach: Reconcile propagated targets to reference intent, diagnostic anatomy, prior fractions, or a documented rule before planning.

5. Plan Selection — Better Coverage, But Why?

Why it matters: Scheduled/on-session recalculated and adapted plans can both appear acceptable while resolving different clinical risks.
General handling approach: Select based on documented target/OAR priorities, composite checks, sensitivity tests, and secondary dose review rather than the most attractive single metric.

6. Delivery / Confirmation Imaging — Settling Toward the Skin

Why it matters: Confirmation imaging should be governed by predefined thresholds, stability checks, or re-verification rather than visual impression or time pressure alone.
General handling approach: Treat only when the approved geometry remains within threshold or can be restored and re-verified; otherwise re-adapt, restart, delay, or cancel according to policy.

BREAST-01 Scoring Sheet

Ms. Serena Seroma • Allowable time: 35 minutes

Workflow step	Choice	Safety penalty	Time penalty	Whammy?	Notes
1. Reference Planning	____	____	____	◻ Yes ◻ No	
2. On-Session Imaging	____	____	____	◻ Yes ◻ No	
3. OAR Contouring	____	____	____	◻ Yes ◻ No	
4. Target Contouring	____	____	____	◻ Yes ◻ No	
5. Plan Selection	____	____	____	◻ Yes ◻ No	
6. Delivery / Confirmation Imaging	____	____	____	◻ Yes ◻ No	

Total safety penalties	__________	**Total time used**	__________ min
Time overage	max(0, time - allowable) = ________	**Whammy resolution**	◻ +10 min choose again ◻ +15 safety
Final score	100 - safety penalties - time overage = ________		
Team	________________________	**Facilitator**	________________________

Scoring reminder: Final score = 100 - safety penalties - time overage. If a whammy is selected, add 10 minutes and choose again, or keep the choice and add 15 safety penalty.

BREAST-01 Answer Key • 1. Reference Planning

Ms. Serena Seroma

Scenario: Clips or Seroma?

Choice A: Default to soft-tissue/seroma registration; it is the visible target.

Scoring: Safety 7; Time 2 min; +3 safety cascade, +2 min cascade; Whammy: No; S/O/D/RPN 4/3/3/36

Explanation: This is tempting because the seroma was the original simulation basis and remains the visible soft-tissue target. The risk is that the seroma has shrunk and the clips now appear shifted relative to it, so seroma-only registration may exclude part of the surgical cavity intended for coverage.

Penalty rationale: High safety penalty because the team chooses one target basis without a documented clip-versus-seroma rule. Low time penalty because the workflow proceeds quickly, but target-basis uncertainty propagates downstream.

Mitigation concept: Do not default to seroma-only when clips and cavity disagree

Choice B: Pause and obtain a physician decision on registration basis before treating.

Scoring: Safety 2; Time 8 min; no cascade; Whammy: No; S/O/D/RPN 4/1/2/8

Explanation: This is safe and appropriate if the clip-versus-seroma disagreement will materially change the high-dose target volume. It clarifies clinical intent before planning, but it may be slower than necessary if a predefined institutional rule exists.

Penalty rationale: Low safety penalty because target-basis ambiguity is escalated before treatment. High time penalty reflects real-time physician involvement during an adaptive session.

Mitigation concept: Escalate clinically meaningful target-basis ambiguity

Choice C: Apply a predefined clip-vs-seroma rule, or if none exists, choose the more inclusive basis and flag for a rule.

Scoring: Safety 1; Time 4 min; no cascade; Whammy: No; S/O/D/RPN 4/1/1/4

Explanation: This is the best balanced choice. It applies an existing rule if available and, if no rule exists, avoids under-defining the cavity by using the more inclusive clinically reasonable basis while flagging the need for a durable rule.

Penalty rationale: Very low safety penalty because the target basis is tied to a rule or conservative inclusive fallback. Moderate time penalty reflects documentation and follow-up clarification.

Mitigation concept: Documented clip-versus-seroma target-basis rule

Choice D: Default to clips today, document the basis, and request a defined clip-vs-seroma rule before next fraction.

Scoring: Safety 4; Time 3 min; +1 safety cascade, +1 min cascade; Whammy: No; S/O/D/RPN 4/2/2/16

Explanation: This is more defensible than an undocumented choice because the basis is recorded, and clips are reliable surgical markers. The limitation is that clips alone may not define the entire cavity, and the rule is still being created after today’s decision.

Penalty rationale: Moderate safety penalty because the target basis remains clip-driven without a predefined rule. Low-moderate time penalty because documentation is added but no real-time clarification occurs.

Mitigation concept: Clips are anchors, not a complete target-definition rule

BREAST-01 Answer Key • 2. On-Session Imaging

Ms. Serena Seroma

Scenario: A Limit Imaging Can't Fix

Choice A: Reacquire CBCT to try to better define the seroma before target work.

Scoring: Safety 4; Time 6 min; +1 safety cascade, +1 min cascade; Whammy: No; S/O/D/RPN 3/2/2/12

Explanation: This is understandable because the seroma boundary is faint, but the limitation is likely CBCT soft-tissue contrast and nearby clip artifact rather than motion or setup error. Reacquiring may reproduce the same uncertainty.

Penalty rationale: Moderate safety penalty because repeat imaging may be mistaken for resolving a target-basis limitation. Moderate-high time penalty because reacquisition consumes on-table time without reliably improving the decision-driving anatomy.

Mitigation concept: Do not use repeat imaging to solve intrinsic soft-tissue contrast limits

Choice B: Use the clips because they image clearly, and do not consider the faint seroma boundary.

Scoring: Safety 8; Time 1 min; +3 safety cascade, +2 min cascade; Whammy: No; S/O/D/RPN 4/3/3/36

Explanation: This is the tempting shortcut because clips are easy to see on CBCT. The risk is that the clinical target may not be defined by clips alone, and disregarding the faint seroma boundary can shift the adaptive target basis without acknowledging the imaging limitation.

Penalty rationale: High safety penalty because the team lets the most visible landmark replace the intended target basis. Low time penalty because the workflow proceeds quickly, but target uncertainty carries forward.

Mitigation concept: Do not let visible clips replace target-definition intent

Choice C: Use the clips as reliable anchors, apply window/level review for any visible seroma detail, and document the seroma-visibility limitation.

Scoring: Safety 1; Time 3 min; no cascade; Whammy: No; S/O/D/RPN 4/1/1/4

Explanation: This is the best balanced response. It uses the clips as reliable anchors without pretending they fully define the target, extracts available seroma detail from optimized review, and documents the residual limitation.

Penalty rationale: Very low safety penalty because the team uses all reliable image information while preserving the distinction between clips and seroma-defined target intent. Low-moderate time penalty reflects focused image review and documentation.

Mitigation concept: Use reliable landmarks while documenting soft-tissue limitation

Choice D: Apply display/contrast optimization and, if the seroma boundary remains uncertain, pause for physician review of the contouring basis.

Scoring: Safety 2; Time 6 min; no cascade; Whammy: No; S/O/D/RPN 4/1/2/8

Explanation: This is safe and appropriate if seroma-boundary uncertainty could change the contouring basis. It is slower than focused team review alone, but it avoids silently converting an imaging limitation into an undocumented target-definition decision.

Penalty rationale: Low safety penalty because unresolved target-basis uncertainty is escalated. Moderate-high time penalty reflects display optimization plus possible physician review.

Mitigation concept: Escalate unresolved target-basis uncertainty

BREAST-01 Answer Key • 3. OAR Contouring

Ms. Serena Seroma

Scenario: High Dose Creeping Toward Skin

Choice A: Perform a focused skin/body-contour review near the cavity and add a max-skin-dose evaluation for downstream plan review.

Scoring: Safety 2; Time 5 min; no cascade; Whammy: No; S/O/D/RPN 3/1/2/6

Explanation: This is a strong mitigation because it recognizes that the cavity-to-skin distance has narrowed and skin has become a decision-driving interface. It also prepares the team to explicitly review maximum skin dose later. The limitation is that it partly jumps ahead to downstream dose evaluation before the contour set itself has been approved.

Penalty rationale: Low safety penalty because the clinically relevant skin/body region is reviewed and a downstream skin-dose check is added. Moderate time penalty reflects focused contour review and added plan-review step.

Mitigation concept: Focused skin-interface review plus downstream max-dose check

Choice B: Verify the skin/body contour and buildup-region representation on today's anatomy before approving the OAR contour set.

Scoring: Safety 1; Time 4 min; no cascade; Whammy: No; S/O/D/RPN 3/1/1/3

Explanation: This is the best balanced choice. At the OAR contouring step, the team is not yet relying on final skin-dose metrics; instead, it verifies that the skin/body contour and buildup region are represented accurately enough for downstream dose calculation and plan comparison.

Penalty rationale: Very low safety penalty because the calculation-relevant skin/body anatomy is verified before planning. Moderate-low time penalty reflects focused review without unnecessary escalation.

Mitigation concept: Verify calculation-relevant skin/body representation before planning

Choice C: Accept the generated skin and chest-wall contours because they look reasonable overall.

Scoring: Safety 7; Time 1 min; +3 safety cascade, +2 min cascade; Whammy: No; S/O/D/RPN 3/3/3/27

Explanation: This is the tempting shortcut. The contours may look reasonable overall, but the cavity has moved closer to the skin, making small skin/body contour or buildup-region errors more important than they were at simulation.

Penalty rationale: High safety penalty because the team approves a decision-driving skin interface based only on overall contour plausibility. Low time penalty because the workflow proceeds quickly, but downstream dose evaluation may be unreliable.

Mitigation concept: Do not approve a limiting interface based only on overall plausibility

Choice D: Request a second-reviewer check of the skin/chest-wall interface before regenerating the plan.

Scoring: Safety 2; Time 7 min; +1 min cascade; Whammy: No; S/O/D/RPN 3/1/2/6

Explanation: This is safe and defensible if the cavity-to-skin distance is very small, the skin/body contour is difficult to judge, or local policy requires escalation for buildup-region uncertainty. It may be slower than necessary if the interface can be confidently reviewed by the primary team.

Penalty rationale: Low safety penalty because uncertainty in the limiting interface is escalated before planning. Higher time penalty reflects additional review while the patient remains on the table.

Mitigation concept: Second-reviewer escalation for limiting skin interface

BREAST-01 Answer Key • 4. Target Contouring

Ms. Serena Seroma

Scenario: Committing the Target

Choice A: Contour to encompass the surgical clips as cavity markers, and document that clip coverage was prioritized.

Scoring: Safety 3; Time 4 min; +1 safety cascade, +1 min cascade; Whammy: No; S/O/D/RPN 4/2/2/16

Explanation: This is clinically defensible because the clips are reliable surgical-cavity markers and may represent tissue at risk even when the seroma has decreased. The limitation is that clips alone may not fully define the intended cavity and may expand the high-dose volume toward skin if used without considering the seroma and documented target rule.

Penalty rationale: Moderate-low safety penalty because cavity coverage is protected, but the target basis may be overly clip-driven. Moderate time penalty reflects contour adjustment and documentation.

Mitigation concept: Use clips as cavity markers, not the sole target definition

Choice B: Contour to the visible seroma only because it is smaller and reduces treated volume.

Scoring: Safety 9; Time 1 min; +3 safety cascade, +3 min cascade; Whammy: YES - Add 10 minutes and choose again, or keep this choice and add 15 safety penalty.; S/O/D/RPN 4/3/3/36

Explanation: This is the unsafe shortcut. It is attractive because the smaller seroma reduces treated volume and may reduce skin dose, but it risks excluding part of the surgical cavity represented by the clips.

Penalty rationale: High safety penalty because the target may be under-contoured by prioritizing tissue sparing over documented cavity coverage. Low time penalty because contouring is quick, but target-miss risk carries into plan selection.

Mitigation concept: Do not reduce target to seroma-only when clips and seroma disagree

Choice C: Apply the documented target-definition rule, or use an inclusive clip-plus-seroma envelope if no narrower rule exists, and document the reasoning.

Scoring: Safety 1; Time 5 min; no cascade; Whammy: No; S/O/D/RPN 4/1/1/4

Explanation: This is the best balanced choice. It applies the predefined contouring basis if available, and if no narrower rule exists, it uses an inclusive clip-plus-seroma envelope to preserve cavity coverage while documenting why the contour was defined that way.

Penalty rationale: Very low safety penalty because target definition is tied to a documented rule or conservative inclusive envelope. Moderate time penalty reflects careful reconciliation of clips, seroma, and skin proximity before planning.

Mitigation concept: Documented cavity-definition rule or inclusive envelope

Choice D: Hold for physician target review before planning because the clip-versus-seroma basis changes the high-dose volume.

Scoring: Safety 2; Time 8 min; no cascade; Whammy: No; S/O/D/RPN 4/1/2/8

Explanation: This is safe and appropriate if the difference between clip-based and seroma-based contouring meaningfully changes the high-dose volume or skin tradeoff. It is slower than applying a documented rule, but it prevents the team from making an undocumented target-definition decision at the console.

Penalty rationale: Low safety penalty because target-basis uncertainty is escalated before planning. High time penalty reflects physician review during the adaptive session.

Mitigation concept: Escalate clinically meaningful target-basis uncertainty

BREAST-01 Answer Key • 5. Plan Selection

Ms. Serena Seroma

Scenario: Better Coverage, But Why?

Choice A: Return to target review to confirm the seroma-based contouring basis before selecting between plans.

Scoring: Safety 2; Time 7 min; no cascade; Whammy: No; S/O/D/RPN 4/1/2/8

Explanation: This is safe and appropriate if the seroma-based target contour has not been clinically justified. It prevents the team from choosing between plans before knowing whether the target basis is valid. The drawback is time, especially late in a 5-fraction APBI course.

Penalty rationale: Low safety penalty because target-basis uncertainty is addressed before plan selection. Higher time penalty reflects returning to target review and possibly regenerating plan comparison.

Mitigation concept: Confirm target basis before plan selection

Choice B: Pick the scheduled/on-session recalculated plan because the target coverage metrics look better.

Scoring: Safety 6; Time 2 min; +2 safety cascade, +2 min cascade; Whammy: No; S/O/D/RPN 4/3/3/36

Explanation: This is tempting because the scheduled/on-session recalculated plan appears to have stronger target coverage metrics. The problem is that the better metrics partly reflect significant over-coverage beyond a possibly under-defined seroma-based target, so the metric may not represent a superior clinical choice.

Penalty rationale: Moderate-high safety penalty because apparent coverage is accepted without resolving whether the target basis and over-coverage are appropriate. Low time penalty because plan selection is quick, but the rationale remains weak.

Mitigation concept: Do not let over-coverage masquerade as better target definition

Choice C: Select according to the documented target-definition basis and OAR review, recording why over-coverage or tighter adaptation was acceptable.

Scoring: Safety 1; Time 4 min; no cascade; Whammy: No; S/O/D/RPN 4/1/1/4

Explanation: This is the best balanced choice if the target-definition basis has been confirmed. It uses the documented cavity basis and skin/OAR review to decide whether the scheduled plan's over-coverage or the adapted plan's tighter OAR-sparing geometry is clinically acceptable.

Penalty rationale: Very low safety penalty because plan selection is anchored to documented target basis and OAR review. Moderate time penalty reflects review of coverage, over-coverage, OAR sparing, and documentation.

Mitigation concept: Target-basis and OAR-driven plan selection

Choice D: Pick the adapted plan because it better spares skin/chest wall and conforms to today's seroma contour.

Scoring: Safety 8; Time 1 min; +3 safety cascade, +2 min cascade; Whammy: No; S/O/D/RPN 4/3/3/36

Explanation: This is the tempting adaptive shortcut. The adapted plan improves OAR sparing and conforms tightly to the seroma, but if the seroma-only target basis is not justified, the plan may be sparing normal tissue by under-covering the intended surgical cavity.

Penalty rationale: High safety penalty because OAR sparing is allowed to substitute for a confirmed target-definition basis. Low time penalty because the team proceeds quickly, but possible under-coverage carries into delivery.

Mitigation concept: Do not select tighter adaptation until target basis is justified

BREAST-01 Answer Key • 6. Delivery / Confirmation Imaging

Ms. Serena Seroma

Scenario: Settling Toward the Skin

Choice A: Apply a positioning correction and re-verify the skin relationship before treating.

Scoring: Safety 2; Time 6 min; no cascade; Whammy: No; S/O/D/RPN 3/1/2/6

Explanation: This is safe and reasonable if the settle appears setup-related and a correction can restore the planned skin relationship. It is less complete than explicitly comparing the change to a defined tolerance, but it addresses the observed shift before delivery.

Penalty rationale: Low safety penalty because the skin relationship is checked again before treatment. Moderate time penalty reflects correction and re-verification.

Mitigation concept: Correct and re-verify skin relationship when setup-related

Choice B: Compare the skin relationship to a predefined tolerance; treat if within, re-adapt if not, document.

Scoring: Safety 1; Time 4 min; no cascade; Whammy: No; S/O/D/RPN 3/1/1/3

Explanation: This is the best balanced choice. It uses a defined threshold to decide whether the small-looking settle is acceptable, rather than relying on visual impression or automatically restarting.

Penalty rationale: Very low safety penalty because the decision is threshold-based and documented. Moderate-low time penalty reflects measurement and documentation, with escalation only if tolerance is exceeded.

Mitigation concept: Threshold-based confirmation skin-interface decision

Choice C: Restart the adaptive session on the settled anatomy.

Scoring: Safety 2; Time 10 min; no cascade; Whammy: No; S/O/D/RPN 3/1/2/6

Explanation: This is safe but may be more than necessary if the settle is within the predefined skin-relationship tolerance. It may be appropriate if the relationship exceeds tolerance or cannot be confidently evaluated.

Penalty rationale: Low safety penalty because the plan is rebuilt on current anatomy. Very high time penalty because restarting the session is disruptive and may worsen patient tolerance.

Mitigation concept: Restart only when confirmation threshold requires it

Choice D: Treat now; the patient is positioned and the shift looks small.

Scoring: Safety 8; Time 1 min; no cascade; Whammy: No; S/O/D/RPN 3/3/3/27

Explanation: This is the unsafe shortcut. The change looks small, but the skin relationship is the decision-driving issue for this fraction, and the team has not checked whether the shift is within tolerance.

Penalty rationale: High safety penalty because visual impression substitutes for a defined confirmation threshold. Low time penalty because treatment proceeds immediately, but a skin hot spot risk may be missed.

Mitigation concept: Do not treat through unmeasured skin-interface drift

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PATIENT CASE FILE • BREAST-02

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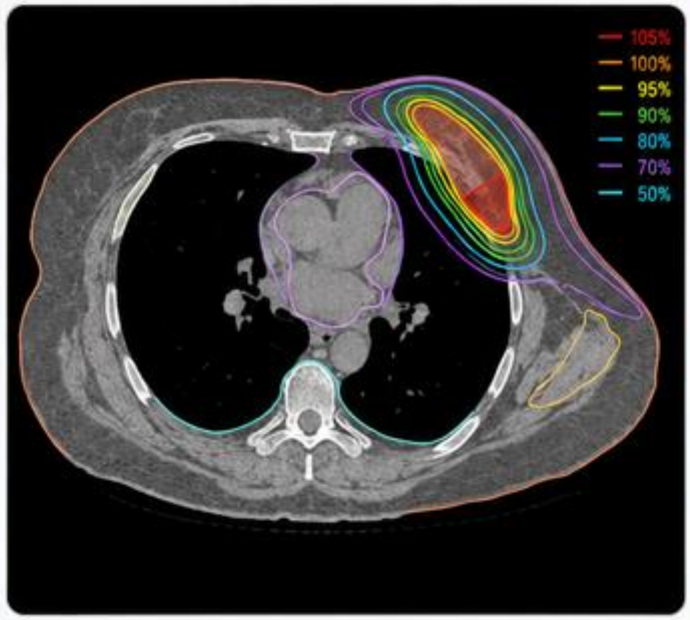
LEARN • CONNECT • ADVANCE

Ms. Aria Armposition

"Not every daily difference deserves adapting to."

1 PATIENT SNAPSHOT

- 48-year-old
- Right breast APBI
- Breast-board setup
- Fraction 2
- Prescription: 30 Gy / 5 fx
- Binding OARs: Skin & rib
- Today's wildcard: Arm/shoulder position differs today
- Allowable time: 35 minutes



2 WHY SHE'S ON THE ADAPTIVE COUCH

Her breast shape and external contour are setup-sensitive on the breast board. Adaptation corrects for daily arm/shoulder positioning while protecting skin, rib, and lung — but the team must tell a correctable setup error apart from a true anatomic change before adapting to it.

3 THE ROOM TODAY

High-volume adaptive program: physicist on console, physician remote, two RTTs. Breast-board APBI is newer to the team, so setup nuances are still being learned.

4 THE CLOCK IS REAL

You get 35 minutes. Overage risks board/arm tolerance and pushes a heart-sparing DIBH patient later, compressing that careful setup.

SAFETY FIRST. TEAMWORK ALWAYS. PATIENTS COUNT ON BOTH.

- **SPEAK UP** Share concerns early.
- **CHECK & VERIFY** Confirm before you commit.
- **FOCUS ON THE GOAL** Right dose, right place, right patient.
- **RESPECT & SUPPORT** We're all part of the same mission.

TURN THE PAGE TO MEET HER SIX DECISIONS.

1 REFERENCE PLANNING

PLAYER CARD

Let's Adapt!

BREAST-02 | Ms. Aria Armposition

Scenario: The Arm Moved

Clinical Situation: Ms. Armposition is on a right-breast APBI course treated on a breast bOARd. Today her ipsilateral arm and shoulder position differ visibly from simulation, which changes the breast shape and the external contour. Bone and breast registration still look workable.

The reference plan assumed a specific arm position but never defined a tolerance for daily arm-position variation or when such variation requires re-setup, so it is unclear how much of today's difference is acceptable.

Decision Prompt: How much arm-position variation is acceptable before the plan no longer applies, and what do you check first?

Choices

A. Proceed; breast-bOARd setups always vary a little and registration is workable.

B. Assess the soft-tissue/external-contour match (not just bone) and document the arm-position difference before proceeding.

C. Assess arm position against a defined tolerance; correct setup or adapt as indicated and document.

D. Re-setup the patient to better match the simulation arm position before continuing.

Decision Log: Why did your team choose this option?

Notes:

2 ON-SESSION IMAGING

PLAYER CARD

Let's Adapt!

BREAST-02 | Ms. Aria Armposition

Scenario: Bone Matches, Contour Doesn't

Clinical Situation: On-session CBCT for Ms. Armposition registers well to bone, but the breast external contour and the arm/axillary region differ from simulation in a beam-entry region. A solid bone match makes the scan look acceptable. Because the changed external contour sits in the beam path, it affects the calculated skin, rib, and lung dose for both the recalculated scheduled and adapted plans - and whether the platform can reliably correct the external representation here is uncertain.

Decision Prompt: Is a good bone match sufficient here, can the external contour be corrected reliably, and if not, what do you do?

Choices

A. Accept; bone registration is solid and that is the primary match.

B. Re-setup to restore the simulation arm position so the external contour represents a known geometry.

C. Verify whether the platform allows a reliable external-contour correction for today's arm geometry; if not, treat plan evaluation as unreliable.

D. Confirm the external contour represents today's arm geometry before trusting dose; if it cannot be corrected reliably, reacquire or re-setup.

Decision Log: Why did your team choose this option?

Notes:

3 OAR CONTOURING

PLAYER CARD

Let’s Adapt!

BREAST-02 | Ms. Aria Armposition

Scenario: Skin and Rib in a Reshaped Field

Clinical Situation: Ms. Armposition’s heart and lung are away from the high-dose region, but today’s arm and shoulder position have changed the breast shape, external contour, and rib/skin geometry near the beam-entry region. The generated skin/body and rib contours look reasonable on quick review, but the geometry closest to the high-dose region is not the same as simulation.

At this step, the team is not yet relying on final skin or rib dose metrics. The decision is whether the skin/body contour, rib contour, buildup region, and external surface near the beam path are accurate enough to approve for downstream dose calculation and plan comparison. A small contour error may matter more today because the arm-position change has altered the skin-entry geometry.

Decision Prompt: Do you approve the reasonable-looking skin/body and rib contours, or perform a focused review of the dose-relevant interface before allowing planning to proceed?

Choices

A. Verify the skin/body contour, rib contour, and buildup-region representation on today’s arm geometry before approving the OAR contour set.

B. Request a second-reviewer check of the skin/rib interface before regenerating the plan.

C. Perform a focused skin/rib review near the beam-entry region and add a downstream skin/rib dose check for plan review.

D. Accept the generated skin and rib contours because heart and lung are away from the field and the contours look reasonable overall.

Decision Log: Why did your team choose this option?

Notes:

4 TARGET CONTOURING

PLAYER CARD

BREAST-02 | Ms. Aria Armposition

Let’s Adapt!

Scenario: Setup Error or Real Change?

Clinical Situation: Ms. Armposition's propagated cavity shifts with the arm-changed breast shape. The team has to decide whether to adapt the target to today's shape or to correct the arm position and restore the planned geometry.

Adapting looks like routine adaptation, but if the shift is really a correctable setup difference, adapting to it would bake a daily setup error into the target.

Decision Prompt: Is today's shift a real anatomic change to adapt to, or a setup difference to correct first?

Choices

A. Apply a defined rule to separate setup difference from anatomic change, then act and document.

B. Adapt the target to today's breast shape; that is what adaptation is for.

C. Hold for physician target review before planning.

D. Attempt arm-position correction first; adapt only if the shift persists, and document.

Decision Log: Why did your team choose this option?

Notes:

5 PLAN SELECTION

PLAYER CARD

Let's Adapt!

BREAST-02 | Ms. Aria Armposition

Scenario: Matches Today, But Why?

Clinical Situation: Ms. Armposition's adapted plan, built on today's arm shape, meets goals. After arm-position correction and re-verification, the on-session recalculated plan also meets goals. Both are acceptable.

The adapted plan matches today's arm-shifted anatomy, but if the shift was a correctable setup difference rather than a true anatomic change, selecting it could normalize a daily setup error across the course.

Decision Prompt: Does selecting the adapted plan risk treating around an unresolved arm-position problem, and how do you decide?

Choices

A. Select per the setup-vs-anatomy determination, confirm with a secondary check, and document.

B. Pick the adapted plan; it matches today and meets goals.

C. Pick the on-session recalculated plan after confirming the arm position is restored, and document.

D. Return to contour editing after arm-position correction, regenerate, then select.

Decision Log: Why did your team choose this option?

Notes:

6 DELIVERY / CONFIRMATION IMAGING

PLAYER CARD

Let's Adapt!

BREAST-02 | Ms. Aria Armposition

Scenario: The Arm Relaxes

Clinical Situation: After approval, Ms. Armposition relaxes her arm slightly, which changes the external contour and nudges the skin-entry geometry. Re-setup is uncomfortable for her, and the change looks minor.

Here a brief re-position back to the approved geometry is often genuinely enough to restore intent, which makes this a case where a proportionate correction - not a restart - is usually the right call.

Decision Prompt: Do you re-position and re-verify, apply a tolerance, restart, or treat on the minor-looking change?

Choices

A. Re-position the arm to the approved geometry and re-verify the skin-entry relationship, then treat.

B. Restart the adaptive session on the relaxed-arm position.

C. Compare the skin-entry change to a predefined tolerance; treat if within, re-adapt if not, document.

D. Treat now; the arm relaxation looks minor and the plan was approved.

Decision Log: Why did your team choose this option?

Notes:

BREAST-02 Explanation Summary

Ms. Aria Armposition • APBI / Breast BOARd Setup Geometry

Patient Context

Clinical Summary	48F, right breast APBI 30 Gy/5 fx, fraction 2. Today the ipsilateral arm/shoulder position on the breast bOARd differs from simulation, changing breast shape and the external contour.
Clinic / Coverage Model	High-volume adaptive program; physicist on console, physician remote, two RTTs; breast-bOARd APBI is newer to the team.
Adaptive Rationale	Breast shape and the external contour are setup-sensitive on the breast bOARd; adaptation corrects for daily arm/shoulder positioning while protecting skin, rib, and lung.
Allowable Time	35 minutes
Time Consequence	Overage risks bOARd/arm tolerance and pushes a heart-sparing DIBH patient later, compressing that careful setup.

Core Lesson

Not every daily difference should be adapted to. Arm and shoulder setup variation can change dose geometry and must be distinguished from true anatomy.

Overall Handling Approach

Make uncertainty visible, verify the anatomy that drives the decision, use documented thresholds or priority rules, and record the rationale before delivery.

1. Reference Planning — The Arm Moved

Why it matters: The reference/treatment directive issue sets the decision rule before the patient is deep into the adaptive session.
General handling approach: Apply or define the missing rule early, document the working basis, and escalate only when the ambiguity affects today's treatment decision.

2. On-Session Imaging — Bone Matches, Contour Doesn't

Why it matters: Image adequacy must be judged against the specific decision the image must support, not only global image quality.
General handling approach: Review the decision-driving anatomy directly. Reacquire or mitigate only when the limitation prevents reliable contouring or plan evaluation.

3. OAR Contouring — Skin and Rib in a Reshaped Field

Why it matters: At this stage the team is approving OAR/influencer anatomy for downstream optimization and plan comparison, not relying on a final DVH.
General handling approach: Focus edits on the OAR/influencer regions that can change optimization, dose calculation, or plan comparison; use second review when uncertainty remains clinically material.

4. Target Contouring — Setup Error or Real Change?

Why it matters: A plausible propagated target does not by itself prove that clinical target intent has been preserved.
General handling approach: Reconcile propagated targets to reference intent, diagnostic anatomy, prior fractions, or a documented rule before planning.

5. Plan Selection — Matches Today, But Why?

Why it matters: Scheduled/on-session recalculated and adapted plans can both appear acceptable while resolving different clinical risks.
General handling approach: Select based on documented target/OAR priorities, composite checks, sensitivity tests, and secondary dose review rather than the most attractive single metric.

6. Delivery / Confirmation Imaging — The Arm Relaxes

Why it matters: Confirmation imaging should be governed by predefined thresholds, stability checks, or re-verification rather than visual impression or time pressure alone.
General handling approach: Treat only when the approved geometry remains within threshold or can be restored and re-verified; otherwise re-adapt, restart, delay, or cancel according to policy.

BREAST-02 Scoring Sheet

Ms. Aria Armposition • Allowable time: 35 minutes

Workflow step	Choice	Safety penalty	Time penalty	Whammy?	Notes
1. Reference Planning	____	____	____	☐ Yes ☐ No	
2. On-Session Imaging	____	____	____	☐ Yes ☐ No	
3. OAR Contouring	____	____	____	☐ Yes ☐ No	
4. Target Contouring	____	____	____	☐ Yes ☐ No	
5. Plan Selection	____	____	____	☐ Yes ☐ No	
6. Delivery / Confirmation Imaging	____	____	____	☐ Yes ☐ No	

Total safety penalties	__________	**Total time used**	__________ min
Time overage	max(0, time - allowable) = ________	**Whammy resolution**	☐ +10 min choose again ☐ +15 safety
Final score	100 - safety penalties - time overage = ________		
Team	________________________	**Facilitator**	________________________

Scoring reminder: Final score = 100 - safety penalties - time overage. If a whammy is selected, add 10 minutes and choose again, or keep the choice and add 15 safety penalty.

BREAST-02 Answer Key • 1. Reference Planning

Ms. Aria Armposition

Scenario: The Arm Moved

Choice A: Proceed; breast-bOARd setups always vary a little and registration is workable.

Scoring: Safety 10; Time 1 min; +8 safety cascade; Whammy: No; S/O/D/RPN 4/4/4/64

Explanation: Tempting because registration is workable, but bone/breast match says little about how the arm change reshaped the external contour and beam paths.

Penalty rationale: High safety, low time: external-contour effect unassessed.

Mitigation concept: Defined arm/external-contour tolerance + soft-tissue check.

Choice B: Assess the soft-tissue/external-contour match (not just bone) and document the arm-position difference before proceeding.

Scoring: Safety 4; Time 3 min; no cascade; Whammy: No; S/O/D/RPN 3/3/3/27

Explanation: Reasonable: checking the external-contour match catches what bone registration misses, though it stops short of a defined tolerance.

Penalty rationale: Low-moderate: targets the real variable.

Mitigation concept: External-contour match assessment + documentation.

Choice C: Assess arm position against a defined tolerance; correct setup or adapt as indicated and document.

Scoring: Safety 2; Time 5 min; no cascade; Whammy: No; S/O/D/RPN 2/2/2/8

Explanation: Best: a tolerance-based decision with correction or adaptation is the durable, reproducible approach.

Penalty rationale: Modest time; safe and reproducible.

Mitigation concept: Tolerance-driven correction or adaptation.

Choice D: Re-setup the patient to better match the simulation arm position before continuing.

Scoring: Safety 2; Time 11 min; +2 min cascade; Whammy: No; S/O/D/RPN 2/2/2/8

Explanation: Re-setup is right for a large mismatch but heavy for arm variation that is within tolerance.

Penalty rationale: High time; reserve for out-of-tolerance setups.

Mitigation concept: Re-setup only beyond tolerance.

BREAST-02 Answer Key • 2. On-Session Imaging

Ms. Aria Armposition

Scenario: Bone Matches, Contour Doesn't

Choice A: Accept; bone registration is solid and that is the primary match.

Scoring: Safety 11; Time 1 min; +8 safety cascade; Whammy: YES - Add 10 minutes and choose again, or keep this choice and add 15 safety penalty.; S/O/D/RPN 5/3/4/60

Explanation: The tempting shortcut: bone match looks definitive, but it can mask an external-contour change that invalidates entry-region skin/rib dose for both the recalculated scheduled and adapted plans.

Penalty rationale: Whammy: accepting a good bone match while external-contour adequacy is questionable in the decision region.

Mitigation concept: Confirm external contour when beam-path geometry changes.

Choice B: Re-setup to restore the simulation arm position so the external contour represents a known geometry.

Scoring: Safety 3; Time 10 min; +2 min cascade; Whammy: No; S/O/D/RPN 2/2/2/8

Explanation: Re-setup restores a known geometry but adds time and board/arm-tolerance cost when confirming/reacquiring the contour would suffice.

Penalty rationale: High time; reserve for large external-contour change.

Mitigation concept: Re-setup when the contour cannot be confirmed or reacquired.

Choice C: Verify whether the platform allows a reliable external-contour correction for today's arm geometry; if not, treat plan evaluation as unreliable.

Scoring: Safety 5; Time 3 min; no cascade; Whammy: No; S/O/D/RPN 3/3/3/27

Explanation: Reasonable first move: check whether the platform can correct the external/body representation at all. If it cannot, this correctly flags plan evaluation as unreliable -- but on its own it does not yet restore a trustworthy calculation.

Penalty rationale: Low-moderate: the right question, but not yet a fix.

Mitigation concept: Verify platform correction capability before relying on it.

Choice D: Confirm the external contour represents today's arm geometry before trusting dose; if it cannot be corrected reliably, reacquire or re-setup.

Scoring: Safety 3; Time 6 min; no cascade; Whammy: No; S/O/D/RPN 2/2/2/8

Explanation: Best: confirm the external contour represents today's arm geometry before trusting dose, and reacquire/re-setup if it cannot be corrected reliably; this grounds both the dose calc and the target/OAR geometry.

Penalty rationale: Moderate time; valid dose plus verified geometry.

Mitigation concept: Confirm-then-reacquire external contour before plan evaluation.

BREAST-02 Answer Key • 3. OAR Contouring

Ms. Aria Armposition

Scenario: Skin and Rib in a Reshaped Field

Choice A: Verify the skin/body contour, rib contour, and buildup-region representation on today's arm geometry before approving the OAR contour set.

Scoring: Safety 1; Time 4 min; no cascade; Whammy: No; S/O/D/RPN 3/1/1/3

Explanation: This is the best balanced choice. At the OAR contouring step, the team is not yet relying on final skin or rib dose metrics. The key task is to verify that the dose-relevant skin/body contour, rib contour, buildup region, and external surface near the beam path accurately represent today's arm-shifted geometry before downstream dose calculation and plan comparison.

Penalty rationale: Very low safety penalty because the calculation-relevant OAR and external-contour geometry is verified before planning. Moderate-low time penalty reflects focused review without unnecessary escalation.

Mitigation concept: Verify dose-relevant skin/rib and buildup representation before planning

Choice B: Request a second-reviewer check of the skin/rib interface before regenerating the plan.

Scoring: Safety 2; Time 7 min; +1 min cascade; Whammy: No; S/O/D/RPN 3/1/2/6

Explanation: This is safe and defensible if the arm-position change has made the skin/rib interface difficult to judge, if the cavity-to-skin or cavity-to-rib distance is small, or if local policy requires escalation for buildup-region uncertainty. It may be slower than necessary if the primary team can confidently verify the interface.

Penalty rationale: Low safety penalty because uncertainty in the dose-relevant skin/rib interface is escalated before planning. Higher time penalty reflects additional review while the patient remains on the table.

Mitigation concept: Second-reviewer escalation for limiting skin/rib interface

Choice C: Perform a focused skin/rib review near the beam-entry region and add a downstream skin/rib dose check for plan review.

Scoring: Safety 2; Time 5 min; no cascade; Whammy: No; S/O/D/RPN 3/1/2/6

Explanation: This is a strong mitigation because it focuses review on the beam-entry region where the arm-position change has altered the external contour and rib/skin geometry. It also prepares the team to explicitly evaluate skin and rib dose later. The limitation is that it partly jumps ahead to downstream dose review before the contour set itself is fully approved.

Penalty rationale: Low safety penalty because the decision-driving skin/rib region is reviewed and a downstream dose check is added. Moderate time penalty reflects focused contour review plus added plan-review burden.

Mitigation concept: Focused skin/rib review plus downstream dose check

Choice D: Accept the generated skin and rib contours because heart and lung are away from the field and the contours look reasonable overall.

Scoring: Safety 8; Time 1 min; +3 safety cascade, +2 min cascade; Whammy: No; S/O/D/RPN 3/3/3/27

Explanation: This is the tempting shortcut. Heart and lung may be away from the high-dose region, but the decision-driving OARs for this fraction are skin and rib near the beam-entry region. Because the arm-position change altered the external contour and buildup geometry, reasonable-looking contours may still be inadequate for reliable downstream dose calculation.

Penalty rationale: High safety penalty because the team approves the limiting skin/rib interface based only on overall plausibility and attention to the wrong OARs. Low time penalty because the workflow proceeds quickly, but downstream plan evaluation may be unreliable.

Mitigation concept: Do not approve a reshaped limiting interface based only on overall plausibility

BREAST-02 Answer Key • 4. Target Contouring

Ms. Aria Armposition

Scenario: Setup Error or Real Change?

Choice A: Apply a defined rule to separate setup difference from anatomic change, then act and document.

Scoring: Safety 2; Time 6 min; no cascade; Whammy: No; S/O/D/RPN 2/2/2/8

Explanation: Also strong: a defined rule separates the two causes and documents the action; marginally more process than B.

Penalty rationale: Modest time; most teachable.

Mitigation concept: Rule-based setup-vs-anatomy decision.

Choice B: Adapt the target to today's breast shape; that is what adaptation is for.

Scoring: Safety 11; Time 1 min; +6 safety cascade; Whammy: No; S/O/D/RPN 5/3/4/60

Explanation: Tempting because it looks like routine adaptation, but adapting to a correctable arm-position difference bakes a daily setup change into the target.

Penalty rationale: High safety, low time: setup difference mistaken for anatomy.

Mitigation concept: Setup-correction-first rule before target adaptation.

Choice C: Hold for physician target review before planning.

Scoring: Safety 3; Time 10 min; +2 min cascade; Whammy: No; S/O/D/RPN 2/2/2/8

Explanation: Physician review helps in ambiguous cases but a setup-first rule resolves most days without escalation.

Penalty rationale: High time; reserve for genuine ambiguity.

Mitigation concept: Setup-vs-anatomy protocol.

Choice D: Attempt arm-position correction first; adapt only if the shift persists, and document.

Scoring: Safety 3; Time 4 min; no cascade; Whammy: No; S/O/D/RPN 3/3/3/27

Explanation: Best: trying arm-position correction first cleanly distinguishes a fixable setup difference from a true change, at low cost.

Penalty rationale: Low combined penalty: correct the cause before adapting.

Mitigation concept: Setup-first, adapt-if-persistent.

BREAST-02 Answer Key • 5. Plan Selection

Ms. Aria Armposition

Scenario: Matches Today, But Why?

Choice A: Select per the setup-vs-anatomy determination, confirm with a secondary check, and document.

Scoring: Safety 2; Time 5 min; no cascade; Whammy: No; S/O/D/RPN 2/2/2/8

Explanation: Also strong: ties selection to the cause determination and a secondary check; marginally more process than B.

Penalty rationale: Modest time; verified and reconstructable.

Mitigation concept: Determination + secondary check + record.

Choice B: Pick the adapted plan; it matches today and meets goals.

Scoring: Safety 9; Time 1 min; +5 safety cascade; Whammy: No; S/O/D/RPN 4/4/4/64

Explanation: Tempting because it matches today, but selecting adapted without resolving the cause can entrench daily arm-position variation across the course.

Penalty rationale: High safety: convenience of 'matches today' over cause resolution.

Mitigation concept: Resolve setup-vs-anatomy before selecting adapted.

Choice C: Pick the on-session recalculated plan after confirming the arm position is restored, and document.

Scoring: Safety 3; Time 3 min; no cascade; Whammy: No; S/O/D/RPN 3/3/3/27

Explanation: Best: choosing the on-session recalculated plan after restoring arm position treats the intended geometry and is quick to confirm.

Penalty rationale: Low combined penalty: cause-resolved selection.

Mitigation concept: Setup-restored recalculated-plan selection.

Choice D: Return to contour editing after arm-position correction, regenerate, then select.

Scoring: Safety 3; Time 11 min; +2 min cascade; Whammy: No; S/O/D/RPN 2/2/2/8

Explanation: Returning to editing after correction is operationally valid but time-heavy if setup-first already resolved it.

Penalty rationale: High time; reserve for unresolved setups.

Mitigation concept: Return-to-editing only when cause remains unclear.

BREAST-02 Answer Key • 6. Delivery / Confirmation Imaging

Ms. Aria Armposition

Scenario: The Arm Relaxes

Choice A: Re-position the arm to the approved geometry and re-verify the skin-entry relationship, then treat.

Scoring: Safety 2; Time 4 min; no cascade; Whammy: No; S/O/D/RPN 2/2/2/8

Explanation: Best here: the change is a correctable arm relaxation, so re-positioning to the approved geometry and re-verifying directly restores intent at low cost -- the safest option is also the efficient one.

Penalty rationale: Low combined penalty: a proportionate fix that resolves the actual cause.

Mitigation concept: Re-position to approved geometry + skin-entry re-verify.

Choice B: Restart the adaptive session on the relaxed-arm position.

Scoring: Safety 3; Time 11 min; no cascade; Whammy: No; S/O/D/RPN 2/2/2/8

Explanation: Justified if re-positioning fails or the change is large, but a full restart for a minor correctable relaxation is heavy on arm tolerance.

Penalty rationale: High time; reserve for uncorrectable change.

Mitigation concept: Restart only when re-position fails.

Choice C: Compare the skin-entry change to a predefined tolerance; treat if within, re-adapt if not, document.

Scoring: Safety 3; Time 6 min; no cascade; Whammy: No; S/O/D/RPN 2/3/2/12

Explanation: Reasonable: a tolerance-based decision is defensible and documented, though for a clearly correctable arm relaxation it adds process beyond simply re-positioning.

Penalty rationale: Moderate time; sound but slightly more than needed here.

Mitigation concept: Documented skin-entry tolerance decision.

Choice D: Treat now; the arm relaxation looks minor and the plan was approved.

Scoring: Safety 10; Time 1 min; no cascade; Whammy: No; S/O/D/RPN 4/3/4/48

Explanation: Tempting because it looks minor and the plan is approved, but treating through an unverified external-contour change is the last-gate failure.

Penalty rationale: High safety, low time: external-contour change unverified at the gate.

Mitigation concept: Verify external-contour change before delivery.

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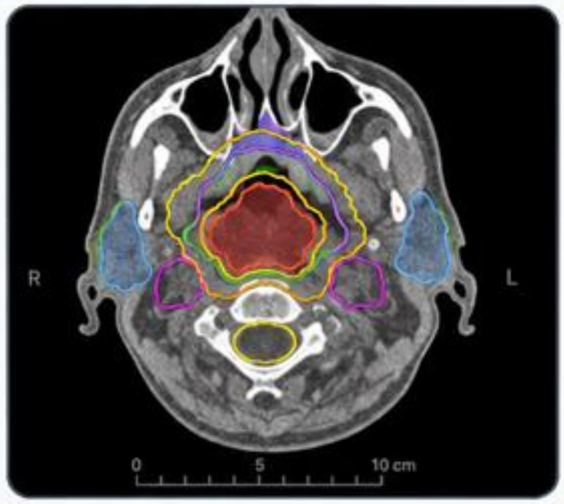
PATIENT CASE FILE • HN-01

Mr. Manny Maskfit

"A closing mask is not a fitting mask."

1 PATIENT SNAPSHOT

56-year-old male	HPV+ oropharynx	Bilateral nodal levels	Fraction 18
Prescription: 70 Gy / 35 fx	Serial OARs: Cord & parotids	Today's wildcard: ~5 kg weight loss; loose mask	Allowable time: 35 minutes



2 WHY HE'S ON THE ADAPTIVE COUCH

Weight loss and nodal response have changed his neck contour and parotid position. Adaptation maintains nodal coverage while sparing parotids and cord — but a loose mask erodes registration confidence with no defined threshold for when that triggers re-sim.

3 THE ROOM TODAY

Academic adaptive H&N program: physicist on console, physician in clinic and reachable, and two RTTs experienced in H&N.
A strong room — the trap here is subtle, not a staffing gap.

4 THE CLOCK IS REAL

You get 35 minutes.
Overage pushes a complex re-sim H&N patient later — rushing a cord-adjacent case downstream is the real hazard.

SAFETY IS ADAPTIVE.
TEAMWORK MAKES IT EFFECTIVE.

VERIFY ACCURATELY | COMMUNICATE CLEARLY | ADAPT WISELY | RESPECT OARs | MANAGE THE MINUTES | SUPPORT THE TEAM

TURN THE PAGE TO MEET HIS SIX DECISIONS. >>

1 REFERENCE PLANNING

PLAYER CARD

HN-01 | Mr. Manny Maskfit

Let's Adapt!

Scenario: A Mask That No Longer Fits

Clinical Situation: Mr. Maskfit is mid-course on a bilateral-nodal head-and-neck plan and has lost about 5 kg since simulation. The mask now closes but fits loosely, and the team has noticed that registration confidence is less consistent than earlier in the course. The reference directive includes parotid and cord constraints but does not define how to handle progressive weight loss, mask-fit degradation, or reduced registration confidence during treatment.

The plan can still be loaded and the mask can still be used, which makes proceeding feel routine. However, a closing mask is not the same as a reproducible immobilization device. Without a defined threshold, the team has no clear trigger for added verification, re-setup, mask modification, or re-simulation.

Decision Prompt: When does cumulative weight loss warrant added verification or re-sim, and what do you do today?

Choices

A. Assess against a defined weight-loss/mask-fit/re-sim threshold; add scrutiny or trigger re-sim as indicated, and document.

B. Proceed; adaptation handles weight loss and the mask still closes.

C. Add registration-confidence and mask-fit checks today, document the weight change, and flag a re-sim threshold for review.

D. Pause for a physician decision on re-simulation before continuing.

Decision Log: Why did your team choose this option?

Notes:

2 ON-SESSION IMAGING

PLAYER CARD

Let’s Adapt!

HN-01 | Mr. Manny Maskfit

Scenario: Spine Matches, Nodes May Not

Clinical Situation: On-session CBCT for Mr. Maskfit registers acceptably to the cervical spine, but the loose mask allows mild rotation and the soft-tissue neck contour differs from simulation. The bony match looks reassuring, but the rotation and contour change most affect the bilateral nodal levels and parotids, not the spine itself.

This is a decision-specific image-registration problem. The registration can be acceptable for spine alignment while still being inadequate for nodal-level and parotid-position confidence. If the team proceeds on the bony match alone, the adaptive plan may be built on a geometry that does not preserve the intended nodal/OAR relationship.

Decision Prompt: Does an acceptable spine match guarantee nodal and parotid position, or do you quantify and correct the rotation first?

Choices

A. Restart with a refit or adjusted mask setup to reduce rotation before imaging again.

B. Accept; spine registration is acceptable and that is the standard match.

C. Quantify the rotation and review nodal/parotid position before planning.

D. Correct the rotation through re-setup or immobilization adjustment, re-verify nodal/parotid alignment, then proceed.

Decision Log: Why did your team choose this option?

Notes:

3 OAR CONTOURING

PLAYER CARD

Let's Adapt!

HN-01 | Mr. Manny Maskfit

Scenario: The Parotid Looks Smaller Than the Patient

Clinical Situation: Mr. Maskfit's propagated parotid contours look plausible on quick review, but the left parotid contour is visibly smaller than the gland on today's planning image. The gland has shifted with weight loss and neck contour change, and the contour under-represents the anatomy most likely to affect later parotid-dose evaluation.

At this step, the team is not yet relying on final parotid mean-dose metrics. The decision is whether the parotid contours are accurate enough to approve as OARs for downstream optimization, dose calculation, cumulative-dose review, and plan comparison. A contour that is too small can later make parotid sparing look better than it really is.

Decision Prompt: Do you approve the reasonable-looking parotid contours, or correct the visibly under-contoured gland before allowing planning to proceed?

Choices

A. Request a parotid second-review before approving the OAR contour set.

B. Accept the parotid contours because they look reasonable overall and the session is time-sensitive.

C. Correct both parotids and compare with prior/reference anatomy before approving the OAR contour set.

D. Correct the left parotid to the visible gland on today's anatomy, then approve the contour set for downstream planning.

Decision Log: Why did your team choose this option?

Notes:

4 TARGET CONTOURING

PLAYER CARD

HN-01 | Mr. Manny Maskfit

Let's Adapt!

Scenario: Level Borders on a Thinner Neck

Clinical Situation: Nodal CTV propagation onto Mr. Maskfit's changed neck contour is plausible, but the level boundaries near the now-thinner neck are uncertain. The propagated CTV looks smooth and reasonable, yet a shifted level boundary could under-cover a nodal level. Over-expanding the CTV, on the other hand, increases dose to the parotids and possibly the cord.

The issue is not simply whether the contour looks acceptable on today's anatomy. The issue is whether today's nodal levels still reflect the reference nodal intent after weight loss, soft-tissue contour change, and registration uncertainty.

Decision Prompt: How do you verify nodal-level coverage when the neck contour and registration geometry have changed?

Choices

A. Reconcile nodal levels to reference intent, adjust as needed, and document the boundary reasoning.

B. Hold for physician nodal-target review before planning.

C. Accept propagation; the nodal CTV looks reasonable on today's neck.

D. Verify level boundaries against the reference nodal intent and adjust, noting changes.

Decision Log: Why did your team choose this option?

Notes:

5 PLAN SELECTION

PLAYER CARD

Let's Adapt!

HN-01 | Mr. Manny Maskfit

Scenario: Parotid Sparing vs. Nodal Coverage

Clinical Situation: Mr. Maskfit's adapted plan improves parotid sparing using the corrected current-anatomy OARs, but it slightly tightens a nodal-level margin. The scheduled/on-session recalculated plan covers the nodes more robustly but raises parotid dose on today's thinner neck geometry. Both appear to meet listed goals.

The adapted plan's parotid sparing looks like the obvious adaptive win, but that benefit is only acceptable if the nodal coverage and level-boundary intent were preserved. A plan should not win because the OAR numbers look better while target robustness silently degrades.

Decision Prompt: How do you weigh nodal robustness against parotid sparing when both plans appear acceptable?

Choices

A. Pick the adapted plan after confirming nodal coverage held, and document the rationale.

B. Return to contour editing to re-confirm nodal coverage, then regenerate the adapted plan before selecting.

C. Select per documented nodal-coverage and cumulative-parotid review, recording the plan-selection rationale.

D. Pick the adapted plan; better parotid sparing is the adaptive win.

Decision Log: Why did your team choose this option?

Notes:

6 DELIVERY / CONFIRMATION IMAGING

PLAYER CARD

Let's Adapt!

HN-01 | Mr. Manny Maskfit

Scenario: A Small Chin-Tuck Shift

Clinical Situation: Confirmation imaging for Mr. Maskfit shows that the loose mask allowed a small chin-tuck change after approval, shifting the cord-to-target relationship. The cord constraint had margin on the approved plan, and the shift looks small visually.

For a serial OAR like the cord, "it had margin" and "the shift looks small" are not enough. The team must verify whether the approved cord-to-target relationship still holds within a defined tolerance before treatment.

Decision Prompt: Do you treat on the small-looking shift, re-position and re-verify, or apply a defined cord tolerance?

Choices

A. Compare the cord relationship to a predefined tolerance; treat if within, re-adapt or correct if not, and document.

B. Re-position/re-immobilize and re-verify the cord relationship before treating.

C. Restart the adaptive session on the current position with attention to immobilization.

D. Treat now; the shift is small and the cord constraint had margin.

Decision Log: Why did your team choose this option?

Notes:

HN-01 Explanation Summary

Mr. Manny Maskfit • Head and Neck / Mask Fit and Weight Loss

Patient Context

Clinical Summary	56M, HPV+ oropharynx, 70 Gy/35 fx with bilateral nodal levels, fraction 18. Approximately 5 kg weight loss since simulation; mask now fits loosely.
Clinic / Coverage Model	Academic adaptive H&N program; physicist on console, physician in clinic reachable, two RTTs experienced in H&N.
Adaptive Rationale	Weight loss and nodal response change neck contour and parotid position; adaptation maintains nodal coverage while sparing parotids and cord.
Allowable Time	35 minutes
Time Consequence	Overage pushes a complex re-sim H&N patient later; rushing a cord-adjacent case downstream is the real hazard.

Core Lesson

Weight loss and a loose mask can make a familiar H&N workflow less reproducible. Adaptation must preserve nodal intent and serial-OAR relationships, not just improve OAR metrics.

Overall Handling Approach

Make uncertainty visible, verify the anatomy that drives the decision, use documented thresholds or priority rules, and record the rationale before delivery.

1. Reference Planning — A Mask That No Longer Fits

Why it matters: The reference/treatment directive issue sets the decision rule before the patient is deep into the adaptive session.
General handling approach: Apply or define the missing rule early, document the working basis, and escalate only when the ambiguity affects today’s treatment decision.

2. On-Session Imaging — Spine Matches, Nodes May Not

Why it matters: Image adequacy must be judged against the specific decision the image must support, not only global image quality.
General handling approach: Review the decision-driving anatomy directly. Reacquire or mitigate only when the limitation prevents reliable contouring or plan evaluation.

3. OAR Contouring — The Parotid Looks Smaller Than the Patient

Why it matters: At this stage the team is approving OAR/influencer anatomy for downstream optimization and plan comparison, not relying on a final DVH.
General handling approach: Focus edits on the OAR/influencer regions that can change optimization, dose calculation, or plan comparison; use second review when uncertainty remains clinically material.

4. Target Contouring — Level Borders on a Thinner Neck

Why it matters: A plausible propagated target does not by itself prove that clinical target intent has been preserved.
General handling approach: Reconcile propagated targets to reference intent, diagnostic anatomy, prior fractions, or a documented rule before planning.

5. Plan Selection — Parotid Sparing vs. Nodal Coverage

Why it matters: Scheduled/on-session recalculated and adapted plans can both appear acceptable while resolving different clinical risks.
General handling approach: Select based on documented target/OAR priorities, composite checks, sensitivity tests, and secondary dose review rather than the most attractive single metric.

6. Delivery / Confirmation Imaging — A Small Chin-Tuck Shift

Why it matters: Confirmation imaging should be governed by predefined thresholds, stability checks, or re-verification rather than visual impression or time pressure alone.
General handling approach: Treat only when the approved geometry remains within threshold or can be restored and re-verified; otherwise re-adapt, restart, delay, or cancel according to policy.

HN-01 Scoring Sheet

Mr. Manny Maskfit • Allowable time: 35 minutes

Workflow step	Choice	Safety penalty	Time penalty	Whammy?	Notes
1. Reference Planning	____	____	____	☐ Yes ☐ No	
2. On-Session Imaging	____	____	____	☐ Yes ☐ No	
3. OAR Contouring	____	____	____	☐ Yes ☐ No	
4. Target Contouring	____	____	____	☐ Yes ☐ No	
5. Plan Selection	____	____	____	☐ Yes ☐ No	
6. Delivery / Confirmation Imaging	____	____	____	☐ Yes ☐ No	

Total safety penalties	__________	**Total time used**	__________ min
Time overage	max(0, time - allowable) = ________	**Whammy resolution**	☐ +10 min choose again ☐ +15 safety
Final score	100 - safety penalties - time overage = ________		
Team	________________________	**Facilitator**	________________________

Scoring reminder: Final score = 100 - safety penalties - time overage. If a whammy is selected, add 10 minutes and choose again, or keep the choice and add 15 safety penalty.

HN-01 Answer Key • 1. Reference Planning

Mr. Manny Maskfit

Scenario: A Mask That No Longer Fits

Choice A: Assess against a defined weight-loss/mask-fit/re-sim threshold; add scrutiny or trigger re-sim as indicated, and document.

Scoring: Safety 2; Time 5 min; no cascade; Whammy: No; S/O/D/RPN 2/2/2/8

Explanation: Strong: threshold-based action (scrutiny vs. re-sim) is durable; marginally more process than B when no threshold exists yet.

Penalty rationale: Modest time; safe and reproducible.

Mitigation concept: Threshold-driven scrutiny or re-sim.

Choice B: Proceed; adaptation handles weight loss and the mask still closes.

Scoring: Safety 10; Time 1 min; +8 safety cascade; Whammy: No; S/O/D/RPN 4/4/4/64

Explanation: Tempting because the mask closes, but a closing mask is not a fitting mask; assuming adaptation 'handles it' lets registration degrade unchecked.

Penalty rationale: High safety, low time: degradation normalized.

Mitigation concept: Defined weight-loss / re-sim thresholds.

Choice C: Add registration-confidence and mask-fit checks today, document the weight change, and flag a re-sim threshold for review.

Scoring: Safety 2; Time 3 min; no cascade; Whammy: No; S/O/D/RPN 2/2/2/8

Explanation: Best: explicit registration/mask checks catch creeping degradation today and flagging starts the threshold fix, without halting.

Penalty rationale: Lowest combined penalty: targets the real variable.

Mitigation concept: Registration-confidence + mask-fit checks + flag.

Choice D: Pause for a physician decision on re-simulation before continuing.

Scoring: Safety 2; Time 12 min; +2 min cascade; Whammy: No; S/O/D/RPN 2/2/2/8

Explanation: Re-sim is right past a threshold, but halting fraction 18 without one over-escalates.

Penalty rationale: Highest time; needs a defined trigger.

Mitigation concept: Pre-agreed re-sim trigger.

HN-01 Answer Key • 2. On-Session Imaging

Mr. Manny Maskfit

Scenario: Spine Matches, Nodes May Not

Choice A: Restart with a refit or adjusted mask setup to reduce rotation before imaging again.

Scoring: Safety 2; Time 11 min; +2 min cascade; Whammy: No; S/O/D/RPN 2/2/2/8

Explanation: A refit mask is ideal but a full restart mid-course is heavy if rotation is correctable.

Penalty rationale: High time; reserve for uncorrectable rotation.

Mitigation concept: Refit only when correction fails.

Choice B: Accept; spine registration is acceptable and that is the standard match.

Scoring: Safety 11; Time 1 min; +8 safety cascade; Whammy: No; S/O/D/RPN 5/4/4/80

Explanation: Tempting because spine match is the usual standard, but a bony match can hide rotation that displaces the structures driving this plan.

Penalty rationale: High safety, low time: rotational risk unquantified.

Mitigation concept: Quantify rotation; verify soft-tissue targets/OARs.

Choice C: Quantify the rotation and review nodal/parotid position before planning.

Scoring: Safety 4; Time 3 min; no cascade; Whammy: No; S/O/D/RPN 3/3/3/27

Explanation: Reasonable: quantifying rotation and checking key structures catches the displacement, though it does not correct the cause.

Penalty rationale: Low-moderate: targeted verification.

Mitigation concept: Rotation quantification + structure review.

Choice D: Correct the rotation through re-setup or immobilization adjustment, re-verify nodal/parotid alignment, then proceed.

Scoring: Safety 2; Time 7 min; no cascade; Whammy: No; S/O/D/RPN 2/2/2/8

Explanation: Best: correcting rotation at the source improves every downstream step for a loose-mask case.

Penalty rationale: Modest time; fixes the cause.

Mitigation concept: Rotation correction + re-verification.

HN-01 Answer Key • 3. OAR Contouring

Mr. Manny Maskfit

Scenario: The Parotid Looks Smaller Than the Patient

Choice A: Request a parotid second-review before approving the OAR contour set.

Scoring: Safety 3; Time 7 min; +2 min cascade; Whammy: No; S/O/D/RPN 2/2/2/8

Explanation: Safe if the parotid boundary is difficult to judge or local policy requires escalation. It may be slower than targeted correction when the under-contour is clear.

Penalty rationale: Low safety penalty because OAR uncertainty is escalated before planning. Higher time penalty reflects second review.

Mitigation concept: Second-reviewer escalation for material parotid uncertainty

Choice B: Accept the parotid contours because they look reasonable overall and the session is time-sensitive.

Scoring: Safety 12; Time 1 min; +9 safety cascade; Whammy: YES - Add 10 minutes and choose again, or keep this choice and add 15 safety penalty.; S/O/D/RPN 5/4/5/100

Explanation: Tempting shortcut: the parotids look reasonable overall, but the left parotid is visibly smaller than the gland on today's image. An undersized OAR contour can make downstream sparing look better than it really is.

Penalty rationale: High safety penalty because a decision-driving OAR is approved based on overall plausibility rather than visible anatomy.

Mitigation concept: Verify OAR contour matches visible anatomy

Choice C: Correct both parotids and compare with prior/reference anatomy before approving the OAR contour set.

Scoring: Safety 2; Time 7 min; no cascade; Whammy: No; S/O/D/RPN 2/2/2/8

Explanation: Robust but more time-consuming. Correcting both parotids and comparing with prior/reference anatomy reduces drift risk but may exceed what is needed if only the left parotid is visibly wrong.

Penalty rationale: Low safety penalty because the OAR set is comprehensively reviewed. Higher time penalty reflects bilateral correction and comparison.

Mitigation concept: Comprehensive bilateral OAR review when anatomy has changed

Choice D: Correct the left parotid to the visible gland on today’s anatomy, then approve the contour set for downstream planning.

Scoring: Safety 3; Time 4 min; no cascade; Whammy: No; S/O/D/RPN 2/3/2/12

Explanation: Best focused correction: correcting the visibly under-contoured left parotid restores meaningful OAR anatomy for downstream optimization, cumulative review, and plan comparison.

Penalty rationale: Very low safety penalty because the decision-driving parotid anatomy is corrected before planning.

Mitigation concept: Correct visible OAR under-contouring before downstream plan review

HN-01 Answer Key • 4. Target Contouring

Mr. Manny Maskfit

Scenario: Level Borders on a Thinner Neck

Choice A: Reconcile nodal levels to reference intent, adjust as needed, and document the boundary reasoning.

Scoring: Safety 2; Time 7 min; no cascade; Whammy: No; S/O/D/RPN 2/2/2/8

Explanation: Best: full reconciliation with recorded reasoning gives the most defensible nodal target.

Penalty rationale: Modest time; lowest residual nodal-coverage risk.

Mitigation concept: Reference reconciliation + reasoning.

Choice B: Hold for physician nodal-target review before planning.

Scoring: Safety 3; Time 10 min; +2 min cascade; Whammy: No; S/O/D/RPN 2/2/2/8

Explanation: Physician nodal review fits true ambiguity but is not a daily default for routine boundaries.

Penalty rationale: High time; reserve for genuine uncertainty.

Mitigation concept: Pre-agreed nodal boundary conventions.

Choice C: Accept propagation; the nodal CTV looks reasonable on today's neck.

Scoring: Safety 10; Time 1 min; +6 safety cascade; Whammy: No; S/O/D/RPN 5/3/4/60

Explanation: Tempting because it looks reasonable, but a 'reasonable-looking' nodal CTV on a changed neck can silently under-cover a level.

Penalty rationale: High safety, low time: nodal coverage unverified.

Mitigation concept: Verify nodal levels against reference intent.

Choice D: Verify level boundaries against the reference nodal intent and adjust, noting changes.

Scoring: Safety 4; Time 4 min; no cascade; Whammy: No; S/O/D/RPN 3/3/3/27

Explanation: Reasonable: checking boundaries against reference intent grounds coverage, though documentation is lighter than C.

Penalty rationale: Low-moderate: targeted verification.

Mitigation concept: Reference-intent boundary verification.

HN-01 Answer Key • 5. Plan Selection

Mr. Manny Maskfit

Scenario: Parotid Sparing vs. Nodal Coverage

Choice A: Pick the adapted plan after confirming nodal coverage held, and document the rationale.

Scoring: Safety 3; Time 3 min; no cascade; Whammy: No; S/O/D/RPN 3/3/3/27

Explanation: Best: choosing adapted after confirming nodal coverage held is substantively correct and quick.

Penalty rationale: Low combined penalty: confirmed basis, recorded.

Mitigation concept: Coverage-confirmed adapted selection.

Choice B: Return to contour editing to re-confirm nodal coverage, then regenerate the adapted plan before selecting.

Scoring: Safety 3; Time 11 min; +2 min cascade; Whammy: No; S/O/D/RPN 2/2/2/8

Explanation: Returning to editing to re-confirm coverage is operationally valid but time-heavy late in an H&N slot.

Penalty rationale: High time; reserve for unresolved coverage.

Mitigation concept: Return-to-editing only when coverage is in doubt.

Choice C: Select per documented nodal-coverage and cumulative-parotid review, recording the plan-selection rationale.

Scoring: Safety 2; Time 5 min; no cascade; Whammy: No; S/O/D/RPN 2/2/2/8

Explanation: Also strong: ties selection to nodal-coverage and cumulative-parotid review; marginally more process than B.

Penalty rationale: Modest time; coherent decision trail.

Mitigation concept: Coverage + cumulative-OAR driven selection.

Choice D: Pick the adapted plan; better parotid sparing is the adaptive win.

Scoring: Safety 9; Time 1 min; +5 safety cascade; Whammy: No; S/O/D/RPN 4/4/4/64

Explanation: Tempting because parotid sparing looks like the win, but trading nodal margin for it without a coverage check risks under-treating disease.

Penalty rationale: High safety: OAR-sparing optics over nodal coverage robustness.

Mitigation concept: Coverage-checked selection mid-course.

HN-01 Answer Key • 6. Delivery / Confirmation Imaging

Mr. Manny Maskfit

Scenario: A Small Chin-Tuck Shift

Choice A: Compare the cord relationship to a predefined tolerance; treat if within, re-adapt or correct if not, and document.

Scoring: Safety 2; Time 6 min; no cascade; Whammy: No; S/O/D/RPN 2/2/2/8

Explanation: Best: a tolerance-based, documented decision is the defensible cord-safety standard.

Penalty rationale: Moderate time; balanced and reconstructable.

Mitigation concept: Documented cord-tolerance decision.

Choice B: Re-position/re-immobilize and re-verify the cord relationship before treating.

Scoring: Safety 4; Time 4 min; no cascade; Whammy: No; S/O/D/RPN 3/3/3/27

Explanation: Reasonable: re-positioning can restore the intended cord relationship without re-adaptation, though it lacks a formal threshold.

Penalty rationale: Low-moderate: proportionate to a small shift.

Mitigation concept: Re-position + cord re-verify.

Choice C: Restart the adaptive session on the current position with attention to immobilization.

Scoring: Safety 2; Time 12 min; no cascade; Whammy: No; S/O/D/RPN 2/2/2/8

Explanation: Restart suits a large shift or failing immobilization but is excessive for a small tuck.

Penalty rationale: Highest time; reserve for tolerance breaches.

Mitigation concept: Restart only beyond tolerance.

Choice D: Treat now; the shift is small and the cord constraint had margin.

Scoring: Safety 10; Time 1 min; no cascade; Whammy: No; S/O/D/RPN 5/3/4/60

Explanation: Tempting because the cord had margin, but for a serial OAR 'had margin' without a tolerance check is exactly the wrong assumption at the last gate.

Penalty rationale: High safety, low time: cord relationship unverified.

Mitigation concept: Predefined cord-relationship tolerance.

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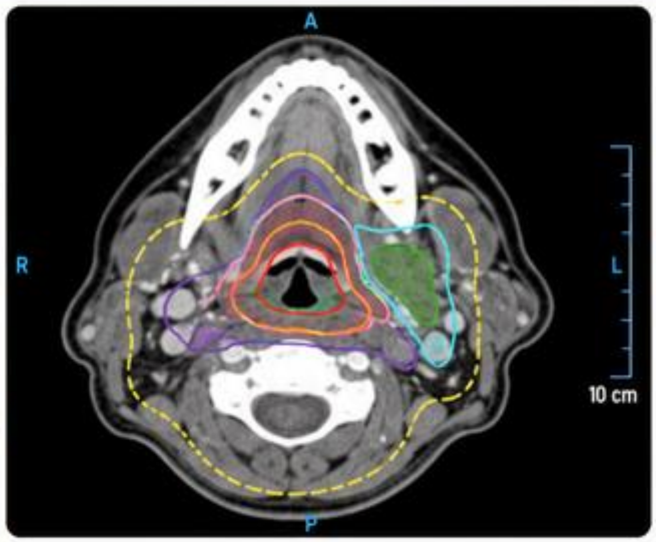
PATIENT CASE FILE • HN-02

Ms. Nina Nodeshift

"A continuous PTV can hide a missing slice."

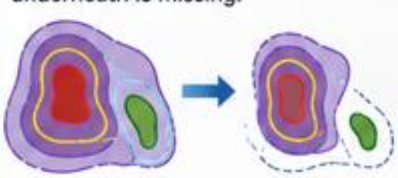
1 PATIENT SNAPSHOT

1. 61-year-old female
2. Larynx / hypopharynx
3. Involved left level III node
4. Fraction 25
5. Prescription: 70 Gy / 35 fx
6. Propagation: Mixed: deformed + rigid
7. Today's wildcard: Marked response; nodal shrinkage
8. Allowable time: 35 minutes



2 WHY SHE'S ON THE ADAPTIVE COUCH

Marked response has moved her primary and involved node while the elective volume stays stable. Mixed propagation tracks the responder and holds elective coverage — but a margin-derived PTV can look continuous even when an involved-node GTV slice underneath is missing.



3 THE ROOM TODAY

Community adaptive H&N program: rotating physicists, physician remote, two RTTs. This is her first adapted (not scheduled) fraction in several days, so more has changed than usual.

4 THE CLOCK IS REAL

You get 35 minutes. Overage compresses the day's schedule — rushing the documentation step or the lower-neck check is the real downstream risk.

**SAFETY IS ADAPTIVE.
TEAMWORK MAKES IT EFFECTIVE.**

VERIFY ACCURATELY | COMMUNICATE CLEARLY | ADAPT WISELY | CHECK CONTINUITY | MANAGE THE MINUTES | SUPPORT THE TEAM

TURN THE PAGE TO MEET HER SIX DECISIONS.

1 REFERENCE PLANNING

PLAYER CARD

HN-02 | Ms. Nina Nodeshift

Let's Adapt!

Scenario: Has the Node Been Verified?

Clinical Situation: Ms. Nodeshift has a larynx/hypopharynx primary with an involved left level III node and has had significant response and nodal shrinkage. The plan uses a deformed high-dose target for the primary and involved node and a rigid elective nodal CTV. The reference directive never states how to confirm involved-node GTV continuity when the node responds substantially.

The margin-derived CTV/PTV can still look continuous even if the underlying involved-node GTV has a slice-level defect. That makes proceeding tempting: the high-dose volume appears covered, but the actual clinical target definition underneath has not been verified.

Decision Prompt: How is involved-node GTV continuity confirmed under major response, and what do you do before planning?

Choices

A. Add a GTV-continuity check, reconcile gaps to reference/diagnostic intent, and document.

B. Pause for physician target re-definition given the degree of response.

C. Add an explicit involved-node GTV-continuity check and document any gaps before proceeding.

D. Proceed; the PTV margin covers minor GTV propagation gaps.

Decision Log: Why did your team choose this option?

Notes:

2 ON-SESSION IMAGING

PLAYER CARD

Let's Adapt!

HN-02 | Ms. Nina Nodeshift

Scenario: Artifact on the Decisive Slices

Clinical Situation: On-session CBCT for Ms. Nodeshift is adequate centrally, but swallowing-motion artifact crosses several slices through the larynx and involved node. Those are exactly the slices needed to confirm GTV continuity and nodal extent.

The image may be adequate for general registration, but the degraded slices are the ones needed to answer today's target-continuity question. Flagging the artifact without resolving it leaves the decisive contouring question unanswered.

Decision Prompt: Can you verify GTV continuity on degraded slices, or do you reacquire to clear them first?

Choices

A. Flag the motion slices as unreliable for GTV/nodal verification before proceeding.

B. Restart the session after coaching to obtain motion-free slices.

C. Proceed; central image quality is adequate for registration.

D. Reacquire with a swallowing-pause/coaching attempt to clear the key slices, then verify.

Decision Log: Why did your team choose this option?

Notes:

3 OAR CONTOURING

PLAYER CARD

Let's Adapt!

HN-02 | Ms. Nina Nodeshift

Scenario: An Overlap Made Ambiguous

Clinical Situation: Ms. Nodeshift's generated or propagated larynx, pharyngeal constrictor, and cord contours look reasonable overall. However, response has changed the larynx-to-target relationship, and the constrictor contour now overlaps ambiguously with the shrunken involved-node region on the key slices.

At this step, the team is not yet relying on final swallowing-OAR or cord dose metrics. The decision is whether the larynx/constrictor/cord contour set is anatomically reliable enough for downstream optimization, dose calculation, and plan comparison. The ambiguous overlap sits exactly where dose will matter most.

Decision Prompt: Do you approve the reasonable-looking OAR contours, or resolve the larynx/constrictor-node overlap on the key slices first?

Choices

A. Request a swallowing-OAR second review before approving the OAR contour set.

B. Accept the generated larynx, constrictor, and cord contours because they look reasonable overall.

C. Resolve the constrictor/larynx-node overlap on the key slices before approving the OAR contour set.

D. Resolve the overlap and cross-check the cord contour/max-dose region on current geometry before proceeding.

Decision Log: Why did your team choose this option?

Notes:

4 TARGET CONTOURING

PLAYER CARD

HN-02 | Ms. Nina Nodeshift

Let's Adapt!

Scenario: A Continuous PTV Hiding a Gap

Clinical Situation: On a motion-affected slice, Ms. Nodeshift's involved-node GTV has a missing superior slice, yet the margin-derived CTV/PTV still appears continuous. The high-dose PTV looks covered, which makes the missing GTV slice easy to miss.

That apparent continuity is exactly the problem: the margin papers over a defect in the underlying target contour without resolving the target-definition uncertainty. A continuous PTV is not proof that the involved-node GTV was correctly propagated.

Decision Prompt: With the PTV looking continuous, what makes you check the involved-node GTV slice underneath, and then what?

Choices

A. Inspect and reconcile the GTV gap to reference/diagnostic imaging, then document the basis.

B. Hold for physician target review of the involved-node GTV gap.

C. Accept; the PTV is continuous so coverage is fine.

D. Inspect the involved-node GTV slice, bridge the gap if real, and document.

Decision Log: Why did your team choose this option?

Notes:

5 PLAN SELECTION

PLAYER CARD

Let’s Adapt!

HN-02 | Ms. Nina Nodeshift

Scenario: Volumes That No Longer Nest

Clinical Situation: Ms. Nodeshift’s adapted plan tightly covers the responded primary/involved-node region, but the rigid elective CTV now overlaps oddly with the shrunken neck geometry. The scheduled/on-session recalculated plan covers elective and lower-neck regions more robustly but over-treats the responded high-dose region. Lower-neck coverage is also in question.

The adapted plan hugs the responded node nicely, which makes it attractive. However, if the deformed high-dose volume and rigid elective volume no longer nest cleanly, the team could under-cover a seam or lower-neck region while focusing on the improved high-dose conformity.

Decision Prompt: How do you select when the deformed node and rigid elective volumes no longer nest cleanly?

Choices

A. Pick the adapted plan; it best fits today’s responded node.

B. Return to contour editing to reconcile the elective/lower-neck seam, then regenerate the adapted plan before selecting.

C. Select based on a composite review of the GTV/elective seam and lower-neck coverage, then document the rationale.

D. Pick the scheduled plan for robust elective/lower-neck coverage and document the over-treatment tradeoff.

Decision Log: Why did your team choose this option?

Notes:

6 DELIVERY / CONFIRMATION IMAGING

PLAYER CARD

HN-02 | Ms. Nina Nodeshift

Let's Adapt!

Scenario: A Shoulder Shift at the Gate

Clinical Situation: Confirmation imaging for Ms. Nodeshift shows that the shoulder and neck position shifted slightly with swallowing after approval, changing the lower-neck and high/elective relationship. The shift looks small, and the plan was just approved.

A small shoulder or neck shift can reopen the mixed-propagation seam or compromise lower-neck coverage. Treating without checking the seam/lower-neck relationship risks repeating the same failure mode at the final gate.

Decision Prompt: Do you treat on the small-looking shift, re-position and re-verify, or apply a defined seam/lower-neck tolerance?

Choices

A. Treat now; the shift is small and the plan was just approved.

B. Restart the adaptive session on the current swallowing/shoulder baseline.

C. Re-position the shoulder/neck and re-verify the lower-neck and high/elective relationship before treating.

D. Compare the seam/lower-neck change to a predefined tolerance; treat if within, re-adapt if not, and document.

Decision Log: Why did your team choose this option?

Notes:

HN-02 Explanation Summary

Ms. Nina Nodeshift • Head and Neck / Nodal Response and Mixed Propagation

Patient Context

Clinical Summary	61F, larynx/hypopharynx with an involved left level III node, 70 Gy/35 fx, fraction 25. Significant response and nodal shrinkage; the elective nodal CTV is rigidly propagated while the primary/involved-node PTV-High is deformed.
Clinic / Coverage Model	Community adaptive H&N program; rotating physicists, physician remote, two RTTs; this is the patient's first adapted rather than scheduled fraction in several days.
Adaptive Rationale	Marked response moves the primary and involved node while the elective nodal volume is more stable; mixed propagation aims to track the primary and involved node while holding elective coverage.
Allowable Time	35 minutes
Time Consequence	Overage compresses the day's schedule; rushing the documentation step or lower-neck check is the real downstream risk.

Core Lesson

A continuous margin-derived CTV/PTV can hide a missing or uncertain involved-node GTV. Mixed propagation must be checked at seams and lower-neck boundaries.

Overall Handling Approach

Make uncertainty visible, verify the anatomy that drives the decision, use documented thresholds or priority rules, and record the rationale before delivery.

1. Reference Planning — Has the Node Been Verified?

Why it matters: The reference/treatment directive issue sets the decision rule before the patient is deep into the adaptive session.
General handling approach: Apply or define the missing rule early, document the working basis, and escalate only when the ambiguity affects today's treatment decision.

2. On-Session Imaging — Artifact on the Decisive Slices

Why it matters: Image adequacy must be judged against the specific decision the image must support, not only global image quality.
General handling approach: Review the decision-driving anatomy directly. Reacquire or mitigate only when the limitation prevents reliable contouring or plan evaluation.

3. OAR Contouring — An Overlap Made Ambiguous

Why it matters: At this stage the team is approving OAR/influencer anatomy for downstream optimization and plan comparison, not relying on a final DVH.
General handling approach: Focus edits on the OAR/influencer regions that can change optimization, dose calculation, or plan comparison; use second review when uncertainty remains clinically material.

4. Target Contouring — A Continuous PTV Hiding a Gap

Why it matters: A plausible propagated target does not by itself prove that clinical target intent has been preserved.
General handling approach: Reconcile propagated targets to reference intent, diagnostic anatomy, prior fractions, or a documented rule before planning.

5. Plan Selection — Volumes That No Longer Nest

Why it matters: Scheduled/on-session recalculated and adapted plans can both appear acceptable while resolving different clinical risks.
General handling approach: Select based on documented target/OAR priorities, composite checks, sensitivity tests, and secondary dose review rather than the most attractive single metric.

6. Delivery / Confirmation Imaging — A Shoulder Shift at the Gate

Why it matters: Confirmation imaging should be governed by predefined thresholds, stability checks, or re-verification rather than visual impression or time pressure alone.
General handling approach: Treat only when the approved geometry remains within threshold or can be restored and re-verified; otherwise re-adapt, restart, delay, or cancel according to policy.

HN-02 Scoring Sheet

Ms. Nina Nodeshift • Allowable time: 35 minutes

Workflow step	Choice	Safety penalty	Time penalty	Whammy?	Notes
1. Reference Planning	____	____	____	□ Yes □ No	
2. On-Session Imaging	____	____	____	□ Yes □ No	
3. OAR Contouring	____	____	____	□ Yes □ No	
4. Target Contouring	____	____	____	□ Yes □ No	
5. Plan Selection	____	____	____	□ Yes □ No	
6. Delivery / Confirmation Imaging	____	____	____	□ Yes □ No	

Total safety penalties	__________	**Total time used**	__________ min
Time overage	max(0, time - allowable) = ________	**Whammy resolution**	□ +10 min choose again □ +15 safety
Final score	100 - safety penalties - time overage = ________		
Team	________________________	**Facilitator**	________________________

Scoring reminder: Final score = 100 - safety penalties - time overage. If a whammy is selected, add 10 minutes and choose again, or keep the choice and add 15 safety penalty.

HN-02 Answer Key • 1. Reference Planning

Ms. Nina Nodeshift

Scenario: Has the Node Been Verified?

Choice A: Add a GTV-continuity check, reconcile gaps to reference/diagnostic intent, and document.

Scoring: Safety 2; Time 5 min; no cascade; Whammy: No; S/O/D/RPN 2/2/2/8

Explanation: Strong: adds reconciliation to reference intent; marginally more process than B.

Penalty rationale: Modest time; safe and reproducible.

Mitigation concept: Continuity check + reference reconciliation.

Choice B: Pause for physician target re-definition given the degree of response.

Scoring: Safety 2; Time 12 min; +2 min cascade; Whammy: No; S/O/D/RPN 2/2/2/8

Explanation: Physician re-definition fits major response, but halting fraction 25 before even a continuity check over-escalates.

Penalty rationale: Highest time; verify before escalating.

Mitigation concept: Stepwise verification before re-definition.

Choice C: Add an explicit involved-node GTV-continuity check and document any gaps before proceeding.

Scoring: Safety 2; Time 3 min; no cascade; Whammy: No; S/O/D/RPN 2/2/2/8

Explanation: Best: a GTV-continuity check surfaces gaps the PTV hides, and documenting starts the fix without halting an established course.

Penalty rationale: Lowest combined penalty: targets the real risk.

Mitigation concept: GTV-continuity check + gap documentation.

Choice D: Proceed; the PTV margin covers minor GTV propagation gaps.

Scoring: Safety 10; Time 1 min; +8 safety cascade; Whammy: No; S/O/D/RPN 4/4/4/64

Explanation: Tempting because the PTV looks continuous, but relying on margin to excuse GTV gaps lets target definition erode invisibly as the node responds.

Penalty rationale: High safety, low time: GTV continuity unverified.

Mitigation concept: Explicit GTV-continuity verification.

HN-02 Answer Key • 2. On-Session Imaging

Ms. Nina Nodeshift

Scenario: Artifact on the Decisive Slices

Choice A: Flag the motion slices as unreliable for GTV/nodal verification before proceeding.

Scoring: Safety 5; Time 2 min; no cascade; Whammy: No; S/O/D/RPN 3/3/3/27

Explanation: Reasonable but not sufficient here: flagging the slices routes caution, but the artifact runs directly through the slices needed to confirm GTV continuity, so a flag alone leaves the decisive question unanswered.

Penalty rationale: Low time, but the decisive slices remain unusable.

Mitigation concept: Flagging helps only when the decisive slices are not the degraded ones.

Choice B: Restart the session after coaching to obtain motion-free slices.

Scoring: Safety 2; Time 12 min; +2 min cascade; Whammy: No; S/O/D/RPN 2/2/2/8

Explanation: A full restart is thorough but heavy if a targeted coached reacquisition clears the key slices.

Penalty rationale: Highest time; reserve for unrecoverable motion.

Mitigation concept: Reacquire before full restart.

Choice C: Proceed; central image quality is adequate for registration.

Scoring: Safety 10; Time 1 min; +7 safety cascade; Whammy: No; S/O/D/RPN 4/4/4/64

Explanation: Tempting because central registration is fine, but adequate registration does not help if the verification slices themselves are blurred.

Penalty rationale: High safety, low time: verification basis degraded.

Mitigation concept: Per-slice adequacy for target verification.

Choice D: Reacquire with a swallowing-pause/coaching attempt to clear the key slices, then verify.

Scoring: Safety 2; Time 8 min; no cascade; Whammy: No; S/O/D/RPN 2/2/2/8

Explanation: Best: because the artifact crosses the exact slices needed to confirm involved-node GTV continuity, a coached swallow-pause reacquisition to clear those slices, then verify, is the corrective action the situation demands.

Penalty rationale: More time, but it restores the slices the decision actually depends on.

Mitigation concept: Reacquire to clear the decision-driving slices, then verify.

HN-02 Answer Key • 3. OAR Contouring

Ms. Nina Nodeshift

Scenario: An Overlap Made Ambiguous

Choice A: Request a swallowing-OAR second review before approving the OAR contour set.

Scoring: Safety 3; Time 8 min; +2 min cascade; Whammy: No; S/O/D/RPN 2/2/2/8

Explanation: Safe when the swallowing-OAR overlap is hard to judge or local policy calls for escalation. It avoids approving ambiguous OAR anatomy before planning.

Penalty rationale: Low safety penalty because material OAR uncertainty is escalated before planning.

Mitigation concept: Second-reviewer escalation for swallowing-OAR uncertainty

Choice B: Accept the generated larynx, constrictor, and cord contours because they look reasonable overall.

Scoring: Safety 10; Time 1 min; +7 safety cascade; Whammy: No; S/O/D/RPN 4/4/4/64

Explanation: Tempting shortcut: the contours look reasonable overall, but the ambiguous overlap occurs on key slices where dose and target/OAR tradeoffs matter most.

Penalty rationale: High safety penalty because the decision-driving overlap is approved on overall plausibility alone.

Mitigation concept: Resolve key-slice OAR/target overlap before planning

Choice C: Resolve the constrictor/larynx-node overlap on the key slices before approving the OAR contour set.

Scoring: Safety 3; Time 4 min; no cascade; Whammy: No; S/O/D/RPN 3/3/2/18

Explanation: Best focused response: resolve the constrictor/larynx-node overlap on the key slices before approving the OAR contour set for downstream optimization and plan comparison.

Penalty rationale: Very low safety penalty because the decision-driving overlap is corrected before planning.

Mitigation concept: Resolve ambiguous OAR overlap at decision-driving slices

Choice D: Resolve the overlap and cross-check the cord contour/max-dose region on current geometry before proceeding.

Scoring: Safety 2; Time 7 min; no cascade; Whammy: No; S/O/D/RPN 2/2/2/8

Explanation: Robust option when overlap and cord high-dose region are both clinically material. It confirms the serial-OAR region on current geometry.

Penalty rationale: Low safety penalty because both overlap and cord region are checked. Higher time penalty reflects broader review.

Mitigation concept: Cross-check serial-OAR region after resolving overlap

HN-02 Answer Key • 4. Target Contouring

Ms. Nina Nodeshift

Scenario: A Continuous PTV Hiding a Gap

Choice A: Inspect and reconcile the GTV gap to reference/diagnostic imaging, then document the basis.

Scoring: Safety 2; Time 6 min; no cascade; Whammy: No; S/O/D/RPN 2/2/2/8

Explanation: Best: reconciling to diagnostic imaging confirms whether the gap is real or artifact -- the most defensible target decision.

Penalty rationale: Modest time; resolves the underlying uncertainty.

Mitigation concept: Diagnostic reconciliation + documentation.

Choice B: Hold for physician target review of the involved-node GTV gap.

Scoring: Safety 3; Time 10 min; +2 min cascade; Whammy: No; S/O/D/RPN 2/2/2/8

Explanation: Physician review fits a genuinely uncertain gap, but a one-slice artifact is often team-resolvable first.

Penalty rationale: High time; reserve for true uncertainty.

Mitigation concept: GTV-continuity protocol.

Choice C: Accept; the PTV is continuous so coverage is fine.

Scoring: Safety 12; Time 1 min; +6 safety cascade; Whammy: YES - Add 10 minutes and choose again, or keep this choice and add 15 safety penalty.; S/O/D/RPN 5/3/5/75

Explanation: The tempting shortcut: a continuous PTV over a missing GTV slice is precisely how a target-definition defect goes undetected; the margin hides it but does not fix it.

Penalty rationale: Whammy: trusting margin-derived PTV continuity to excuse a missing involved-node GTV slice.

Mitigation concept: Always verify involved-node GTV continuity, not just PTV.

Choice D: Inspect the involved-node GTV slice, bridge the gap if real, and document.

Scoring: Safety 3; Time 3 min; no cascade; Whammy: No; S/O/D/RPN 3/3/2/18

Explanation: Reasonable: inspecting and bridging the GTV gap restores continuity, though it does not confirm whether the gap was real or artifact.

Penalty rationale: Low-moderate: fixes the underlying target.

Mitigation concept: GTV inspection + bridge + documentation.

HN-02 Answer Key • 5. Plan Selection

Ms. Nina Nodeshift

Scenario: Volumes That No Longer Nest

Choice A: Pick the adapted plan; it best fits today's responded node.

Scoring: Safety 9; Time 1 min; +5 safety cascade; Whammy: No; S/O/D/RPN 4/4/4/64

Explanation: Tempting because it fits the node, but ignoring the odd rigid-elective overlap and lower-neck match can under- or over-cover electives unpredictably.

Penalty rationale: High safety: seam mismatch driving selection unseen.

Mitigation concept: Composite check across mixed-propagation volumes.

Choice B: Return to contour editing to reconcile the elective/lower-neck seam, then regenerate the adapted plan before selecting.

Scoring: Safety 3; Time 11 min; +2 min cascade; Whammy: No; S/O/D/RPN 2/2/2/8

Explanation: Returning to editing to reconcile the seam is operationally valid but time-heavy this late.

Penalty rationale: High time; reserve for genuine nesting conflicts.

Mitigation concept: Return-to-editing for true seam conflicts.

Choice C: Select per the resolved GTV/seam status and a composite coverage check, including lower neck, recording rationale.

Scoring: Safety 2; Time 5 min; no cascade; Whammy: No; S/O/D/RPN 2/2/2/8

Explanation: Best: ties selection to GTV/seam status and a composite coverage check including lower neck -- the most reconstructable.

Penalty rationale: Modest time; coherent decision trail.

Mitigation concept: Seam-status + composite-driven selection.

Choice D: Pick the scheduled plan for robust elective/lower-neck coverage and document the over-treatment tradeoff.

Scoring: Safety 4; Time 3 min; no cascade; Whammy: No; S/O/D/RPN 3/3/3/27

Explanation: Reasonable: robust elective/lower-neck coverage with a documented over-treatment tradeoff is defensible.

Penalty rationale: Low-moderate: coverage-first, recorded.

Mitigation concept: Elective-coverage-first with documentation.

HN-02 Answer Key • 6. Delivery / Confirmation Imaging

Ms. Nina Nodeshift

Scenario: A Shoulder Shift at the Gate

Choice A: Treat now; the shift is small and the plan was just approved.

Scoring: Safety 10; Time 1 min; no cascade; Whammy: No; S/O/D/RPN 4/3/4/48

Explanation: Tempting because approval feels recent, but a small shoulder/neck shift can open the seam or compromise lower-neck coverage; treating without a check repeats the seam failure at the gate.

Penalty rationale: High safety, low time: seam/lower-neck relationship unverified.

Mitigation concept: Predefined seam/lower-neck tolerance at confirmation.

Choice B: Restart the adaptive session on the current swallowing/shoulder baseline.

Scoring: Safety 2; Time 12 min; no cascade; Whammy: No; S/O/D/RPN 2/2/2/8

Explanation: Restart suits persistent motion but is excessive for a transient shift.

Penalty rationale: Highest time; reserve for tolerance breaches.

Mitigation concept: Restart only beyond tolerance.

Choice C: Re-position the shoulder/neck and re-verify the lower-neck and high/elective relationship before treating.

Scoring: Safety 3; Time 4 min; no cascade; Whammy: No; S/O/D/RPN 3/3/3/27

Explanation: Best: a shoulder/neck re-position and re-verify can confirm the seam and lower-neck are intact without full re-adaptation, at low cost.

Penalty rationale: Low combined penalty: proportionate to a small shift.

Mitigation concept: Re-position + seam/lower-neck re-verify.

Choice D: Compare the seam/lower-neck change to a predefined tolerance; treat if within, re-adapt if not, and document.

Scoring: Safety 2; Time 6 min; no cascade; Whammy: No; S/O/D/RPN 2/2/2/8

Explanation: Also strong: tolerance-based and documented; marginally more time than B for a formal record.

Penalty rationale: Moderate time; balanced and reconstructable.

Mitigation concept: Documented seam/lower-neck tolerance decision.

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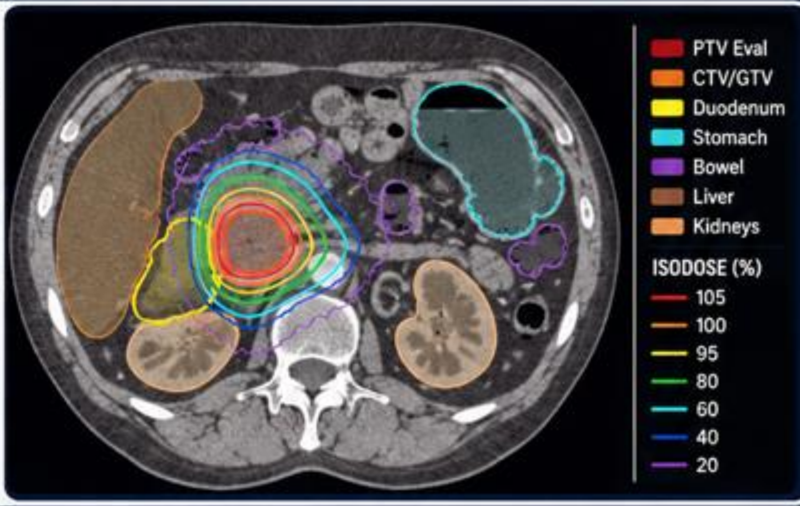
PATIENT CASE FILE • ABD-01

Ms. Dora Duodenum

"The duodenum sets a hard limit — and it moved."

1 PATIENT SNAPSHOT

67-year-old	Pancreatic head SBRT
Unresectable	2/5 Fraction 2
Prescription: 40 Gy / 5 fx	Hard limit: Duodenum D0.5cc
Today's wildcard: Told to fast; had a light meal	Allowable time: 45 minutes



2 WHY SHE'S ON THE ADAPTIVE COUCH

Her stomach and duodenum fill differently each day and abut the high-dose region. Adaptation is required to keep luminal GI within tolerance — and the duodenal D0.5cc is a hard limit, hypersensitive to a millimeter of contour at the interface.

3 THE ROOM TODAY

Adaptive GI-SBRT program: physicist on console, physician reachable, two RTTs. Long sessions are expected for this site — the extra time is clinical, not indulgent.

4 THE CLOCK IS REAL

You get 45 minutes. More than the others — but bowel/stomach motion, OAR proximity, and repeat imaging eat it legitimately. The longer clock is not a free pass.

SAFETY IS OUR PLAN.
TEAMWORK IS OUR POWER.

SPEAK UP
See something. Say something.

SHARE THE PLAN
Say it. Confirm it. Own it.

DOUBLE CHECK
Patient. Plan. Position.

SUPPORT EACH OTHER
Assume good intent. Deliver safe care.

WE ALL WIN
Every patient. Every time.

TURN THE PAGE TO MEET HER SIX DECISIONS. →

1 REFERENCE PLANNING

PLAYER CARD

ABD-01 | Ms. Dora Duodenum

Let's Adapt!

Scenario: A Light Meal Before Pancreas SBRT

Clinical Situation: Ms. Duodenum is on a pancreatic head SBRT course where the duodenum and stomach abut the target. She was instructed to fast but reports a light recent meal, so today's gastric and duodenal filling differ from the fasted simulation state. The reference plan sets a hard duodenum D0.5cc limit but does not define how to handle non-fasting or variable filling.

The hard limit is documented, which can make the team assume the optimizer will simply respect it. But if today's non-fasted anatomy places stomach or duodenum differently at the target interface, the workflow needs a defined fasting-deviation pathway before trusting plan generation.

Decision Prompt: Is the fasted-state plan still applicable today, and what does your local fasting-deviation pathway call for?

Choices

A. Follow the local fasting-deviation protocol: wait and reassess, repeat imaging, or reschedule if the OAR relationship is not acceptable; document.

B. Proceed; the duodenum limit is hard and the optimizer will respect it.

C. Verify gastric/duodenal filling and confirm the limit can be evaluated on today's anatomy before proceeding.

D. Stop and reschedule to restore the fasted state per protocol.

Decision Log: Why did your team choose this option?

Notes:

2 ON-SESSION IMAGING

PLAYER CARD

Let's Adapt!

ABD-01 | Ms. Dora Duodenum

Scenario: The OAR You Can't Quite See

Clinical Situation: On-session CBCT for Ms. Duodenum shows a distended stomach, and the duodenum is poorly distinguished from adjacent bowel near the target. The target itself is visible.

The dose-limiting luminal OAR is exactly the structure the image renders least reliably today. Trusting the optimizer to respect a hard limit on a structure that cannot be confidently localized is the central risk. Ad hoc hydration is not a neutral fix because it can further change stomach or bowel filling.

Decision Prompt: Is the image adequate to localize the dose-limiting duodenum, and if not, what does your protocol call for?

Choices

A. Restart the session, following protocol, if the duodenum cannot be localized.

B. Per protocol, allow brief settling and reassess or repeat imaging to localize the duodenum before planning.

C. Use display/contrast optimization and manually localize the duodenum before planning.

D. Proceed; the target is visible and the optimizer handles the OAR.

Decision Log: Why did your team choose this option?

Notes:

3 OAR CONTOURING

PLAYER CARD

Let's Adapt!

ABD-01 | Ms. Dora Duodenum

Scenario: Millimeters That Decide a Hard Limit

Clinical Situation: Ms. Duodenum's generated duodenum contour is plausible overall, but its luminal boundary near the target is uncertain on several key slices. A 1–2 mm boundary difference at this interface could meaningfully change the later D0.5cc evaluation.

At this step, the team is not yet relying on the final D0.5cc result. The decision is whether the duodenal boundary is accurate enough to approve for downstream optimization, dose calculation, sensitivity review, and plan comparison. A plausible contour is not enough when the hard-limit metric is hypersensitive to the uncertain boundary.

Decision Prompt: How precise must the duodenal boundary be before the team approves it for downstream dose evaluation?

Choices

A. Request a luminal-GI second review before approving the duodenum contour.

B. Accept the generated duodenum contour because it looks plausible overall.

C. Refine the luminal boundary on the key target-facing slices before approving the OAR contour set.

D. Refine the boundary and plan a small-boundary-expansion sensitivity check for downstream dose review.

Decision Log: Why did your team choose this option?

Notes:

4 TARGET CONTOURING

PLAYER CARD

ABD-01 | Ms. Dora Duodenum

Let's Adapt!

Scenario: Coverage Meets the Duodenum

Clinical Situation: Ms. Duodenum's target propagation is reasonable, but the GTV-duodenum interface is where coverage and OAR sparing directly conflict. The interface has shifted with today's stomach/duodenum filling. Every millimeter of GTV at this boundary trades against a hard duodenal limit.

The propagated target may look reasonable, but accepting it without reviewing the interface can hide where coverage was traded for luminal-GI sparing. The team must decide deliberately how the target/OAR priority hierarchy applies at this boundary.

Decision Prompt: How do you set the target at the GTV-duodenum interface without silently sacrificing coverage or breaching the limit?

Choices

A. Hold for physician target/OAR review of the interface before planning.

B. Review the GTV-duodenum interface explicitly and set it per the priority hierarchy, noting the tradeoff.

C. Accept propagation; it balances coverage and OAR automatically.

D. Review the interface, apply the priority hierarchy, and document the coverage/limit tradeoff accepted.

Decision Log: Why did your team choose this option?

Notes:

5 PLAN SELECTION

PLAYER CARD

ABD-01 | Ms. Dora Duodenum

Let's Adapt!

Scenario: Coverage Is King?

Clinical Situation: Ms. Duodenum's adapted plan respects the hard duodenal limit on the refined current-anatomy contours, but it does so by trimming GTV coverage at the duodenal interface. The scheduled/on-session recalculated plan preserves more GTV coverage, but on today's distended anatomy the duodenal D0.5cc is borderline and changes with small differences in the duodenal boundary.

This is not a simple "coverage versus sparing" preference. For pancreatic SBRT, maintaining ablative coverage matters, but a hard luminal-GI limit is also not optional. The team must decide whether the borderline duodenal result is robust enough to accept, whether the interface needs more contour review, or whether the adapted plan's coverage tradeoff is the safer documented choice for today's anatomy.

Decision Prompt: How do you weigh ablative coverage against a borderline hard duodenal limit on today's anatomy?

Choices

A. Pick the scheduled plan; coverage is king for ablative SBRT.

B. Return to contour editing to refine the interface, then regenerate the adapted plan before selecting.

C. Select per the duodenal sensitivity test and secondary dose check, recording the coverage/limit rationale.

D. Pick the adapted plan to respect the hard duodenal limit and document the coverage tradeoff.

Decision Log: Why did your team choose this option?

Notes:

6 DELIVERY / CONFIRMATION IMAGING

PLAYER CARD

ABD-01 | Ms. Dora Duodenum

Let's Adapt!

Scenario: A Plan-Invalidating Drift

Clinical Situation: Confirmation imaging for Ms. Duodenum shows that the duodenum has filled and shifted into the high-dose region since approval. A quick re-measure indicates the hard D0.5cc limit would now be exceeded, so the approved plan no longer reflects a safe luminal-GI relationship.

The session is already long and the plan was approved, so treating now is tempting. But once the re-measure shows the hard limit would be exceeded, the approved plan is no longer valid for delivery.

Decision Prompt: With the hard limit now exceeded, do you treat, settle and re-check, or restart the adaptive workflow before delivery?

Choices

A. Briefly settle and re-verify the duodenal relationship before deciding.

B. Compare the change to the predefined tolerance and document; the re-measure already shows it is exceeded.

C. Restart the adaptive workflow by returning to contour editing and regenerating on current GI anatomy before any delivery.

D. Treat now; the session is already long and the plan was approved.

Decision Log: Why did your team choose this option?

Notes:

ABD-01 Explanation Summary

Ms. Dora Duodenum • Pancreas SBRT / Duodenum at Risk

Patient Context

Clinical Summary	67F, unresectable pancreatic head tumor, SBRT 40 Gy/5 fx, fraction 2. Duodenum and stomach abut the target; the patient was instructed to fast but reports a light recent meal.
Clinic / Coverage Model	Adaptive GI-SBRT program; physicist on console, physician reachable, two RTTs; long sessions are expected for this site.
Adaptive Rationale	Stomach/duodenum filling and bowel position change daily and abut the high-dose region; adaptation is required to keep luminal GI within tolerance.
Allowable Time	45 minutes
Time Consequence	Overage matters, but abdominal cases legitimately need more time: bowel/stomach motion, OAR proximity, and repeat imaging/review are inherent, not a luxury.

Core Lesson

For pancreas SBRT, a hard luminal-GI limit requires reliable GI localization, contour sensitivity awareness, and an explicit coverage-versus-limit rationale.

Overall Handling Approach

Make uncertainty visible, verify the anatomy that drives the decision, use documented thresholds or priority rules, and record the rationale before delivery.

1. Reference Planning — A Light Meal Before Pancreas SBRT

Why it matters: The reference/treatment directive issue sets the decision rule before the patient is deep into the adaptive session.
General handling approach: Apply or define the missing rule early, document the working basis, and escalate only when the ambiguity affects today's treatment decision.

2. On-Session Imaging — The OAR You Can't Quite See

Why it matters: Image adequacy must be judged against the specific decision the image must support, not only global image quality.
General handling approach: Review the decision-driving anatomy directly. Reacquire or mitigate only when the limitation prevents reliable contouring or plan evaluation.

3. OAR Contouring — Millimeters That Decide a Hard Limit

Why it matters: At this stage the team is approving OAR/influencer anatomy for downstream optimization and plan comparison, not relying on a final DVH.
General handling approach: Focus edits on the OAR/influencer regions that can change optimization, dose calculation, or plan comparison; use second review when uncertainty remains clinically material.

4. Target Contouring — Coverage Meets the Duodenum

Why it matters: A plausible propagated target does not by itself prove that clinical target intent has been preserved.
General handling approach: Reconcile propagated targets to reference intent, diagnostic anatomy, prior fractions, or a documented rule before planning.

5. Plan Selection — Coverage Is King?

Why it matters: Scheduled/on-session recalculated and adapted plans can both appear acceptable while resolving different clinical risks.
General handling approach: Select based on documented target/OAR priorities, composite checks, sensitivity tests, and secondary dose review rather than the most attractive single metric.

6. Delivery / Confirmation Imaging — A Plan-Invalidating Drift

Why it matters: Confirmation imaging should be governed by predefined thresholds, stability checks, or re-verification rather than visual impression or time pressure alone.
General handling approach: Treat only when the approved geometry remains within threshold or can be restored and re-verified; otherwise re-adapt, restart, delay, or cancel according to policy.

ABD-01 Scoring Sheet

Ms. Dora Duodenum • Allowable time: 45 minutes

Workflow step	Choice	Safety penalty	Time penalty	Whammy?	Notes
1. Reference Planning	____	____	____	◻ Yes ◻ No	
2. On-Session Imaging	____	____	____	◻ Yes ◻ No	
3. OAR Contouring	____	____	____	◻ Yes ◻ No	
4. Target Contouring	____	____	____	◻ Yes ◻ No	
5. Plan Selection	____	____	____	◻ Yes ◻ No	
6. Delivery / Confirmation Imaging	____	____	____	◻ Yes ◻ No	

Total safety penalties	__________	**Total time used**	__________ min
Time overage	max(0, time - allowable) = ________	**Whammy resolution**	◻ +10 min choose again ◻ +15 safety
Final score	100 - safety penalties - time overage = ________		
Team	________________________	**Facilitator**	________________________

Scoring reminder: Final score = 100 - safety penalties - time overage. If a whammy is selected, add 10 minutes and choose again, or keep the choice and add 15 safety penalty.

ABD-01 Answer Key • 1. Reference Planning

Ms. Dora Duodenum

Scenario: A Light Meal Before Pancreas SBRT

Choice A: Follow the local fasting-deviation protocol: wait and reassess, repeat imaging, or reschedule if the OAR relationship is not acceptable; document.

Scoring: Safety 2; Time 8 min; no cascade; Whammy: No; S/O/D/RPN 2/2/2/8

Explanation: Best: following the local fasting-deviation protocol (wait/reassess, repeat imaging, or reschedule) is the durable, defensible approach; it legitimately takes time.

Penalty rationale: Moderate time; appropriate for this site.

Mitigation concept: Protocol-driven fasting-deviation handling.

Choice B: Proceed; the duodenum limit is hard and the optimizer will respect it.

Scoring: Safety 11; Time 1 min; +8 safety cascade; Whammy: No; S/O/D/RPN 5/3/4/60

Explanation: Tempting because the limit is hard, but a hard limit means nothing if today's distended GI sits where the optimizer cannot both cover and spare.

Penalty rationale: High safety, low time: filling deviation unassessed.

Mitigation concept: Defined fasting-deviation policy.

Choice C: Verify gastric/duodenal filling and confirm the limit can be evaluated on today's anatomy before proceeding.

Scoring: Safety 4; Time 4 min; no cascade; Whammy: No; S/O/D/RPN 3/3/3/27

Explanation: Reasonable: verifying filling and re-confirming the limit on today's anatomy is the core adaptive check, though it stops short of a protocol path.

Penalty rationale: Low-moderate: inherent to GI-SBRT.

Mitigation concept: Filling verification + on-anatomy limit check.

Choice D: Stop and reschedule to restore the fasted state per protocol.

Scoring: Safety 2; Time 14 min; +2 min cascade; Whammy: No; S/O/D/RPN 2/2/2/8

Explanation: Rescheduling restores the simulated state but is heavy if today's anatomy is treatable within limits per protocol.

Penalty rationale: Highest time; reserve for untreatable filling.

Mitigation concept: Reschedule only when limits cannot be met.

ABD-01 Answer Key • 2. On-Session Imaging

Ms. Dora Duodenum

Scenario: The OAR You Can't Quite See

Choice A: Restart the session, following protocol, if the duodenum cannot be localized.

Scoring: Safety 2; Time 14 min; +3 min cascade; Whammy: No; S/O/D/RPN 2/2/2/8

Explanation: A protocol restart is thorough but heavy if lesser, protocol-consistent measures suffice.

Penalty rationale: Highest time; reserve for non-mitigable conspicuity.

Mitigation concept: Graded measures before full restart.

Choice B: Per protocol, allow brief settling and reassess or repeat imaging to localize the duodenum before planning.

Scoring: Safety 2; Time 9 min; no cascade; Whammy: No; S/O/D/RPN 2/2/2/8

Explanation: Best: per protocol, brief settling and reassessment (or repeat imaging) gives a reliable duodenal image -- appropriately time-consuming, not a luxury.

Penalty rationale: Moderate time; inherent to the site.

Mitigation concept: Protocol settling/reassessment + targeted reimaging.

Choice C: Use display/contrast optimization and manually localize the duodenum before planning.

Scoring: Safety 4; Time 5 min; no cascade; Whammy: No; S/O/D/RPN 3/3/3/27

Explanation: Reasonable: display optimization and manual localization can salvage duodenal definition, though it does not address filling itself.

Penalty rationale: Low-moderate: legitimate added time.

Mitigation concept: Display optimization + manual localization.

Choice D: Proceed; the target is visible and the optimizer handles the OAR.

Scoring: Safety 12; Time 1 min; +9 safety cascade; Whammy: No; S/O/D/RPN 5/3/4/60

Explanation: Tempting because the target images well, but trusting the optimizer for an OAR you cannot reliably see is the central abdominal failure mode.

Penalty rationale: Highest safety, low time: limiting OAR unlocalized.

Mitigation concept: Reliable luminal-GI localization before planning.

ABD-01 Answer Key • 3. OAR Contouring

Ms. Dora Duodenum

Scenario: Millimeters That Decide a Hard Limit

Choice A: Request a luminal-GI second review before approving the duodenum contour.

Scoring: Safety 3; Time 8 min; +2 min cascade; Whammy: No; S/O/D/RPN 2/2/2/8

Explanation: Safe when the duodenal boundary is difficult to identify or policy requires luminal-GI second review. Slower than targeted refinement.

Penalty rationale: Low safety penalty because luminal-GI uncertainty is escalated before planning.

Mitigation concept: Second-reviewer escalation for luminal-GI boundary uncertainty

Choice B: Accept the generated duodenum contour because it looks plausible overall.

Scoring: Safety 12; Time 1 min; +9 safety cascade; Whammy: No; S/O/D/RPN 5/4/5/100

Explanation: Tempting shortcut: the duodenum looks plausible overall, but the target-facing luminal boundary is uncertain and 1-2 mm can change later D0.5cc interpretation.

Penalty rationale: High safety penalty because the hard-limit boundary is approved on plausibility alone.

Mitigation concept: Boundary precision proportional to hard-limit sensitivity

Choice C: Refine the luminal boundary on the key target-facing slices before approving the OAR contour set.

Scoring: Safety 4; Time 5 min; no cascade; Whammy: No; S/O/D/RPN 3/3/2/18

Explanation: Best focused correction: refine the target-facing duodenal boundary before approving the OAR set so downstream optimization and dose review are meaningful.

Penalty rationale: Very low safety penalty because the clinically sensitive boundary is corrected before planning.

Mitigation concept: Refine target-facing luminal boundary before planning

Choice D: Refine the boundary and plan a small-boundary-expansion sensitivity check for downstream dose review.

Scoring: Safety 2; Time 8 min; no cascade; Whammy: No; S/O/D/RPN 2/2/2/8

Explanation: Robust and forward-looking: boundary refinement plus a downstream sensitivity check helps the team understand whether the duodenal limit is fragile.

Penalty rationale: Low safety penalty because uncertainty is corrected and bounded.

Mitigation concept: Boundary refinement plus sensitivity check

ABD-01 Answer Key • 4. Target Contouring

Ms. Dora Duodenum

Scenario: Coverage Meets the Duodenum

Choice A: Hold for physician target/OAR review of the interface before planning.

Scoring: Safety 3; Time 11 min; +2 min cascade; Whammy: No; S/O/D/RPN 2/2/2/8

Explanation: Physician interface review fits true conflict but is not a daily default if a hierarchy exists.

Penalty rationale: High time; reserve for genuine conflict.

Mitigation concept: Pre-agreed interface priority hierarchy.

Choice B: Review the GTV-duodenum interface explicitly and set it per the priority hierarchy, noting the tradeoff.

Scoring: Safety 3; Time 5 min; no cascade; Whammy: No; S/O/D/RPN 3/3/3/27

Explanation: Best: explicit interface setting per the hierarchy makes the tradeoff deliberate and visible, at modest cost.

Penalty rationale: Low-moderate: legitimate focused time.

Mitigation concept: Hierarchy-based interface setting.

Choice C: Accept propagation; it balances coverage and OAR automatically.

Scoring: Safety 10; Time 1 min; +6 safety cascade; Whammy: No; S/O/D/RPN 5/3/4/60

Explanation: Tempting because 'automatic balance' sounds safe, but it hides where coverage was traded for sparing at the decisive interface.

Penalty rationale: High safety, low time: interface tradeoff invisible.

Mitigation concept: Explicit interface review against priority hierarchy.

Choice D: Review the interface, apply the priority hierarchy, and document the coverage/limit tradeoff accepted.

Scoring: Safety 2; Time 8 min; no cascade; Whammy: No; S/O/D/RPN 2/2/2/8

Explanation: Also strong: documents the accepted tradeoff at the interface; slightly more process than B.

Penalty rationale: Moderate time; reconstructable decision.

Mitigation concept: Interface review + documented tradeoff.

ABD-01 Answer Key • 5. Plan Selection

Ms. Dora Duodenum

Scenario: Coverage Is King?

Choice A: Pick the scheduled plan; coverage is king for ablative SBRT.

Scoring: Safety 11; Time 1 min; +5 safety cascade; Whammy: YES - Add 10 minutes and choose again, or keep this choice and add 15 safety penalty.; S/O/D/RPN 5/3/4/60

Explanation: The tempting shortcut: 'coverage is king' is a reflex that here overrides a hard luminal limit -- the highest-consequence GI-SBRT error.

Penalty rationale: Whammy: prioritizing ablative coverage past a borderline hard duodenal limit.

Mitigation concept: Hard-limit-first selection for luminal GI.

Choice B: Return to contour editing to refine the interface, then regenerate the adapted plan before selecting.

Scoring: Safety 3; Time 12 min; +2 min cascade; Whammy: No; S/O/D/RPN 2/2/2/8

Explanation: Returning to editing to refine the interface is operationally valid but time-heavy; legitimate for this site when needed.

Penalty rationale: High time; reserve for true coverage/limit conflict.

Mitigation concept: Return-to-editing when both can be met with effort.

Choice C: Select per the duodenal sensitivity test and secondary dose check, recording the coverage/limit rationale.

Scoring: Safety 2; Time 7 min; no cascade; Whammy: No; S/O/D/RPN 2/2/2/8

Explanation: Also strong: ties selection to the sensitivity test and secondary check; marginally more process than B.

Penalty rationale: Moderate time; verified decision trail.

Mitigation concept: Sensitivity + secondary-check driven selection.

Choice D: Pick the adapted plan to respect the hard duodenal limit and document the coverage tradeoff.

Scoring: Safety 3; Time 4 min; no cascade; Whammy: No; S/O/D/RPN 3/3/3/27

Explanation: Best: respecting the hard limit with a documented coverage tradeoff is the defensible default and quick.

Penalty rationale: Low combined penalty: limit-first, recorded.

Mitigation concept: Hard-limit-first with documented tradeoff.

ABD-01 Answer Key • 6. Delivery / Confirmation Imaging

Ms. Dora Duodenum

Scenario: A Plan-Invalidating Drift

Choice A: Briefly settle and re-verify the duodenal relationship before deciding.

Scoring: Safety 6; Time 5 min; no cascade; Whammy: No; S/O/D/RPN 3/3/3/27

Explanation: Reasonable but insufficient here: a settle and re-verify is sensible for borderline change, but the re-measure already shows the hard limit exceeded, so a brief settle will not make the approved plan deliverable.

Penalty rationale: Moderate: under-responds to a confirmed breach.

Mitigation concept: Settle helps only for borderline, not confirmed, breaches.

Choice B: Compare the change to the predefined tolerance and document; the re-measure already shows it is exceeded.

Scoring: Safety 5; Time 6 min; no cascade; Whammy: No; S/O/D/RPN 3/2/2/12

Explanation: Near-miss: the tolerance comparison is the right instinct and documenting is good, but the facts already answer it -- the change is out of tolerance, so this cannot end in delivery of the approved plan.

Penalty rationale: Moderate: correct framework, but the data already mandate re-derivation.

Mitigation concept: Apply the tolerance result decisively when it is exceeded.

Choice C: Restart the adaptive workflow by returning to contour editing and regenerating on current GI anatomy before any delivery.

Scoring: Safety 1; Time 12 min; no cascade; Whammy: No; S/O/D/RPN 2/2/1/4

Explanation: Best: with the hard duodenal limit now exceeded on today's anatomy, the approved plan is invalid, so returning to contour editing and regenerating on current GI anatomy before any delivery is the expensive but correct action.

Penalty rationale: Realistic high time, lowest safety: re-derivation is genuinely warranted.

Mitigation concept: Re-derive when the hard limit is exceeded on current anatomy.

Choice D: Treat now; the session is already long and the plan was approved.

Scoring: Safety 12; Time 1 min; no cascade; Whammy: No; S/O/D/RPN 5/3/4/60

Explanation: Tempting because of session fatigue, but treating when the re-measure shows the hard limit would be exceeded is the gate failure this step exists to prevent.

Penalty rationale: High safety, low time: a hard-limit breach delivered anyway.

Mitigation concept: Do not deliver a plan that breaches a hard luminal-GI limit.

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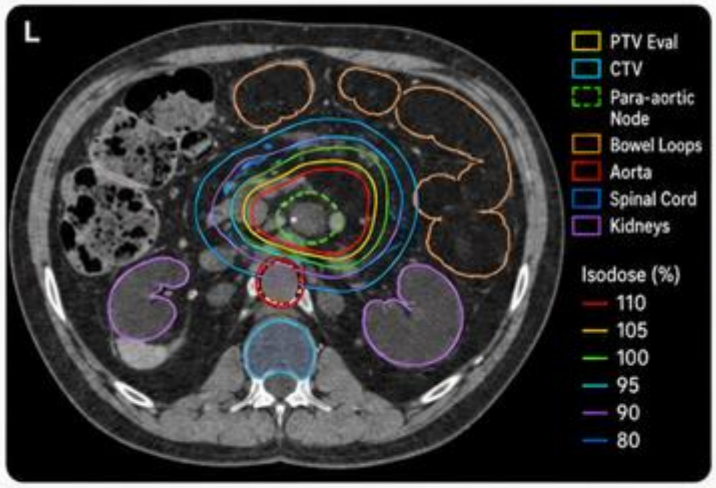
PATIENT CASE FILE • ABD-02

Mr. Gus Gasbubble

"The bowel loop won't hold still."

1 PATIENT SNAPSHOT

59-year-old man	Para-aortic node SBRT
Oligometastatic	Fraction 3
Prescription: 35 Gy / 5 fx	Daily-limiting OAR: Mobile small bowel
Today's wildcard: A loop migrated toward high dose	Allowable time: 45 minutes



2 WHY HE'S ON THE ADAPTIVE COUCH

Small-bowel position is the daily-limiting factor near his para-aortic node. Adaptation exists specifically to account for unpredictable loop migration — but deformable propagation is least reliable exactly where loops migrate toward the target.

3 THE ROOM TODAY

Adaptive abdominal SBRT program: physicist on console, physician reachable, two RTTs. The team knows deformable bowel propagation is unreliable — the question is when to distrust it.

4 THE CLOCK IS REAL

You get 45 minutes. Bowel-motion assessment, deformable distrust, and repeat imaging make the longer session clinically necessary — not a luxury.

SAFETY IS OUR PLAN.
TEAMWORK IS OUR POWER.

SPEAK UP
See something. Say something.

SHARE THE PLAN
Say it. Confirm it. Own it.

DOUBLE CHECK
Patient. Plan. Position.

SUPPORT EACH OTHER
Assume good intent. Deliver safe care.

WE ALL WIN
Every patient. Every time.

TURN THE PAGE TO MEET HIS SIX DECISIONS.

1 REFERENCE PLANNING

PLAYER CARD

ABD-02 | Mr. Gus Gasbubble

Let’s Adapt!

Scenario: When to Distrust the Deformable Result

Clinical Situation: Mr. Gasbubble is on an oligometastatic para-aortic node SBRT course where small bowel position is the daily-limiting factor. Today’s on-session anatomy shows a bowel loop closer to the high-dose region than on prior fractions. The reference workflow uses deformable propagation for bowel, but the directive does not define when migrated bowel should trigger manual verification.

On many fractions, deformable propagation may provide a reasonable starting contour. Today is different because the loop that moved is also the loop closest to the target. If the deformable result is wrong by only a few millimeters at that target-facing wall, the downstream bowel-dose evaluation and plan choice could change. The team must decide whether today’s anatomy still falls within routine deformable-propagation review, or whether the migrated loop is a trigger for manual verification before planning.

Decision Prompt: When is the deformable bowel result trustworthy, and how do you decide whether to verify it manually today?

Choices

A. Use deformable propagation as a starting point but manually verify migrated loops and request a defined verification trigger.

B. Apply a defined trigger for manual bowel verification, verify the migrated loop, and document.

C. Trust the deformable bowel result; it is the specified method.

D. Return to contour editing for manual bowel delineation of the target-facing region before regenerating the plan.

Decision Log: Why did your team choose this option?

Notes:

PLAYER CARD

2 ON-SESSION IMAGING

Let's Adapt!

ABD-02 | Mr. Gus Gasbubble

Scenario: Blur on the Limiting Distance

Clinical Situation: On-session CBCT for Mr. Gasbubble shows the migrated bowel loop, but peristalsis blur and gas make the loop's nearest approach to the target uncertain. The gas also affects local density representation.

The dose-limiting distance is the loop-to-target relationship, and that is exactly what the image blurs. Trusting the optimizer to keep bowel dose acceptable rests on a measurement the team cannot make cleanly unless the image is adequate or the uncertainty is deliberately estimated and documented.

Decision Prompt: Is the image adequate to measure the loop-to-target distance and represent density, and if not, what do you do?

Choices

A. Proceed; the loop is visible and the optimizer will keep bowel dose down.

B. Allow brief settling or anti-peristaltic measures per protocol, reacquire, and re-measure the approach.

C. Restart the session to obtain a clearer bowel image.

D. Conservatively estimate the loop's nearest approach, then document the gas/density and distance uncertainty..

Decision Log: Why did your team choose this option?

Notes:

3 OAR CONTOURING

PLAYER CARD

ABD-02 | Mr. Gus Gasbubble

Let’s Adapt!

Scenario: A Wall a Few Millimeters Off

Clinical Situation: The propagated bowel loop for Mr. Gasbubble is roughly in the right location, but the wall nearest the target is visibly mispositioned by a few millimeters compared with the bowel loop on today’s image. That target-facing wall is the boundary that will drive later bowel max-dose evaluation and plan comparison.

At this step, the team is not yet relying on the final bowel max-dose result. The decision is whether the bowel wall contour is accurate enough to approve for downstream optimization, dose calculation, sensitivity testing, and plan selection. A few millimeters may determine whether the plan is safe.

Decision Prompt: Do you approve the propagated bowel loop, or correct the target-facing wall before allowing planning to proceed?

Choices

A. Accept the propagated bowel loop because it is roughly in the right location.

B. Request a bowel second-review before approving the OAR contour set.

C. Reposition the target-facing wall and plan a small-loop-expansion sensitivity check for downstream dose review.

D. Manually reposition the target-facing wall on today’s image before approving the OAR contour set.

Decision Log: Why did your team choose this option?

Notes:

4 TARGET CONTOURING

PLAYER CARD

ABD-02 | Mr. Gus Gasbubble

Let's Adapt!

Scenario: Trim, Hold, or Address the Loop

Clinical Situation: Mr. Gasbubble's target propagation is stable, but the node's high-dose margin facing the migrated loop is where coverage and bowel sparing conflict today. The conflict is positional: the node is the same target, but a bowel loop has moved closer.

Trimming the node margin facing the loop spares bowel and can look reasonable, but it may quietly under-cover an oligometastatic target intended for ablation. The team must decide whether to adjust the target, manage the bowel/plan relationship, or escalate the target/OAR conflict.

Decision Prompt: Do you trim the margin facing the loop, hold it and manage bowel another way, or escalate the conflict?

Choices

A. Hold for physician target/OAR review of the margin-loop conflict.

B. Trim the node margin facing the loop; bowel sparing comes first.

C. Decide per the priority hierarchy on today's verified geometry and document the tradeoff.

D. Hold the node margin and manage bowel via the plan/loop position, documenting the approach.

Decision Log: Why did your team choose this option?

Notes:

5 PLAN SELECTION

PLAYER CARD

ABD-02 | Mr. Gus Gasbubble

Let’s Adapt!

Scenario: Coverage vs. a Borderline Bowel Limit

Clinical Situation: Mr. Gasbubble’s adapted plan keeps bowel under its limit by trimming node coverage near the migrated loop. The scheduled/on-session recalculated plan covers the node fully, but the bowel max-dose is borderline on today’s loop position, and the loop may move again.

Full ablative node coverage is the goal, which makes the scheduled plan tempting. However, a borderline bowel limit on a mobile loop is not stable reassurance. The plan-selection rationale must explicitly account for bowel sensitivity, motion uncertainty, and the accepted coverage tradeoff.

Decision Prompt: How do you weigh ablative node coverage against a borderline bowel limit on a loop that may move again?

Choices

A. Pick the scheduled plan; full ablative node coverage is the goal.

B. Pick the adapted plan to keep bowel within limit and document the coverage tradeoff.

C. Select per the bowel sensitivity test and secondary check, recording the coverage/limit rationale.

D. Return to contour editing to refine the loop/margin, then regenerate the adapted plan before selecting.

Decision Log: Why did your team choose this option?

Notes:

6 DELIVERY / CONFIRMATION IMAGING

PLAYER CARD

ABD-02 | Mr. Gus Gasbubble

Let's Adapt!

Scenario: A Loop That Won't Hold Still

Clinical Situation: Confirmation imaging for Mr. Gasbubble shows the bowel loop has migrated substantially into the high-dose region since approval, well beyond the predefined tolerance, and a quick re-measure shows it is still moving. The approved plan no longer represents the bowel relationship.

Time pressure tempts a quick fix, but the migration is large and ongoing. Treating is unsafe, and immediately replanning on this instant's anatomy may simply chase a loop that has not stabilized.

Decision Prompt: With a large, still-moving, out-of-tolerance loop, do you treat, restart immediately, or stop and reassess for stability?

Choices

A. Do not treat. Pause and reassess for stability; restart adaptation only if the bowel relationship stabilizes, otherwise delay or cancel the fraction.

B. Confirm the out-of-tolerance shift and re-measure once; if still out of tolerance, do not deliver the approved plan.

C. Treat now; the plan was approved and bowel motion is expected.

D. Restart adaptation immediately on current anatomy because the loop is out of tolerance.

Decision Log: Why did your team choose this option?

Notes:

ABD-02 Explanation Summary

Mr. Gus Gasbubble • Para-Aortic SBRT / Mobile Bowel

Patient Context

Clinical Summary	59M, oligometastatic para-aortic node, SBRT 35 Gy/5 fx, fraction 3. Small bowel loops migrate near the target; today a loop sits closer to the high-dose region than on prior fractions.
Clinic / Coverage Model	Adaptive abdominal SBRT program; physicist on console, physician reachable, two RTTs; deformable bowel propagation is known to be unreliable.
Adaptive Rationale	Small-bowel position is the daily-limiting factor; adaptation exists specifically to account for unpredictable loop migration near the node.
Allowable Time	45 minutes
Time Consequence	Overage matters, but bowel-motion assessment, deformable-propagation distrust, and repeat imaging make longer sessions clinically necessary, not indulgent.

Core Lesson

Mobile bowel is the daily-limiting factor. Deformable propagation must be distrusted when a migrated loop is near target, and delivery requires a stable bowel relationship.

Overall Handling Approach

Make uncertainty visible, verify the anatomy that drives the decision, use documented thresholds or priority rules, and record the rationale before delivery.

1. Reference Planning — When to Distrust the Deformable Result

Why it matters: The reference/treatment directive issue sets the decision rule before the patient is deep into the adaptive session.
General handling approach: Apply or define the missing rule early, document the working basis, and escalate only when the ambiguity affects today's treatment decision.

2. On-Session Imaging — Blur on the Limiting Distance

Why it matters: Image adequacy must be judged against the specific decision the image must support, not only global image quality.
General handling approach: Review the decision-driving anatomy directly. Reacquire or mitigate only when the limitation prevents reliable contouring or plan evaluation.

3. OAR Contouring — A Wall a Few Millimeters Off

Why it matters: At this stage the team is approving OAR/influencer anatomy for downstream optimization and plan comparison, not relying on a final DVH.
General handling approach: Focus edits on the OAR/influencer regions that can change optimization, dose calculation, or plan comparison; use second review when uncertainty remains clinically material.

4. Target Contouring — Trim, Hold, or Address the Loop

Why it matters: A plausible propagated target does not by itself prove that clinical target intent has been preserved.
General handling approach: Reconcile propagated targets to reference intent, diagnostic anatomy, prior fractions, or a documented rule before planning.

5. Plan Selection — Coverage vs. a Borderline Bowel Limit

Why it matters: Scheduled/on-session recalculated and adapted plans can both appear acceptable while resolving different clinical risks.
General handling approach: Select based on documented target/OAR priorities, composite checks, sensitivity tests, and secondary dose review rather than the most attractive single metric.

6. Delivery / Confirmation Imaging — A Loop That Won't Hold Still

Why it matters: Confirmation imaging should be governed by predefined thresholds, stability checks, or re-verification rather than visual impression or time pressure alone.
General handling approach: Treat only when the approved geometry remains within threshold or can be restored and re-verified; otherwise re-adapt, restart, delay, or cancel according to policy.

ABD-02 Scoring Sheet

Mr. Gus Gasbubble • Allowable time: 45 minutes

Workflow step	Choice	Safety penalty	Time penalty	Whammy?	Notes
1. Reference Planning	____	____	____	□ Yes □ No	
2. On-Session Imaging	____	____	____	□ Yes □ No	
3. OAR Contouring	____	____	____	□ Yes □ No	
4. Target Contouring	____	____	____	□ Yes □ No	
5. Plan Selection	____	____	____	□ Yes □ No	
6. Delivery / Confirmation Imaging	____	____	____	□ Yes □ No	

Total safety penalties	__________	**Total time used**	__________ min
Time overage	max(0, time - allowable) = ________	**Whammy resolution**	□ +10 min choose again □ +15 safety
Final score	100 - safety penalties - time overage = ________		
Team	________________________	**Facilitator**	________________________

Scoring reminder: Final score = 100 - safety penalties - time overage. If a whammy is selected, add 10 minutes and choose again, or keep the choice and add 15 safety penalty.

ABD-02 Answer Key • 1. Reference Planning

Mr. Gus Gasbubble

Scenario: When to Distrust the Deformable Result

Choice A: Use deformable propagation as a starting point but manually verify migrated loops and request a defined verification trigger.

Scoring: Safety 2; Time 5 min; no cascade; Whammy: No; S/O/D/RPN 2/2/2/8

Explanation: Best: deformable-as-start with manual verification of migrated loops is the realistic best practice, and requesting a trigger starts the durable fix.

Penalty rationale: Low combined penalty: legitimate verification time.

Mitigation concept: Deformable start + manual loop verification + trigger request.

Choice B: Apply a defined trigger for manual bowel verification, verify the migrated loop, and document.

Scoring: Safety 3; Time 6 min; no cascade; Whammy: No; S/O/D/RPN 2/2/2/8

Explanation: Strong: a defined trigger makes verification consistent; marginally more process than B when no trigger exists yet.

Penalty rationale: Moderate time; reproducible and defensible.

Mitigation concept: Trigger-based manual verification.

Choice C: Trust the deformable bowel result; it is the specified method.

Scoring: Safety 11; Time 1 min; +9 safety cascade; Whammy: No; S/O/D/RPN 5/3/4/60

Explanation: Tempting because deformable is the specified method, but mobile loops are exactly where it fails; trusting it blindly near the target is the core hazard.

Penalty rationale: High safety, low time: unreliable propagation trusted.

Mitigation concept: Defined deformable-distrust/verification trigger.

Choice D: Return to contour editing for manual bowel delineation of the target-facing region before regenerating the plan.

Scoring: Safety 2; Time 13 min; +2 min cascade; Whammy: No; S/O/D/RPN 2/2/2/8

Explanation: Full manual delineation is safest for bowel but time-heavy if targeted verification suffices.

Penalty rationale: Highest time; reserve for extensive migration.

Mitigation concept: Targeted verification before full manual.

ABD-02 Answer Key • 2. On-Session Imaging

Mr. Gus Gasbubble

Scenario: Blur on the Limiting Distance

Choice A: Proceed; the loop is visible and the optimizer will keep bowel dose down.

Scoring: Safety 12; Time 1 min; +8 safety cascade; Whammy: No; S/O/D/RPN 5/3/4/60

Explanation: Tempting because the loop is visible, but trusting the optimizer when you cannot measure the limiting distance (or trust the gas-affected density) is the abdominal trap.

Penalty rationale: Highest safety, low time: limiting distance and density unverified.

Mitigation concept: Reliable loop-to-target measurement before planning.

Choice B: Allow brief settling or anti-peristaltic measures per protocol, reacquire, and re-measure the approach.

Scoring: Safety 2; Time 9 min; no cascade; Whammy: No; S/O/D/RPN 2/2/2/8

Explanation: Best: settling/anti-peristaltic measures plus reacquisition give a measurable distance and better density -- appropriately time-consuming.

Penalty rationale: Moderate time; inherent to the site.

Mitigation concept: Settling + targeted reacquisition.

Choice C: Restart the session to obtain a clearer bowel image.

Scoring: Safety 2; Time 14 min; +3 min cascade; Whammy: No; S/O/D/RPN 2/2/2/8

Explanation: A full restart is thorough but heavy if settling/reacquisition suffices.

Penalty rationale: Highest time; reserve for non-mitigable blur.

Mitigation concept: Graded measures before full restart.

Choice D: Bound the loop's nearest approach conservatively and note the gas/density and distance uncertainty.

Scoring: Safety 4; Time 5 min; no cascade; Whammy: No; S/O/D/RPN 3/3/3/27

Explanation: Reasonable: a conservative bound on nearest approach is a legitimate way to proceed under blur, though it does not improve density representation.

Penalty rationale: Low-moderate: documented conservatism.

Mitigation concept: Conservative approach-bounding + documentation.

ABD-02 Answer Key • 3. OAR Contouring

Mr. Gus Gasbubble

Scenario: A Wall a Few Millimeters Off

Choice A: Accept the propagated bowel loop because it is roughly in the right location.

Scoring: Safety 12; Time 1 min; +9 safety cascade; Whammy: YES - Add 10 minutes and choose again, or keep this choice and add 15 safety penalty.; S/O/D/RPN 5/4/5/100

Explanation: Tempting shortcut: the propagated loop is roughly located, but the target-facing bowel wall is visibly mispositioned by a few millimeters, exactly where downstream dose evaluation will matter.

Penalty rationale: High safety penalty because a few-millimeter wall error can determine whether the plan is safe.

Mitigation concept: Do not approve migrated bowel loops based on rough position alone

Choice B: Request a bowel second-review before approving the OAR contour set.

Scoring: Safety 3; Time 8 min; +2 min cascade; Whammy: No; S/O/D/RPN 2/2/2/8

Explanation: Safe when the bowel wall is difficult to judge or policy requires second review. Slower than manually correcting a clear wall-position error.

Penalty rationale: Low safety penalty because material bowel uncertainty is escalated before planning.

Mitigation concept: Second-reviewer escalation for bowel-wall uncertainty

Choice C: Reposition the target-facing wall and plan a small-loop-expansion sensitivity check for downstream dose review.

Scoring: Safety 2; Time 8 min; no cascade; Whammy: No; S/O/D/RPN 2/2/2/8

Explanation: Robust and forward-looking: correct the wall and plan a downstream small-loop-expansion sensitivity check to understand how fragile the bowel limit is.

Penalty rationale: Low safety penalty because the wall is corrected and sensitivity is bounded.

Mitigation concept: Correct target-facing wall plus sensitivity check

Choice D: Manually reposition the target-facing wall on today’s image before approving the OAR contour set.

Scoring: Safety 3; Time 5 min; no cascade; Whammy: No; S/O/D/RPN 3/3/2/18

Explanation: Best focused correction: manually reposition the target-facing bowel wall on today's image before downstream optimization and plan selection.

Penalty rationale: Very low safety penalty because the target-facing wall is corrected before planning.

Mitigation concept: Manually correct target-facing bowel wall before planning

ABD-02 Answer Key • 4. Target Contouring

Mr. Gus Gasbubble

Scenario: Trim, Hold, or Address the Loop

Choice A: Hold for physician target/OAR review of the margin-loop conflict.

Scoring: Safety 3; Time 11 min; +2 min cascade; Whammy: No; S/O/D/RPN 2/2/2/8

Explanation: Physician review fits true conflict but a hierarchy resolves most days without daily escalation.

Penalty rationale: High time; reserve for genuine conflict.

Mitigation concept: Pre-agreed margin-vs-bowel hierarchy.

Choice B: Trim the node margin facing the loop; bowel sparing comes first.

Scoring: Safety 9; Time 1 min; +6 safety cascade; Whammy: No; S/O/D/RPN 4/3/4/48

Explanation: Tempting because bowel sparing feels safe-first, but reflexively trimming the node margin can under-cover an oligometastatic target meant to be ablated.

Penalty rationale: Moderate-high safety: coverage quietly sacrificed.

Mitigation concept: Hierarchy-based margin decision, not reflex.

Choice C: Decide per the priority hierarchy on today's verified geometry and document the tradeoff.

Scoring: Safety 2; Time 8 min; no cascade; Whammy: No; S/O/D/RPN 2/2/2/8

Explanation: Best: applying the hierarchy on verified geometry with documentation is the most defensible.

Penalty rationale: Moderate time; reconstructable decision.

Mitigation concept: Hierarchy on verified geometry + documentation.

Choice D: Hold the node margin and manage bowel via the plan/loop position, documenting the approach.

Scoring: Safety 4; Time 5 min; no cascade; Whammy: No; S/O/D/RPN 3/3/3/27

Explanation: Reasonable: holding the margin while managing bowel through plan/position is a legitimate, documented approach.

Penalty rationale: Low-moderate: deliberate tradeoff.

Mitigation concept: Margin-hold + documented bowel management.

ABD-02 Answer Key • 5. Plan Selection

Mr. Gus Gasbubble

Scenario: Coverage vs. a Borderline Bowel Limit

Choice A: Pick the scheduled plan; full ablative node coverage is the goal.

Scoring: Safety 11; Time 1 min; +5 safety cascade; Whammy: No; S/O/D/RPN 5/3/4/60

Explanation: The tempting shortcut: chasing full coverage past a borderline bowel limit on a mobile loop risks a serious bowel event.

Penalty rationale: Whammy-adjacent risk: ablative coverage overriding a borderline bowel limit (whammy sits at OAR contouring in this packet).

Mitigation concept: Bowel-limit-first selection for mobile loops.

Choice B: Pick the adapted plan to keep bowel within limit and document the coverage tradeoff.

Scoring: Safety 3; Time 4 min; no cascade; Whammy: No; S/O/D/RPN 3/3/3/27

Explanation: Best: keeping bowel within limit with a documented coverage tradeoff is the defensible default and quick.

Penalty rationale: Low combined penalty: limit-first, recorded.

Mitigation concept: Bowel-limit-first with documented tradeoff.

Choice C: Select per the bowel sensitivity test and secondary check, recording the coverage/limit rationale.

Scoring: Safety 2; Time 7 min; no cascade; Whammy: No; S/O/D/RPN 2/2/2/8

Explanation: Also strong: ties selection to the sensitivity test and secondary check; marginally more process than B.

Penalty rationale: Moderate time; verified decision trail.

Mitigation concept: Sensitivity + secondary-check driven selection.

Choice D: Return to contour editing to refine the loop/margin, then regenerate the adapted plan before selecting.

Scoring: Safety 3; Time 12 min; +2 min cascade; Whammy: No; S/O/D/RPN 2/2/2/8

Explanation: Returning to editing to refine the loop/margin is operationally valid but time-heavy; legitimate here when needed.

Penalty rationale: High time; reserve for true coverage/limit conflict.

Mitigation concept: Return-to-editing when both can be met with effort.

ABD-02 Answer Key • 6. Delivery / Confirmation Imaging

Mr. Gus Gasbubble

Scenario: A Loop That Won't Hold Still

Choice A: Do not treat. Pause and reassess for stability; restart adaptation only if the bowel relationship stabilizes, otherwise delay or cancel the fraction.

Scoring: Safety 1; Time 12 min; no cascade; Whammy: No; S/O/D/RPN 2/2/1/4

Explanation: Best: with a large, out-of-tolerance, still-moving loop, the safe path is to NOT treat, pause and reassess for stability, and restart adaptation only if the relationship stabilizes -- otherwise delay or cancel. This avoids both treating a bad plan and chasing unstable anatomy.

Penalty rationale: Realistic high time, lowest safety: the expensive but correct action for moving anatomy.

Mitigation concept: Pause for stability; restart only if stable, else delay/cancel.

Choice B: Confirm the out-of-tolerance shift and re-measure once; if still out of tolerance, do not deliver the approved plan.

Scoring: Safety 4; Time 4 min; no cascade; Whammy: No; S/O/D/RPN 3/2/2/12

Explanation: Reasonable and near-best: confirming the out-of-tolerance shift and re-measuring once correctly establishes that the approved plan must not be delivered. It stops short of saying what to do next, which is why the explicit stability-pause is stronger.

Penalty rationale: Low-moderate: correctly halts delivery but does not resolve next steps.

Mitigation concept: Confirm out-of-tolerance, then do not deliver the approved plan.

Choice C: Treat now; the plan was approved and bowel motion is expected.

Scoring: Safety 12; Time 1 min; no cascade; Whammy: No; S/O/D/RPN 5/3/4/60

Explanation: Treating through a large, out-of-tolerance migration because motion is 'expected' is the gate failure this step exists to prevent.

Penalty rationale: High safety, low time: a clearly out-of-tolerance relationship treated anyway.

Mitigation concept: Do not deliver a plan that no longer represents the bowel relationship.

Choice D: Restart adaptation immediately on current anatomy because the loop is out of tolerance.

Scoring: Safety 4; Time 11 min; no cascade; Whammy: No; S/O/D/RPN 3/2/2/12

Explanation: Near-miss: restarting is better than treating, but restarting immediately on a loop that is still moving re-derives to an instant that will not hold, so stability must be confirmed first.

Penalty rationale: High time; restarting into ongoing motion can repeat the problem.

Mitigation concept: Confirm stability before restarting on current anatomy.